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(54) **COMPOSITIONS AND METHODS FOR
IMPROVING MITOCHONDRIAL FUNCTION**

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continuation of application No. 16/293,789, filed on
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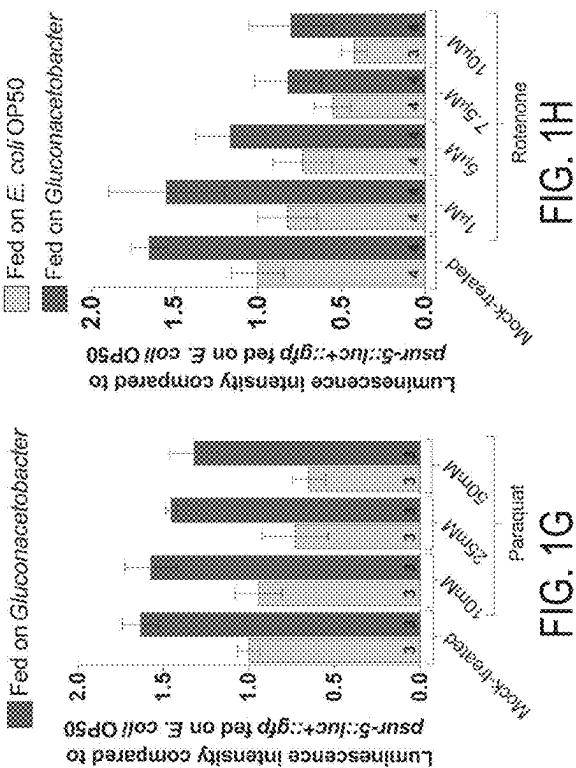
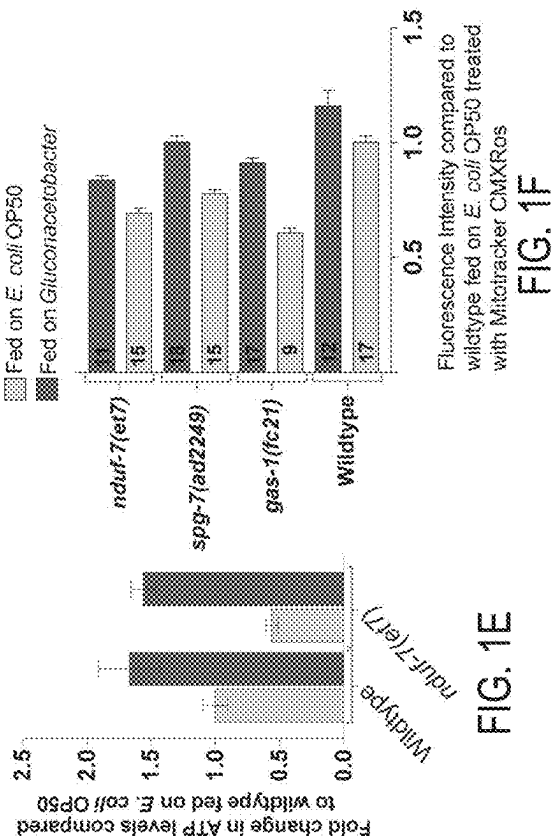
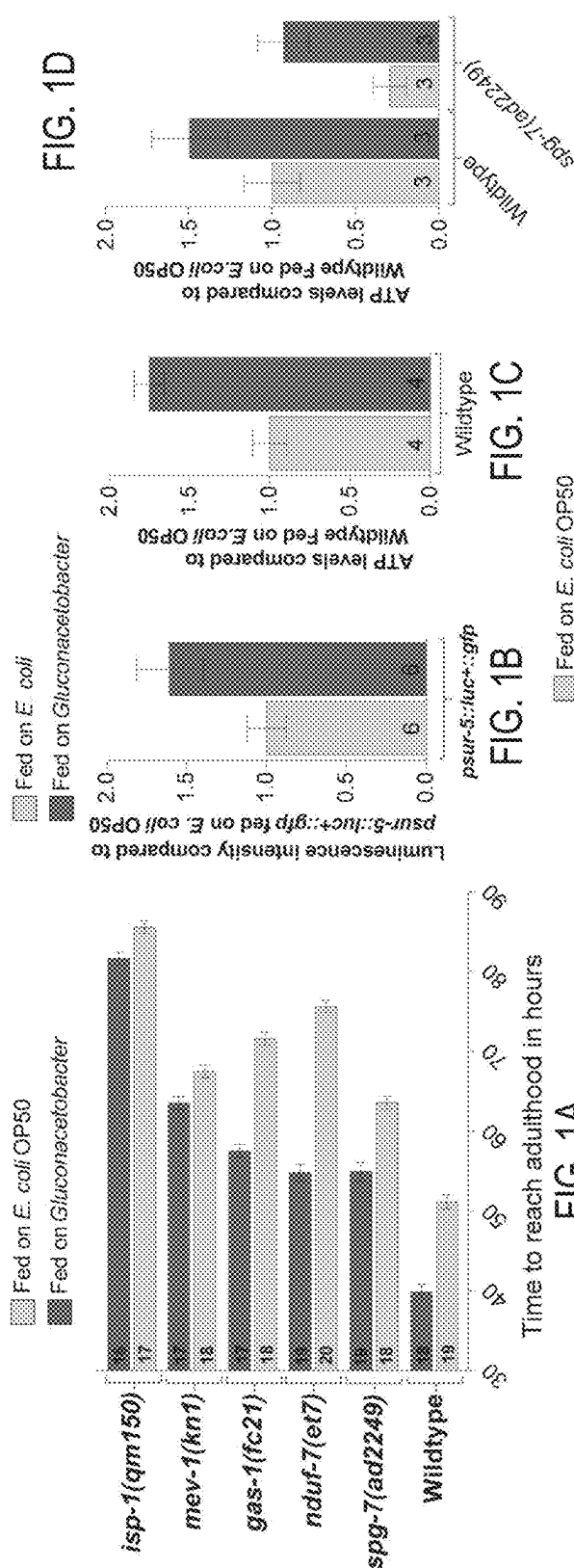
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(2016.05); *A23L 2/52* (2013.01); *A23L 33/135*
(2016.08); *A61K 9/0053* (2013.01); *A61K*
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C12N 1/20 (2013.01); *C12N 9/0006*
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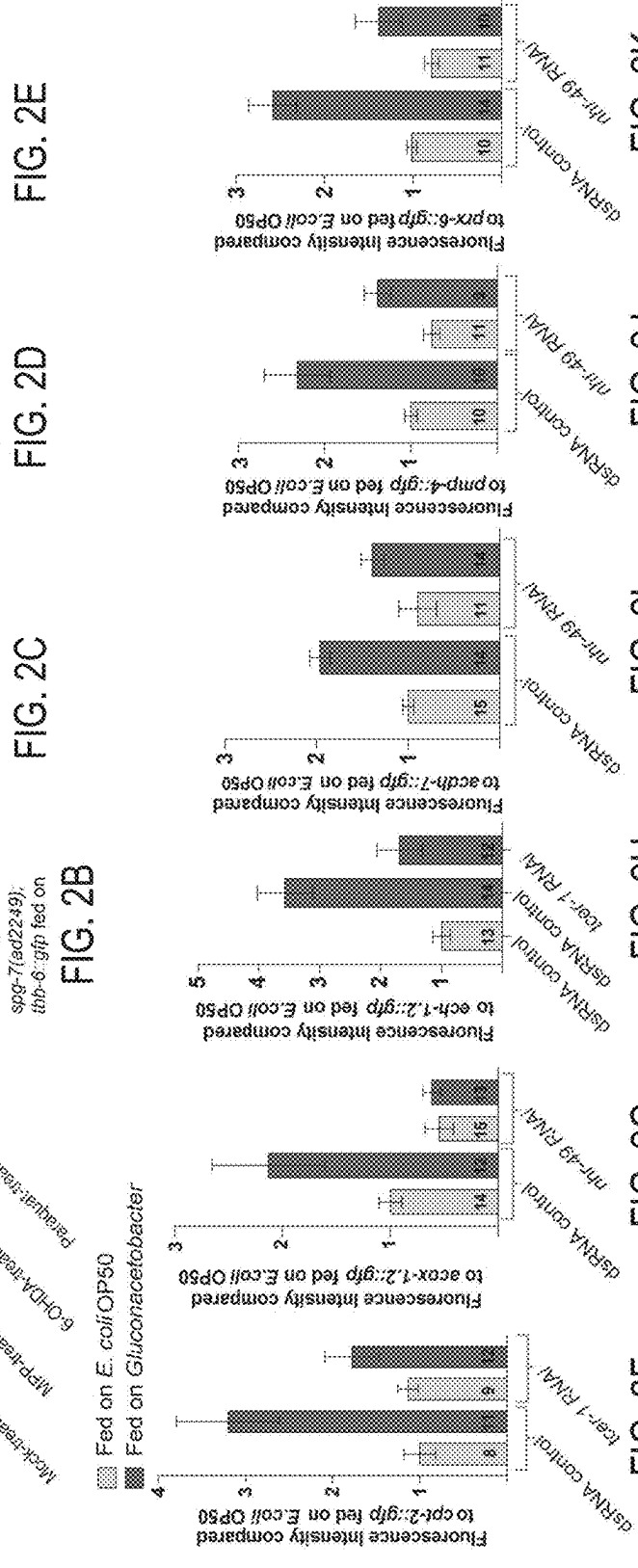
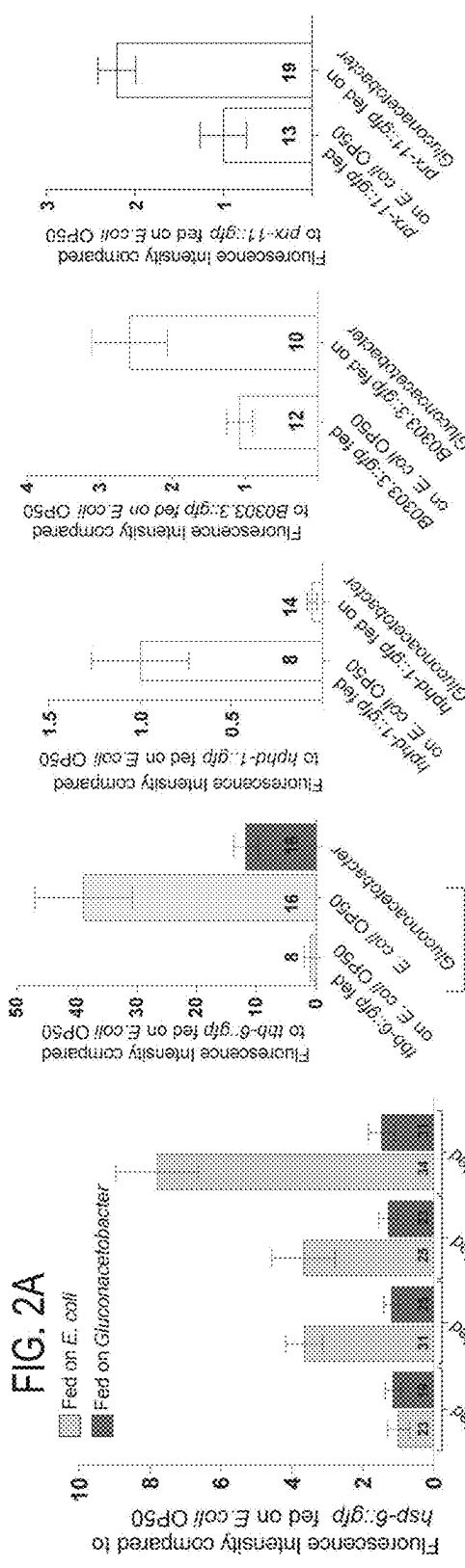
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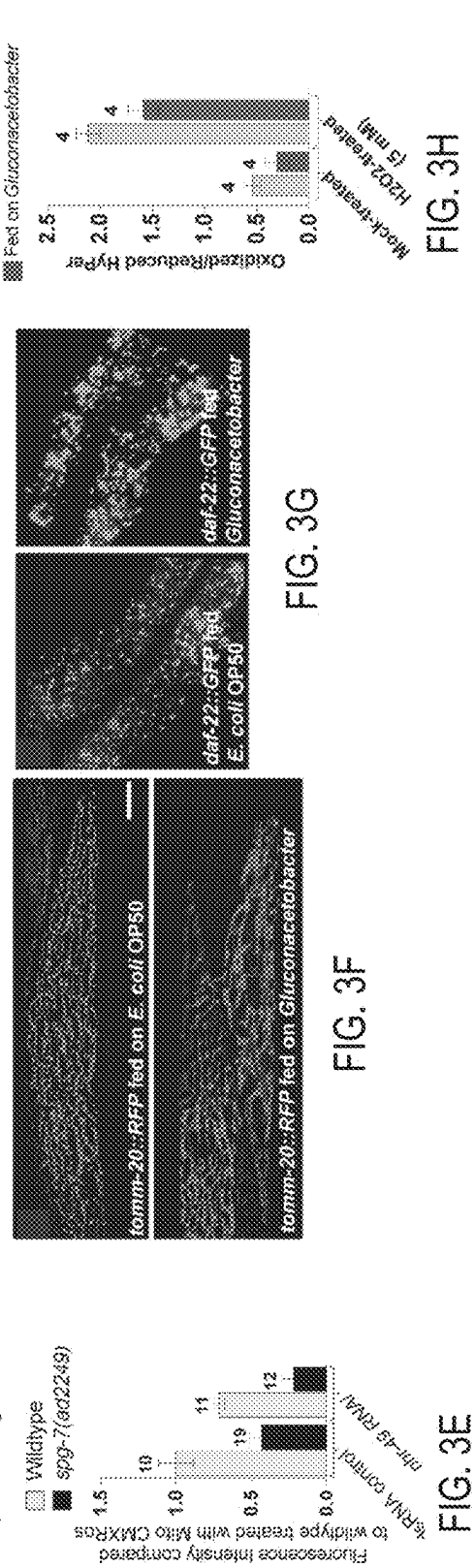
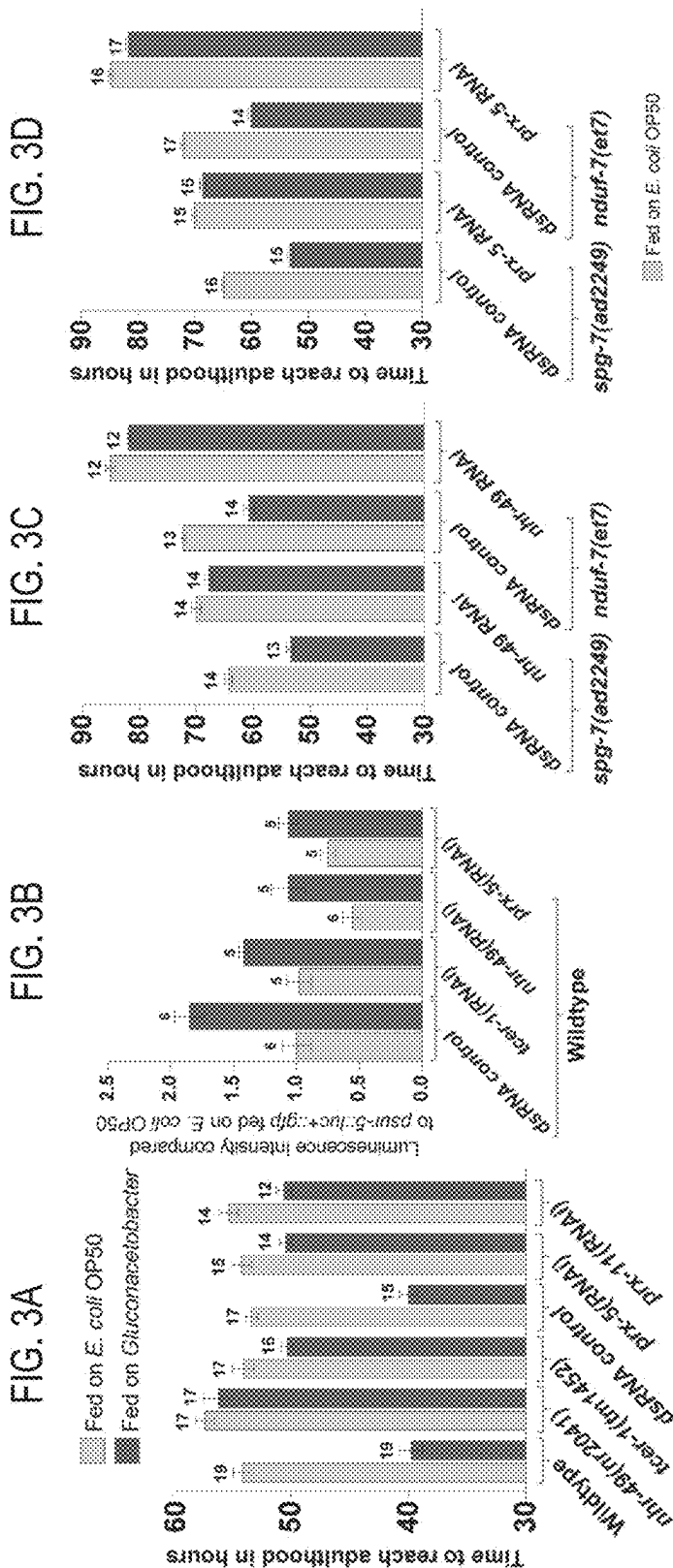
ABSTRACT

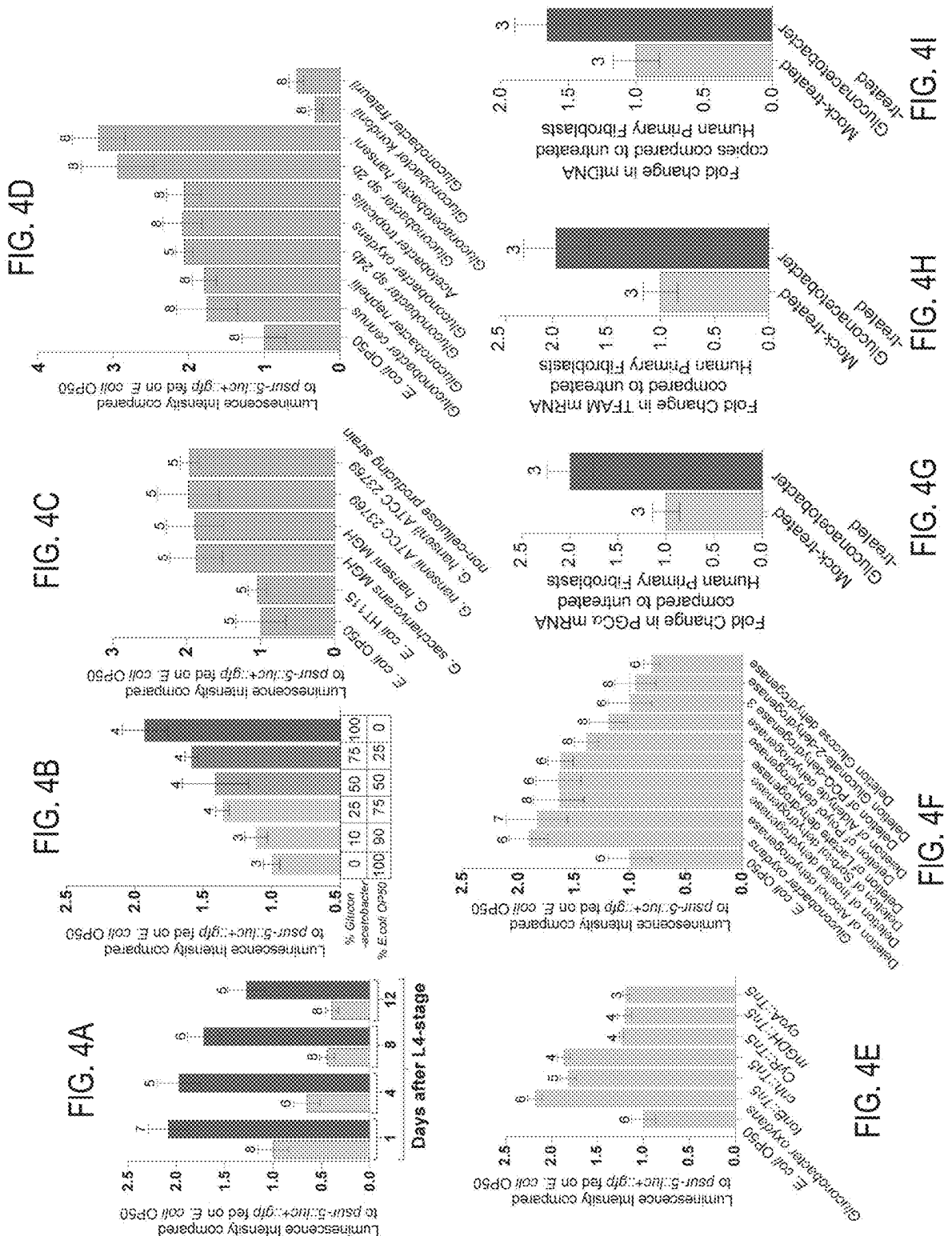
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bacterium or a metabolite thereof for enhancing mitochon-
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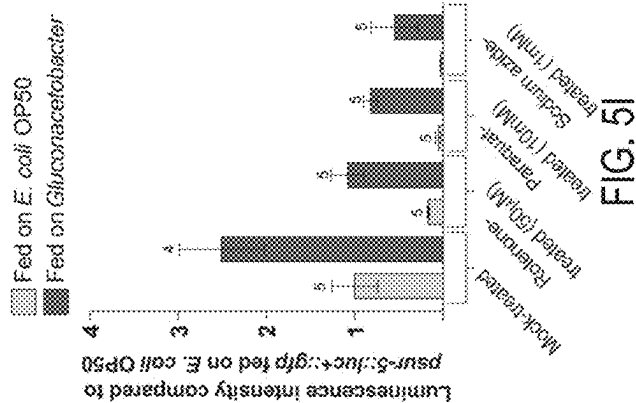
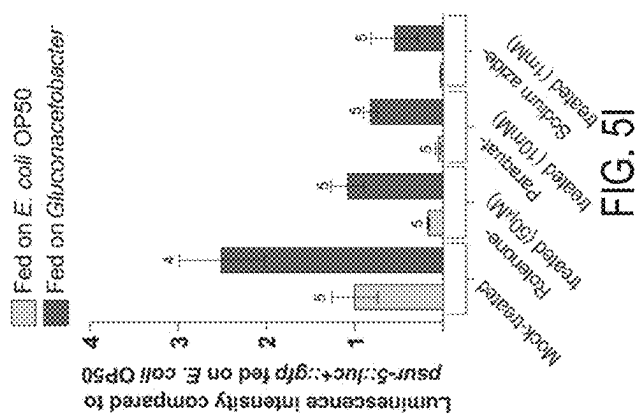
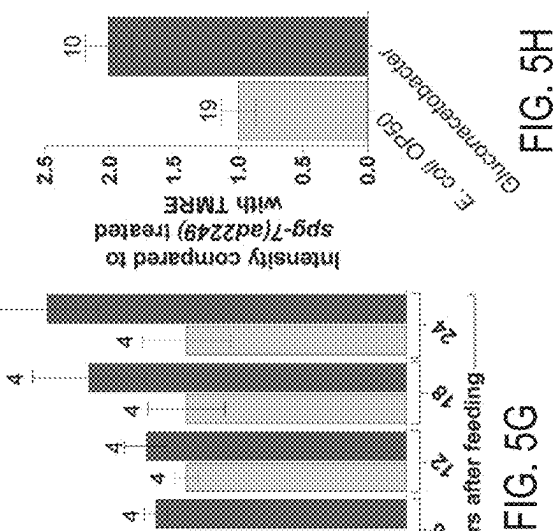
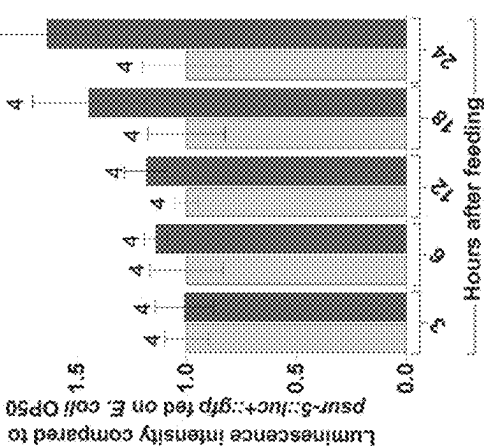
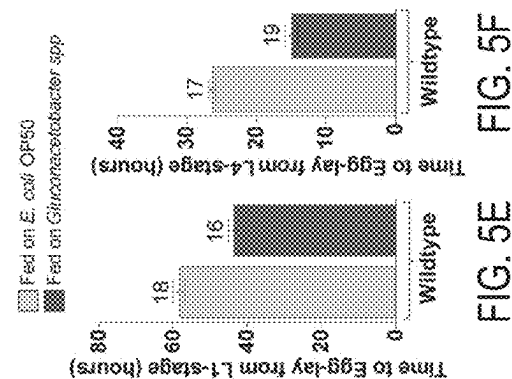
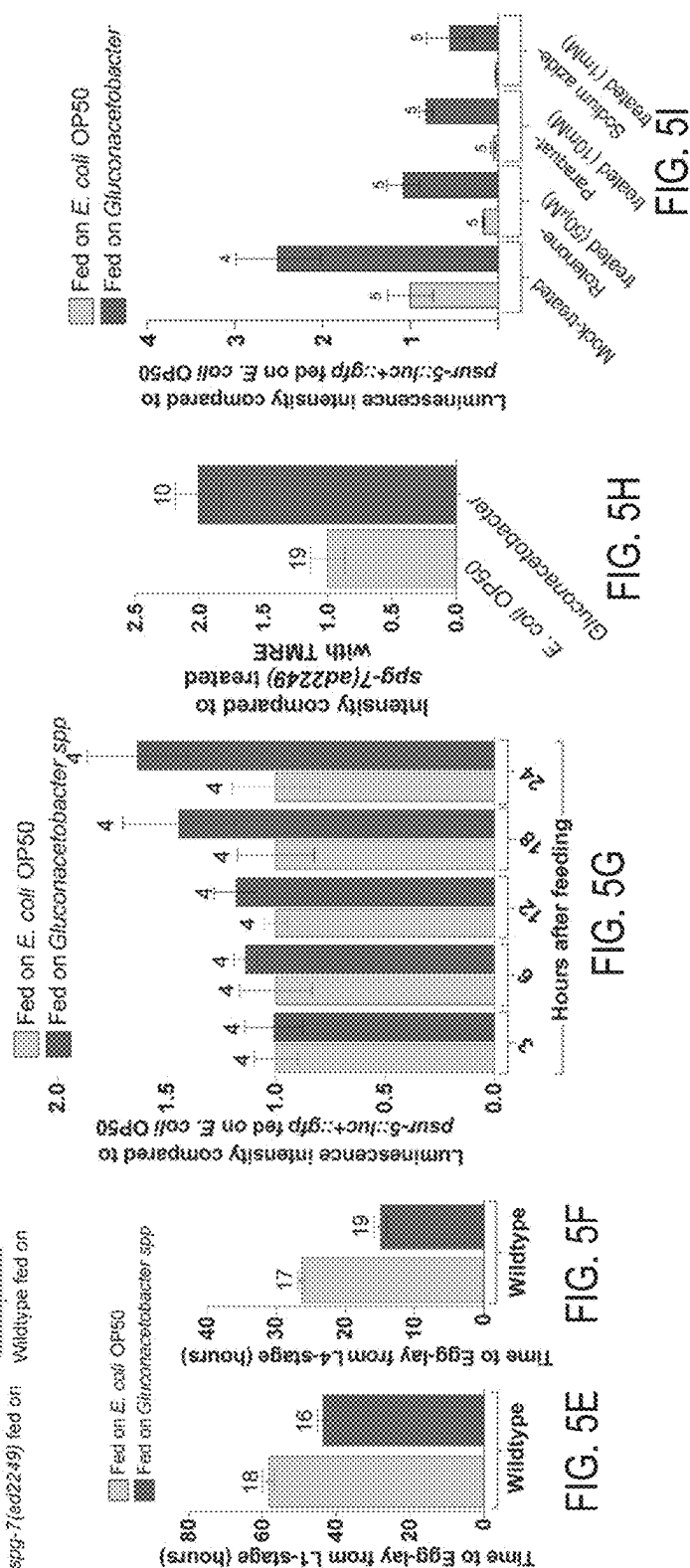
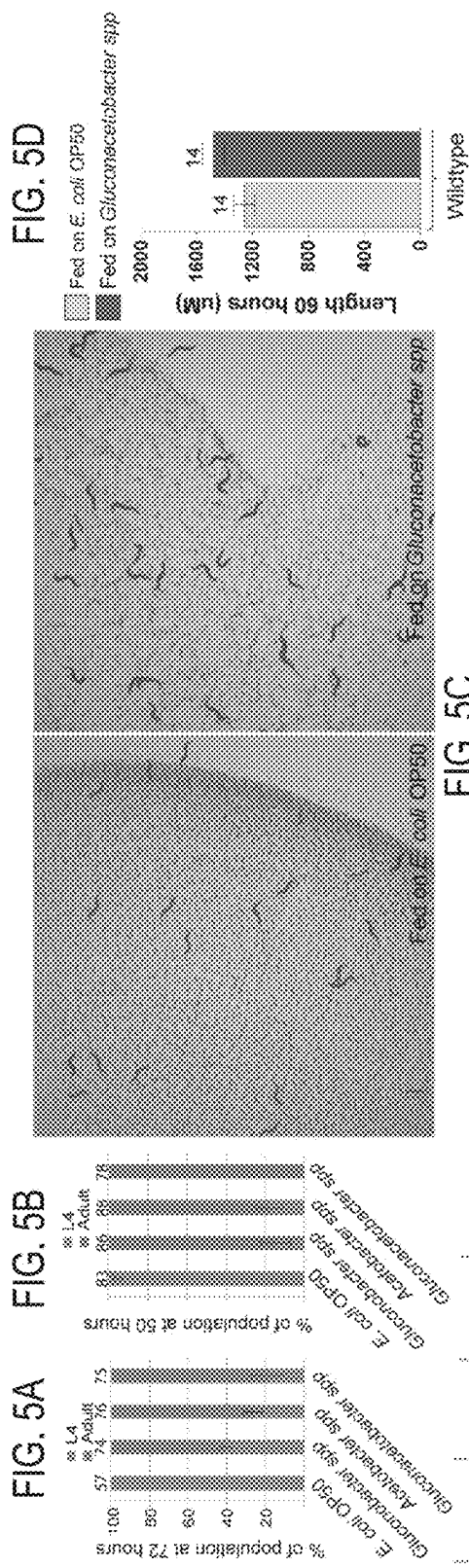
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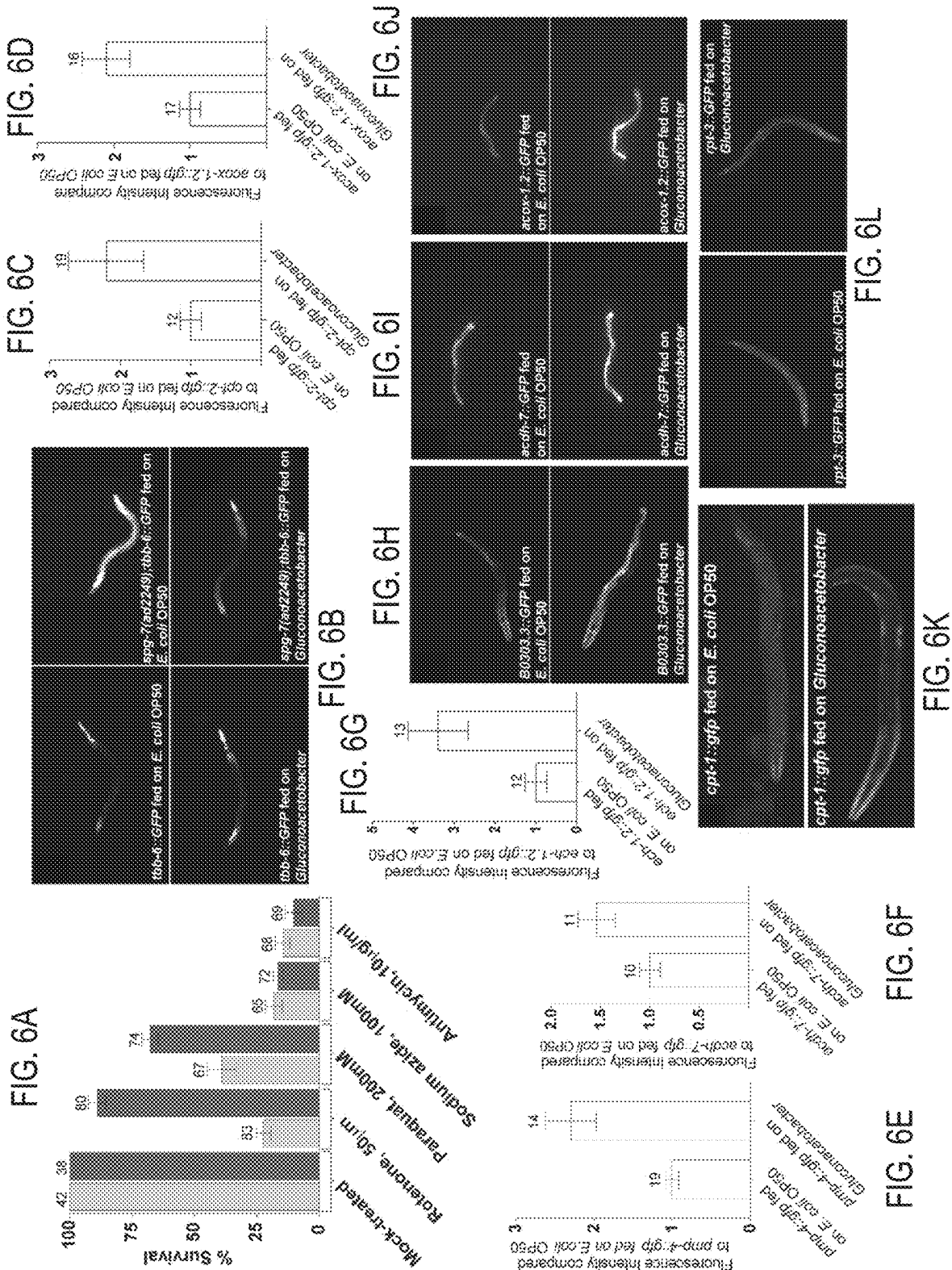


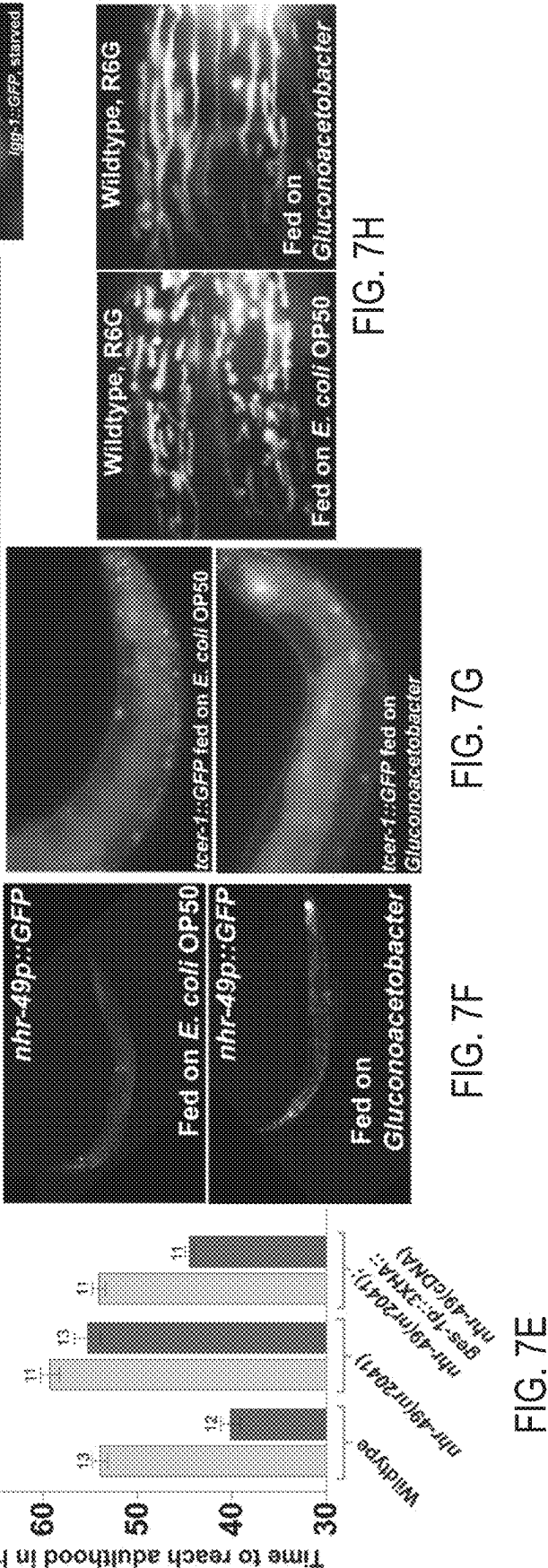
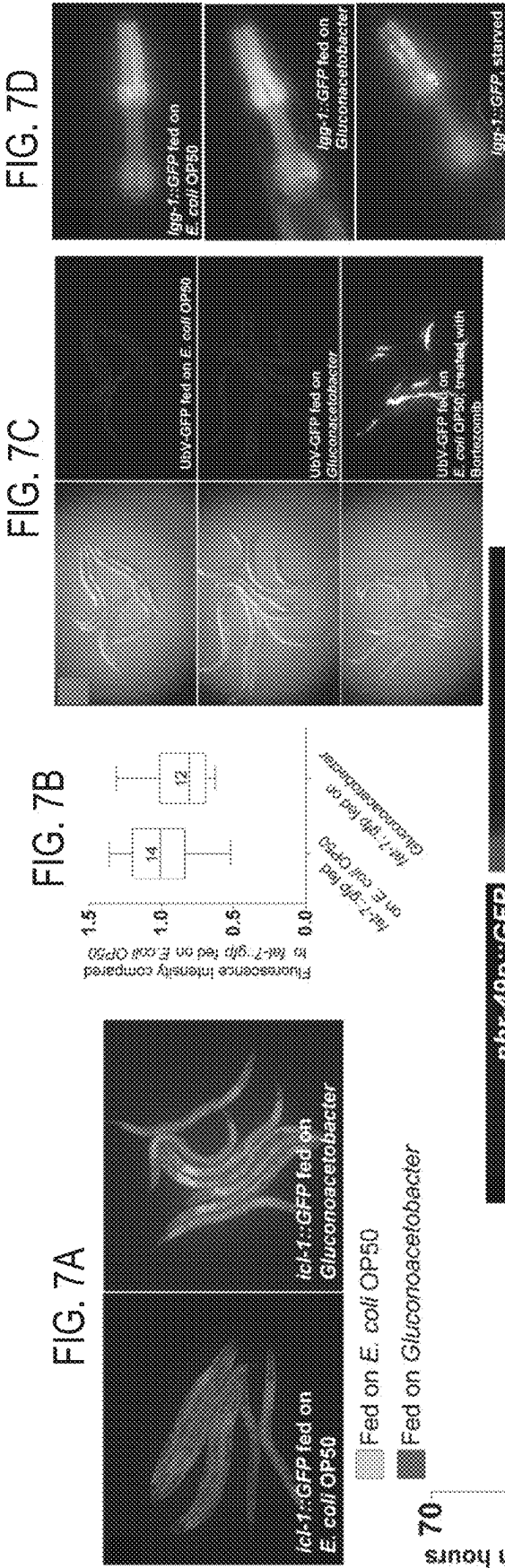












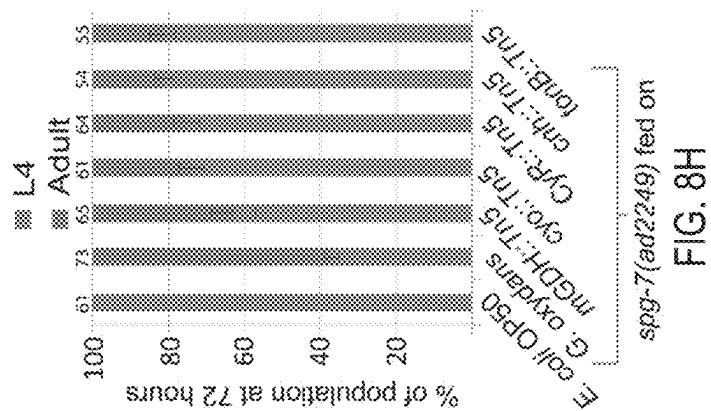
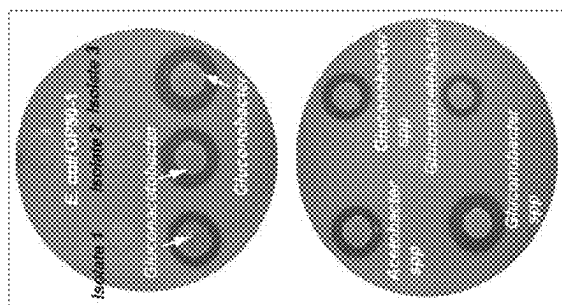
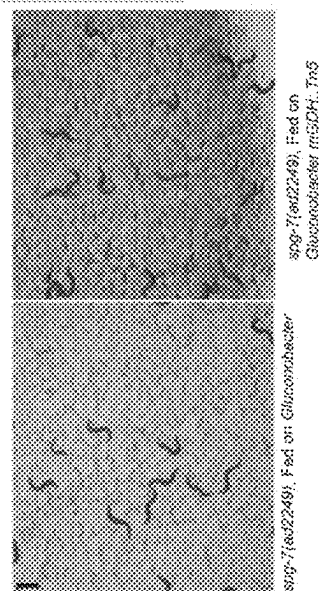
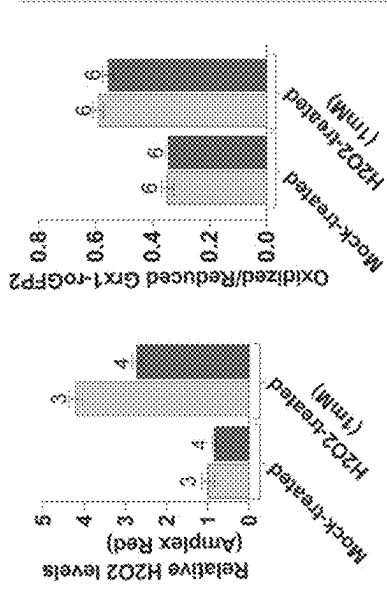
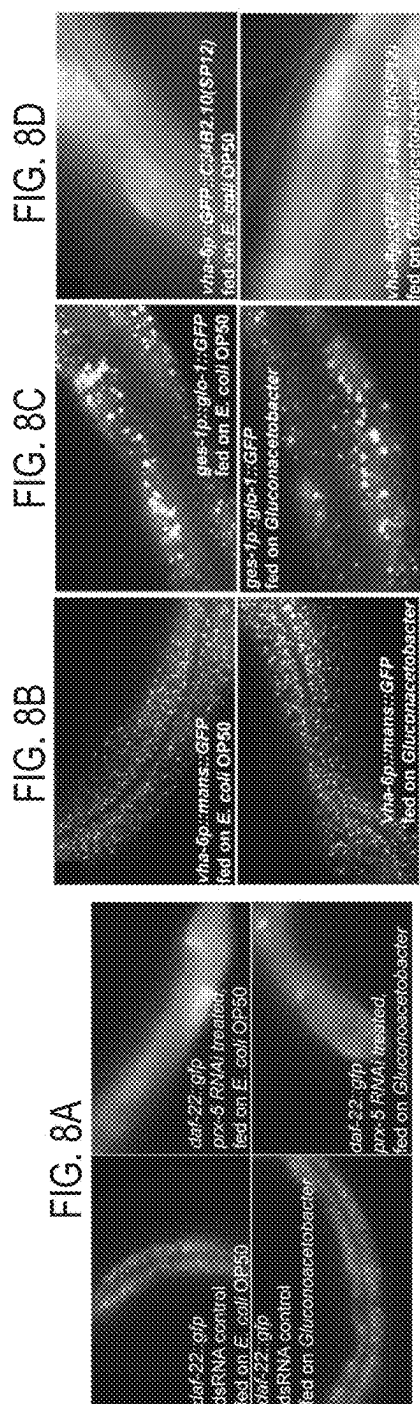


FIG. 9A

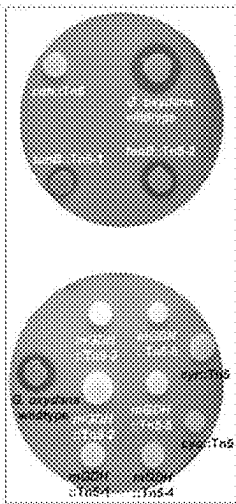
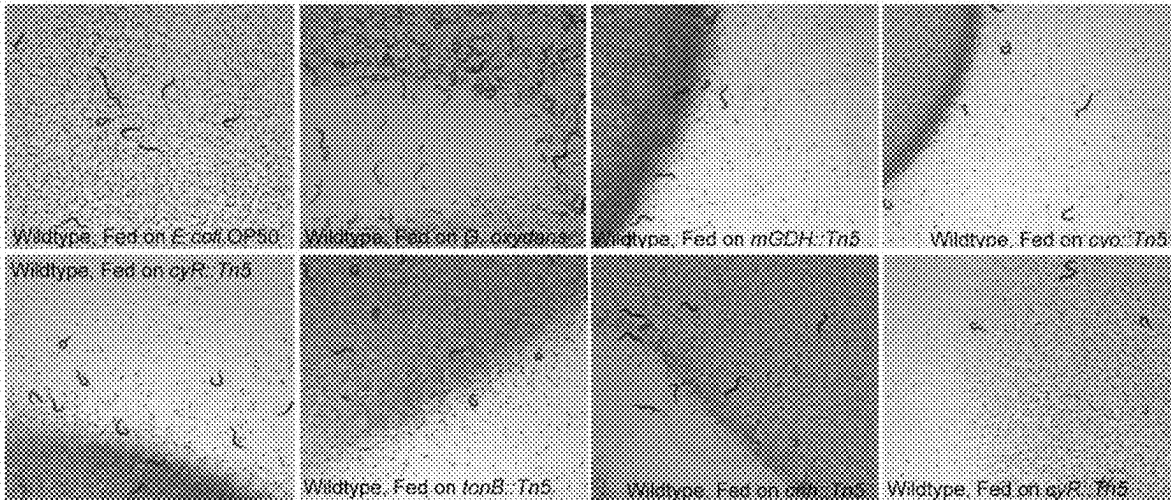


FIG. 9B

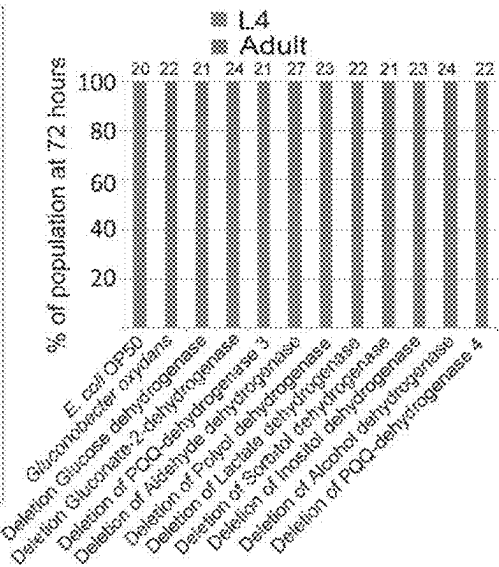


FIG. 9C

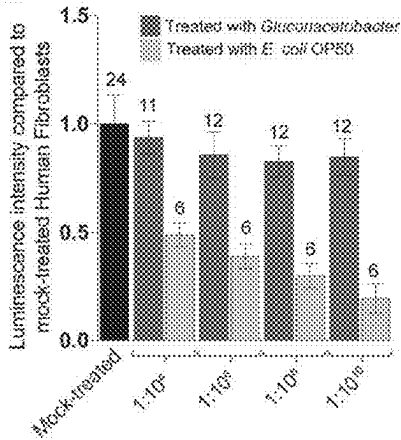


FIG. 9D

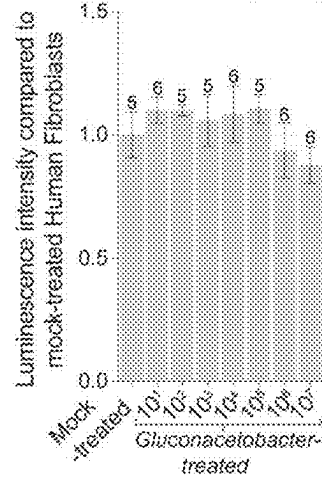


FIG. 9E

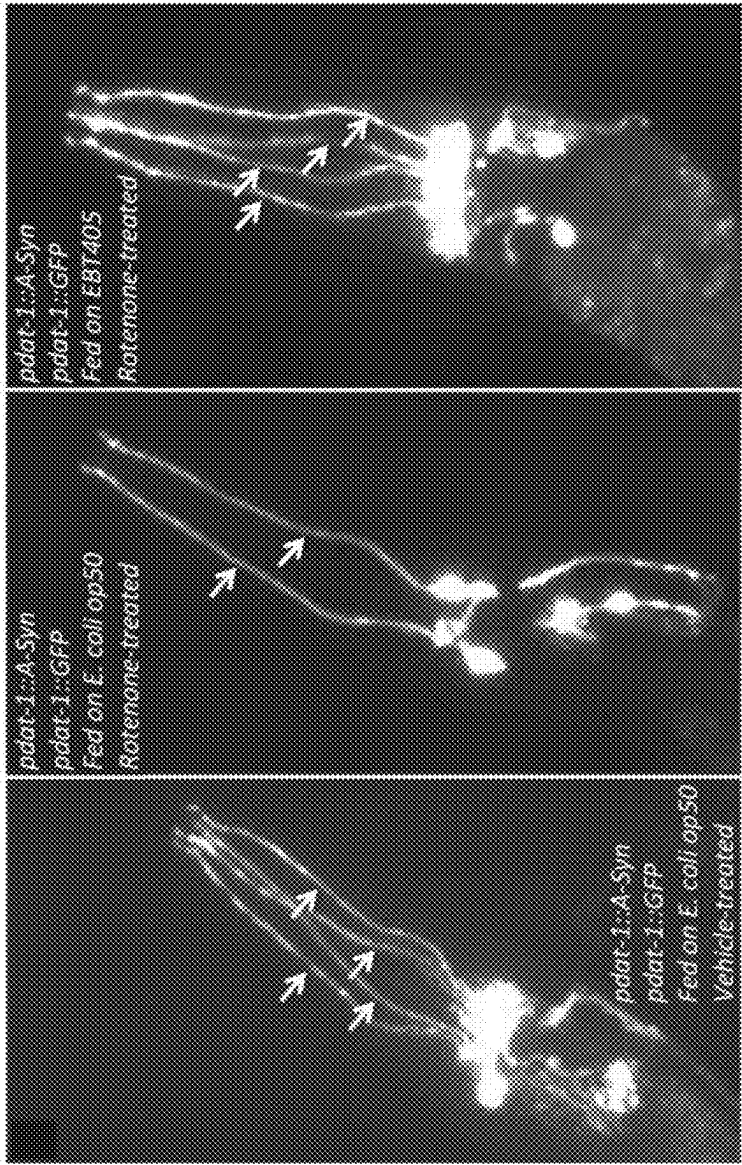


FIG. 10A

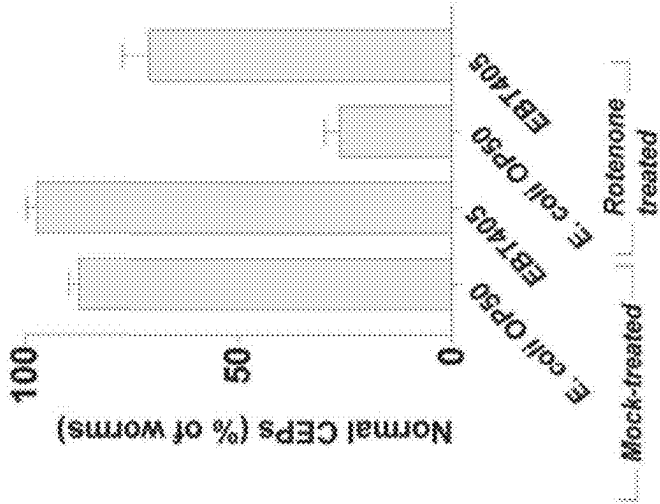


FIG. 10B

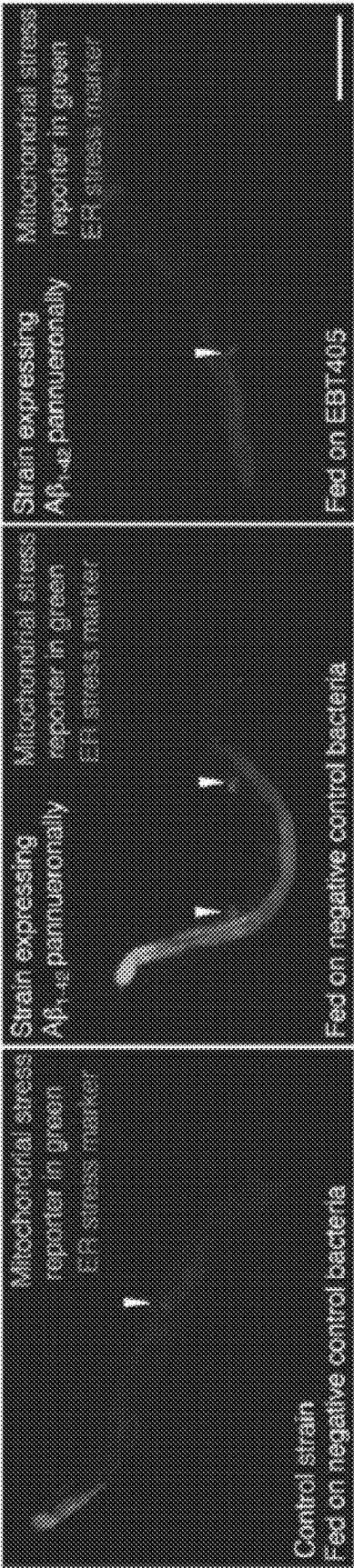


FIG. 11

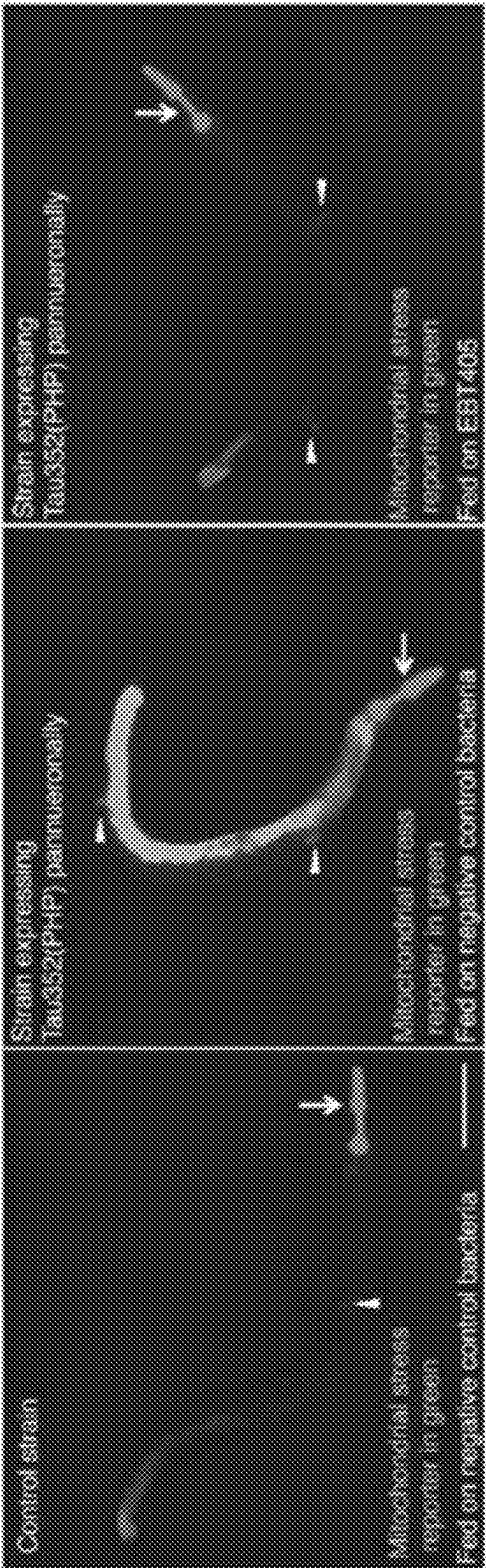


FIG. 12

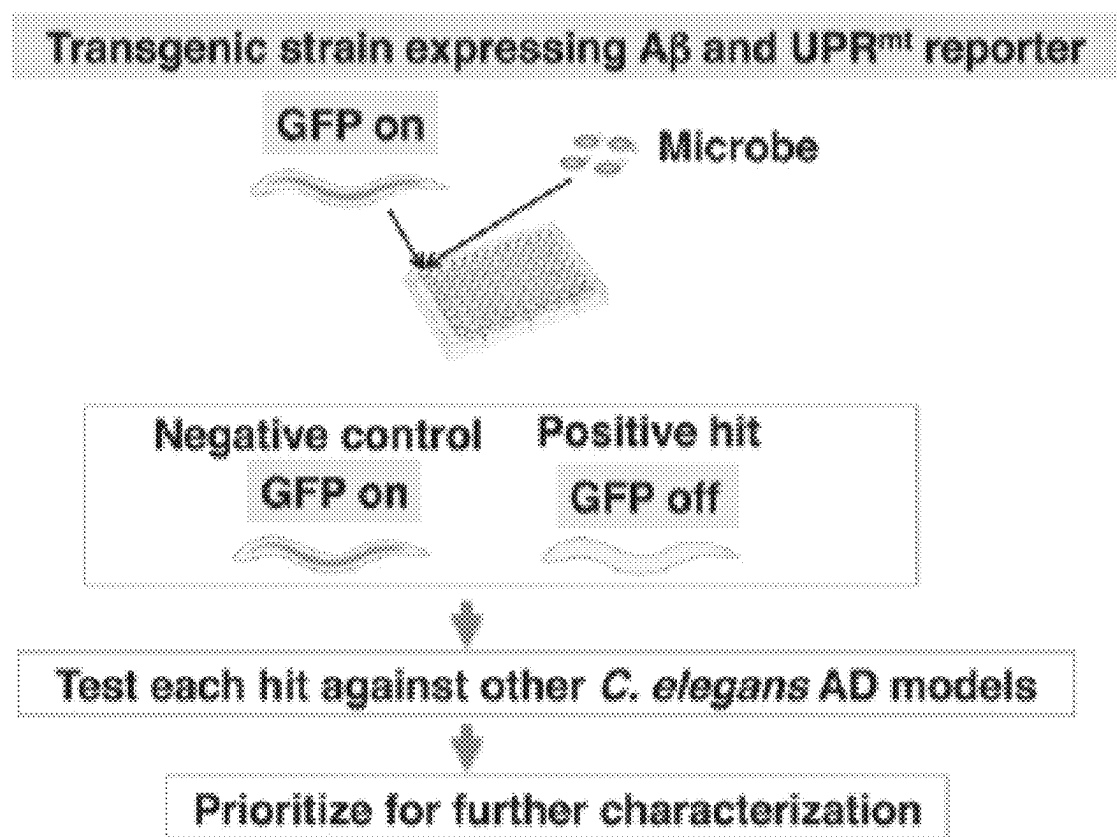


FIG. 13

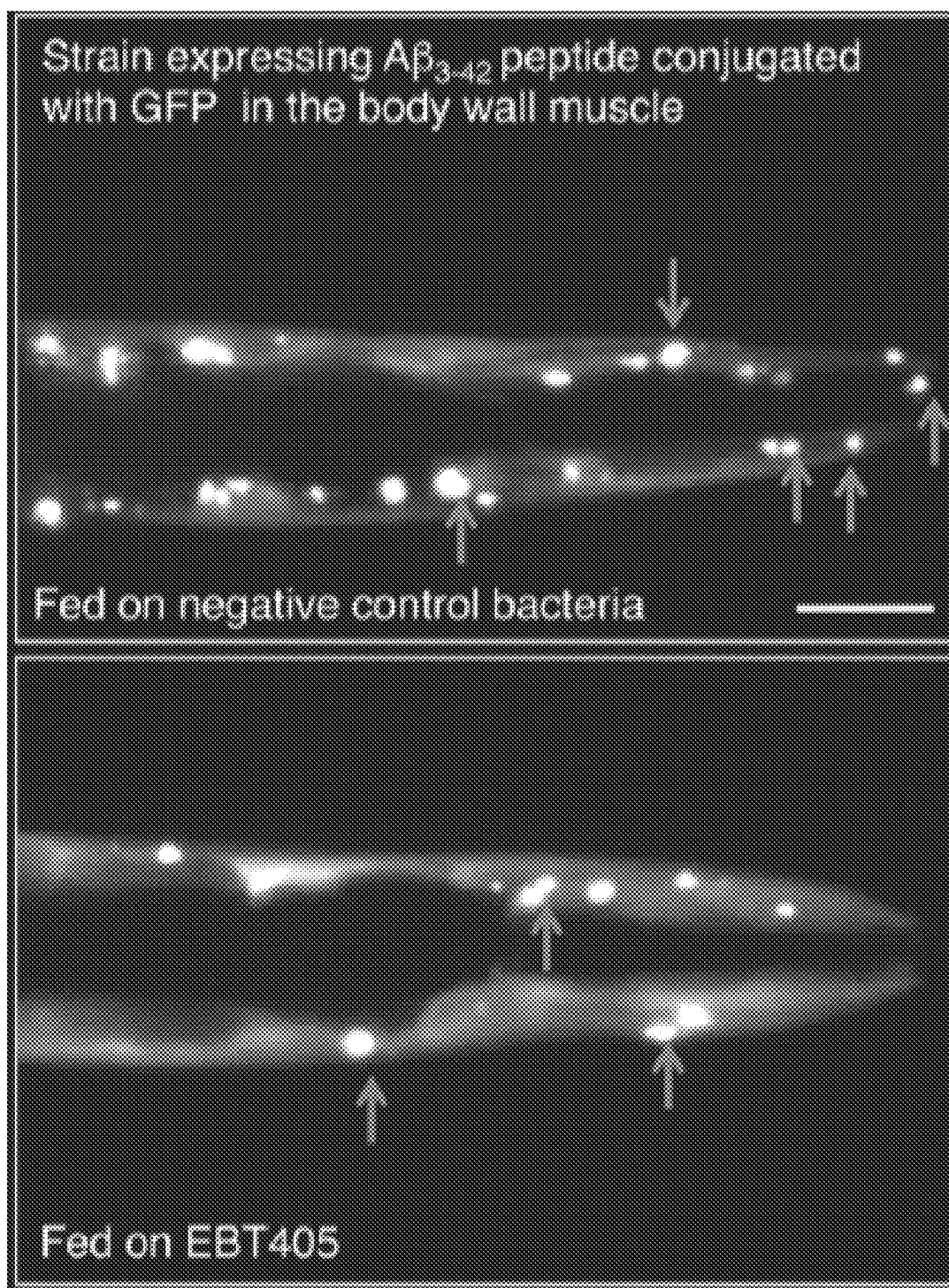


FIG. 14

COMPOSITIONS AND METHODS FOR IMPROVING MITOCHONDRIAL FUNCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of U.S. application Ser. No. 18/351,198, filed Jul. 12, 2023, which is a continuation of U.S. application Ser. No. 17/196,271, filed Mar. 9, 2021, now U.S. Pat. No. 11,738,056, which is a continuation of U.S. application Ser. No. 16/293,789, filed Mar. 6, 2019, now U.S. Pat. No. 10,973,860, which is a continuation of PCT Application No. PCT/US2019/014725, filed Jan. 23, 2019, which claims benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Application No. 62/620,641, filed Jan. 23, 2018 and U.S. Provisional Application No. 62/721,979, filed Aug. 23, 2018, the contents of each of which are incorporated herein by reference in their entirety.

GOVERNMENT SUPPORT

[0002] This invention was made with government support under Grant No. AG043181-16A, awarded by the National Institutes of Health. The government has certain rights in the invention.

SEQUENCE LISTING

[0003] The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jul. 11, 2023, is named 030258-091760USC2_SL.xml and is 69,318 bytes in size.

FIELD OF THE INVENTION

[0004] The field of the invention relates to bacterial compositions or bacterial extracts for the treatment of a symptom of a disease associated with mitochondrial dysfunction.

BACKGROUND

[0005] Mitochondria play a crucial role in ATP generation, fatty acid β -oxidation, catabolism of amino acids, ketogenesis, generation of reactive oxygen species (ROS) and calcium homeostasis. Mitochondrial dysfunction is associated with a number of human diseases, including neurological and muscular degeneration, cardiovascular disorders, obesity, diabetes, and conditions such as aging. Mitochondria contribute to aging through decreased ATP production, increased ROS production, and activation of apoptotic pathways. Decreases in mitochondrial energy production combined with increased oxidative stress, play a causal role in several neurodegenerative disease including Alzheimer's disease, Parkinson's disease, Huntington's disease, and amyotrophic lateral sclerosis. Also, aging is by far the greatest risk factor for several of these neurodegenerative diseases. Regardless of whether a primary or secondary cause, mitochondrial dysfunction holds promise as a target for identifying therapeutics for neurodegenerative diseases. With the increasing average life expectancy worldwide, the number of people affected by aging related decline in mitochondrial function will rise considerably in the near future. Thus, there is a significant clinical unmet need for new therapeutic approaches that not only can be used to slow

neurodegenerative diseases but also as preventive measures for protecting or improving mitochondrial function in the aging population.

SUMMARY

[0006] The methods, compositions and treatments described herein are based, in part, on the discovery of a bacterial extract or bacterial composition that improves mitochondrial function by, for example, increasing production of intracellular energy (e.g., ATP). Accordingly, provided herein are bacterial extracts and compositions for the treatment of mitochondrial dysfunction associated with metabolic disorders, neurodegenerative disorders (e.g., Parkinson's disease, among others) and peroxisomal disorders.

[0007] The present invention relates generally to the use of members of live or inactivated Acetobacteriaceae family microbes, a microbial combination, an extract, active fraction or metabolites, or a combination thereof for promoting, enhancing and/or restoring mitochondrial function. Such compositions can be used for treatment or prevention of diseases, developmental delays, and symptoms related to mitochondrial dysfunction, such as aging, metabolic disorders, muscular disorders and neurodegenerative diseases.

[0008] Provided herein in one aspect is a composition comprising a measured amount of one or more bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0009] In one embodiment of this aspect and all other aspects described herein, the bacterium is from the Acetobacteriaceae family.

[0010] In another embodiment of this aspect and all other aspects described herein, the bacterium is *Gluconobacter* spp., *Acetobacter* spp., *Gluconoacetobacter* spp., *Acidomonas* spp., *Ameyamaea* spp., *Asaia* spp., *Granulibacter* spp., *Kozakia* spp., *Neoasaia* spp., *Neokomagataea* spp., *Saccharibacter* spp., *Swaminathania* spp., or *Tanticharoenia* spp. or combination of two or more of these (e.g., a combination of three or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, or 12 or more of these).

[0011] In another embodiment of this aspect and all other aspects described herein, the bacterium is *Gluconobacter albidus*, *Gluconobacter cerinus*, *Gluconobacter frateruii*, *Gluconobacter japonicus*, *Gluconobacter kondonii*, *Gluconobacter nephelii*, *Gluconobacter oxydans*, *Gluconoacetobacter diazotrophicus*, *Gluconoacetobacter hansenii*, *Gluconoacetobacter saccharivorans*, *Acetobacter aceti* or *Acetobacter malorum*. In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* EBT 405. In another embodiment, the composition comprises at least two or more, at least three or more, at least four or more, at least five or more, at least six or more, at least seven or more, at least eight or more, at least nine or more, at least ten or more, at least eleven or more, at least twelve (or each of) these bacterial species.

[0012] In another embodiment of this aspect and all other aspects described herein, one or more of the nucleic acid sequences are exogenous nucleic acid sequences.

[0013] In another embodiment of this aspect and all other aspects described herein, the bacterium comprises and

expresses nucleic acid sequences encoding each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0014] In another embodiment of this aspect and all other aspects described herein, the measured amount of the one or more bacterium is lyophilized.

[0015] In another embodiment of this aspect and all other aspects described herein, the measured amount is an amount effective to induce expression and/or activity of mitochondrial transcription factor A (TFAM), or peroxisome proliferator activated receptor gamma coactivator 1 (PGC) in a human cell upon administration to a subject in need thereof.

[0016] In another embodiment of this aspect and all other aspects described herein, the amount effective to induce expression and/or activity of TFAM, or PGC in a human cell is 1×10^6 bacteria.

[0017] In another embodiment of this aspect and all other aspects described herein, the composition further comprises one or more added bacterial metabolites selected from the group consisting of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0018] In another embodiment of this aspect and all other aspects described herein, the bacterium is viable, attenuated or heat-inactivated.

[0019] In another embodiment of this aspect and all other aspects described herein, the composition is formulated as a food, a beverage, a feed composition, a probiotic, a nutritional supplement, or a pharmaceutical composition.

[0020] In another embodiment of this aspect and all other aspects described herein, the composition further comprises a prebiotic.

[0021] In another embodiment of this aspect and all other aspects described herein, the prebiotic comprises a fructooligosaccharide, an inulin, an isomaltooligosaccharide, lactitol, lactosucrose, lactulose, a soy oligosaccharide, a transgalactooligosaccharide or a xylooligosaccharide,

[0022] In another embodiment of this aspect and all other aspects described herein, the composition further comprises a pharmaceutically acceptable carrier.

[0023] In another embodiment of this aspect and all other aspects described herein, the composition is formulated for oral administration.

[0024] In another embodiment of this aspect and all other aspects described herein, the composition is an enteric-coated formulation.

[0025] w

[0026] Another aspect provided herein relates to a composition comprising a therapeutically effective amount of an extract or fraction derived from at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding an enzyme selected from the group consisting of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0027] In one embodiment of this aspect and all other aspects provided herein, the at least one bacterium comprises and expresses each of the enzymes membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0028] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction comprises one or more metabolites selected from gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0029] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction is from bacterial cells cultured in a medium comprising: (i) standard Lysogeny broth (1% tryptone, 0.5% yeast extract, and 1% sodium chloride) containing 1% glucose, (ii) standard Hestrin-Schramm broth (D-glucose 2%, 0.5% yeast extract, 0.5% peptone, 0.27% disodium phosphate, 0.115% citric acid) or (iii) CaCO_3 medium comprising 8% glucose, 0.5% yeast extract, 0.2% mannitol, 0.05% magnesium sulphate, and 10% calcium carbonate.

[0030] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction comprises metabolites or bacterial byproducts that promote ATP production in at least one cell type of a host mammal administered the composition.

[0031] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction does not comprise viable bacterial cells.

[0032] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction lacks detectable bacteria.

[0033] In another embodiment of this aspect and all other aspects provided herein, the extract or fraction further comprises attenuated or heat-inactivated bacteria.

[0034] In another embodiment of this aspect and all other aspects provided herein, the bacterium is from the Acetobacteriaceae family.

[0035] In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* spp., *Acetobacter* spp., *Gluconoacetobacter* spp., *Acidomonas* spp., *Ameyamaea* spp., *Asaia* spp., *Granulibacter* spp., *Kozakia* spp., *Neoasaia* spp., *Neokomagataea* spp., *Saccharibacter* spp., *Swaminathania* spp., or *Tanticharoenia* spp. or combination of two or more of these (e.g., a combination of three or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 11 or more, or 12 or more of these).

[0036] In another embodiment of this aspect and all other aspects described herein, the bacterium is *Gluconobacter albidus*, *Gluconobacter cerinus*, *Gluconobacter frateruii*, *Gluconobacter japonicus*, *Gluconobacter kondonii*, *Gluconobacter nephelii*, *Gluconobacter oxydans*, *Gluconoacetobacter diazotrophicus*, *Gluconoacetobacter hansenii*, *Gluconoacetobacter saccharivorans*, *Acetobacter aceti* or *Acetobacter malorum*. In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* EBT 405. In another embodiment, the composition comprises at least two or more, at least three or more, at least four or more, at least five or more, at least six or more, at least seven or more, at least eight or more, at least nine or more, at least ten or more, at least eleven or more, at least twelve (or each of) these bacterial species.

[0037] In another embodiment of this aspect and all other aspects provided herein, one or more of the nucleic acid sequences is/are an exogenous nucleic acid sequence(s).

[0038] In another embodiment of this aspect and all other aspects provided herein, the composition is formulated as a food, a beverage, a feed composition, a nutritional supplement, or a pharmaceutical composition.

[0039] In another embodiment of this aspect and all other aspects provided herein, the composition further comprises a pharmaceutically acceptable carrier.

[0040] In another embodiment of this aspect and all other aspects provided herein, the composition is formulated for oral administration.

[0041] Also provided herein, in another aspect, is a method for increasing cellular ATP production in at least one cell type of a subject in need thereof, the method comprising administering to the subject a composition comprising an amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, effective to increase cellular ATP production in at least one cell type.

[0042] In one embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0043] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0044] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0045] In another embodiment of this aspect and all other aspects provided herein, one or more of the nucleic acid sequences is exogenous to the bacterium.

[0046] In another embodiment of this aspect and all other aspects provided herein, the activity of complex I and/or complex II of the mitochondrial electron transport chain is increased in the one or more cell types.

[0047] In another embodiment of this aspect and all other aspects provided herein, the administering increases mitochondrial membrane potential.

[0048] In another embodiment of this aspect and all other aspects provided herein, the subject is human.

[0049] In another embodiment of this aspect and all other aspects provided herein, expression of peroxisome proliferator-activated receptor gamma coactivator 1- α (PGC-1 α), and/or mitochondrial transcription factor A (TFAM) is increased.

[0050] In another embodiment of this aspect and all other aspects provided herein, AMP-activated protein kinase (AMPK) phosphorylation levels, nuclear respiratory factor-2 (Nrf2) protein levels, PGC α mRNA levels TFAM mRNA levels and/or mitochondrial DNA replication is increased.

[0051] In another embodiment of this aspect and all other aspects provided herein, the mitochondrial DNA copy number (mtDNA) is increased.

[0052] In another embodiment of this aspect and all other aspects provided herein, the method increases the developmental growth rate of a subject carrying a mutations in mitochondrial respiratory Complex I NADH: ubiquinone reductase.

[0053] In another embodiment of this aspect and all other aspects provided herein, the expression of at least one mitochondrial B-oxidation enzyme is increased.

[0054] In another embodiment of this aspect and all other aspects provided herein, the at least one mitochondrial B-oxidation enzyme is B0303.3, cpt-2, cpt-1, ech-1.2, or acdh-7.

[0055] In another embodiment of this aspect and all other aspects provided herein, the longevity of the subject is increased.

[0056] In another embodiment of this aspect and all other aspects provided herein, mitochondrial biogenesis is maintained or increased.

[0057] In another embodiment of this aspect and all other aspects provided herein, cellular ATP production is increased by at least 10% compared to the cellular ATP production prior to administration of the composition.

[0058] Another aspect provided herein relates to a method for making a bacterial extract, the method comprising culturing at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II in a medium comprising: (i) standard Lysogeny broth (1% tryptone, 0.5% yeast extract, and 1% sodium chloride) containing 1% glucose, (ii) standard Hestrin-Schramm broth (D-glucose 2%, 0.5% yeast extract, 0.5% peptone, 0.27% disodium phosphate, 0.115% citric acid) or (iii) CaCO₃ medium comprising 8% glucose, 0.5% yeast extract, 0.2% mannitol, 0.05% magnesium sulphate, and 10% calcium carbonate.

[0059] In one embodiment of this aspect and all other aspects provided herein, the at least one bacterium comprises and expresses nucleic acid sequences encoding each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0060] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0061] Also provided herein, in another aspect, is a composition comprising an amount of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid effective to increase cellular ATP production in at least one cell type of a host mammal administered the composition.

[0062] Another aspect provided herein relates to a composition comprising an amount of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid effective to increase the expression and/or activity of mitochondrial transcription factor A (TFAM), or peroxisome proliferator activated receptor gamma coactivator 1 (PGC) in at least one cell type of a human subject administered the composition.

[0063] In one embodiment of this aspect and all other aspects provided herein, the composition comprises a bacterial extract or active fraction thereof.

[0064] In another embodiment of this aspect and all other aspects provided herein, the composition is formulated as a food, a beverage, a feed composition, a nutritional supplement, or a pharmaceutical composition.

[0065] Provided herein, in another aspect, is a method of treating Parkinson's disease, the method comprising administering to a subject having Parkinson's disease, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of Parkinson's disease.

[0066] In one embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0067] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0068] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0069] In another embodiment of this aspect and all other aspects provided herein, the at least one symptom is selected from the group consisting of: tremor, sleep disturbance, mobility impairment, involuntary movement, muscle rigidity, rhythmic muscle contraction, slow bodily movement, slow shuffling gait, fatigue, dizziness, impaired balance, restlessness, amnesia, confusion, dementia, cognitive impairment, impaired speech, anxiety, apathy, distorted or loss of sense of smell, urinary incontinence, reduced facial expression, weight loss and constipation.

[0070] In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* EBT 405.

[0071] Another aspect provided herein relates to a method of treating a mitochondrial electron transport chain disorder, the method comprising administering to a subject having a mitochondrial electron transport chain disorder, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of the mitochondrial electron transport chain disorder.

[0072] In one embodiment of this aspect and all other aspects provided herein, the mitochondrial electron transport chain disorder comprises a disorder or impaired activity in Complex I and/or Complex II.

[0073] In another embodiment of this aspect and all other aspects provided herein, the mitochondrial electron transport chain disorder is NADH dehydrogenase (NADH-CoQ reductase) deficiency, succinate dehydrogenase deficiency, Leigh Disease, mitochondrial DNA depletion, or mitochondrial insufficiency.

[0074] In another embodiment of this aspect and all other aspects provided herein, the at least one symptom is selected

from the group consisting of: myopathy, mitochondrial encephalomyopathy, failure to thrive, developmental delay, hypotonia, lethargy, respiratory failure, ataxia, myoclonus, lactic acidosis, seizures, fatigue, nystagmus, poor reflexes, difficulty eating or swallowing, breathing difficulties, ataxia, congenital myopathy, infantile myopathy and hepatopathy.

[0075] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0076] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0077] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0078] Another aspect provided herein relates to a method of treating a peroxisomal disorder, the method comprising administering to a subject having a peroxisomal disorder, a composition comprising a therapeutically effective amount of at least one bacterium that comprises one or more nucleic acid sequences such that the bacterium expresses the following enzymes: membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or a bacterial extract thereof, thereby reducing at least one symptom of the peroxisomal disorder.

[0079] In another embodiment of this aspect and all other aspects provided herein, the peroxisomal disorder is Zellweger syndrome spectrum (PBD-ZSD), or rhizomelic chondrodysplasia punctata type 1 (RCDP1).

[0080] In another embodiment of this aspect and all other aspects provided herein, the PBD-ZSD is infantile Refsum disease, neonatal adrenoleukodystrophy, or Zellweger syndrome.

[0081] In another embodiment of this aspect and all other aspects provided herein, the at least one symptom is selected from the group consisting of: skeletal and craniofacial dysmorphism, liver dysfunction, progressive sensorineural hearing loss and retinopathy.

[0082] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0083] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0084] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0085] Also provided herein, in another aspect, is a method for increasing the biogenesis of cellular mitochondria or peroxisomes, the method comprising administering to a subject, a composition comprising an amount of at least one bacterium that comprises and expresses one or more

nucleic acid sequences encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or a fraction or extract thereof, effective to increase the biogenesis of cellular mitochondria or peroxisomes.

[0086] In one embodiment of this aspect and all other aspects provided herein, the size and/or number of peroxisomes is increased.

[0087] In one embodiment of this aspect and all other aspects provided herein, mitochondrial activity is increased.

[0088] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0089] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0090] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0091] Also provided herein, in another aspect, is a method of treating Alzheimer's disease, the method comprising administering to a subject having Alzheimer's disease, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of Alzheimer's disease.

[0092] In one embodiment of this aspect and all other aspects provided herein, the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0093] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0094] In another embodiment of this aspect and all other aspects provided herein, the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.

[0095] In another embodiment of this aspect and all other aspects provided herein, the at least one symptom is selected from the group consisting of: cognitive decline, confusion, delusion, disorientation, forgetfulness, difficulty concentrating, inability to generate new memories, inability to do simple math, inability to recognize common items, aggression, agitation, irritability, meaningless repetition of own words, personality changes, restlessness, lack of restraint, wandering, anger, apathy, general discontent, loneliness, mood swings, depression, hallucination, paranoia, loss of appetite, inability to combine muscle movements and jumbled speech.

[0096] In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* EBT 405.

[0097] Another aspect provided herein relates to the use of a composition comprising gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid for the treatment of Parkinson's disease or Alzheimer's disease.

[0098] Another aspect provided herein relates to the use of composition as described herein in the treatment of Parkinson's disease or Alzheimer's disease.

[0099] Another aspect provided herein relates to the use of a composition comprising gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid for increasing cellular ATP production.

[0100] Another aspect provided herein relates to the use of a composition as described herein for increasing cellular ATP production.

[0101] Another aspect provided herein relates to the use of a composition comprising gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid for treating a mitochondrial electron transport chain disorder or a peroxisomal disorder.

[0102] Another aspect provided herein relates to the use of a composition as described herein for the treatment of a mitochondrial electron transport chain disorder or a peroxisomal disorder.

[0103] Another aspect provided herein relates to the use of a composition comprising gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid for increasing cellular biogenesis of mitochondria or peroxisomes.

[0104] Another aspect provided herein relates to the use of a composition as described herein for increasing cellular biogenesis of mitochondria or peroxisomes.

[0105] Another aspect provided herein relates to the use of a composition comprising gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid for increasing the expression and/or activity of TFAM or peroxisome proliferator activated receptor gamma coactivator 1 (PGC-1).

[0106] Another aspect provided herein relates to the use of a composition of as described herein for increasing the expression and/or activity of TFAM or peroxisome proliferator activated receptor gamma coactivator 1 (PGC-1).

[0107] Another aspect provided herein relates to the use of a composition comprising a measured amount of one or more bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II for the treatment or prevention of Alzheimer's disease or Parkinson's disease.

[0108] In one embodiment of this aspect and all other aspects provided herein, the one or more bacterium comprises *Gluconobacter* EBT 405.

Definitions

[0109] The terms "patient," "subject" and "individual" are used interchangeably herein, and refer to an animal, particularly a human, to whom treatment, including prophylactic treatment is provided. The term "subject" as used herein refers to human and non-human animals. The term "non-

human animals” and “non-human mammals” are used interchangeably herein and includes all vertebrates, e.g., mammals, such as non-human primates, (particularly higher primates), sheep, dog, rodent (e.g. mouse or rat), guinea pig, goat, pig, cat, rabbits, cows, and non-mammals such as chickens, amphibians, reptiles etc. In one embodiment, the subject is human. In another embodiment, the subject is an experimental animal or animal substitute as a disease model. In another embodiment, the subject is a domesticated animal including companion animals (e.g., dogs, cats, rats, guinea pigs, hamsters etc.). It is specifically contemplated herein that a subject can be of any developmental age including, but not limited to, a fetus, a neonate, an infant, a toddler, a child, an adolescent, an adult, post-menopausal, or a geriatric subject.

[0110] As used herein, a “prebiotic” refers to an ingredient that allows or promotes specific changes, both in the composition and/or activity in the gastrointestinal microbiota that may (or may not) confer benefits upon the host. In some embodiments, a prebiotic can include one or more of the following: fructooligosaccharide, galactooligosaccharides, hemicelluloses (e.g., arabinoxylan, xylan, xyloglucan, and glucomannan), inulin, chitin, lactulose, mannan oligosaccharides, oligofructose-enriched inulin, gums (e.g., guar gum, gum arabic and carrageenan), oligofructose, oligodextrose, tagatose, resistant maltodextrins (e.g., resistant starch), trans-galactooligosaccharide, pectins (e.g., xylogalactouronan, citrus pectin, apple pectin, and rhamnogalacturonan-1), dietary fibers (e.g., soy fiber, sugar beet fiber, pea fiber, corn bran, and oat fiber) and xylooligosaccharides.

[0111] As used herein, the terms “administering,” “introducing” and “transplanting” are used interchangeably in the context of the placement of cells, e.g., a bacterial composition, as described herein into a subject, by a method or route which results in at least partial localization of the introduced cells at a desired site, such as the intestines or a region thereof, such that a desired effect(s) is produced (e.g., increased mitochondrial ATP production). The cells can be administered by any appropriate route which results in delivery to a desired location in the subject where at least a portion of the delivered cells or components of the cells remain viable. The period of viability of the cells after administration to a subject can be as short as a few hours, e.g., six to twenty-four hours, to a few days, to as long as several years, i.e., long-term engraftment. In some embodiments, the term “administering” refers to the administration of a bacterial extract or preparation comprising one or more bacterial metabolites and/or byproducts but lacking fully viable bacterial cells.

[0112] As used herein “preventing” or “prevention” refers to any methodology where the disease state does not occur due to the actions of the methodology (such as, for example, administration of a composition as described herein). In one aspect, it is understood that prevention can also mean that the disease is not established to the extent that occurs in untreated controls. For example, there can be a 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90, or 100% reduction in the establishment of disease frequency relative to untreated controls. Accordingly, prevention of a disease encompasses a reduction in the likelihood that a subject will develop the disease, relative to an untreated subject (e.g. a subject who is not treated with a composition as described herein).

[0113] “Synergy” or “synergistic interactions” refers to the interaction or cooperation of two or more microbes or

metabolites to produce a combined effect greater than the sum of their separate effects. For example, in one embodiment, “synergy” between two or more microbes can result from a first microbe secreting a waste product or metabolite that the second microbe uses to fuel growth or other processes.

[0114] The terms “decrease”, “reduced”, “reduction”, or “inhibit” are all used herein to mean a decrease or lessening of a property, level, or other parameter by a statistically significant amount. In some embodiments, “reduce,” “reduction” or “decrease” or “inhibit” typically means a decrease by at least 10% as compared to a reference level (e.g., the absence of a given treatment) and can include, for example, a decrease by at least about 10%, at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 98%, at least about 99%, or more. As used herein, “reduction” or “inhibition” does not encompass a complete inhibition or reduction as compared to a reference level. “Complete inhibition” is a 100% inhibition as compared to a reference level. A decrease can be preferably down to a level accepted as within the range of normal for an individual without a given disorder.

[0115] The terms “increased,” “increase” or “enhance” or “activate” are all used herein to generally mean an increase of a property, level, or other parameter by a statistically significant amount; for the avoidance of any doubt, the terms “increased”, “increase” or “enhance” or “activate” means an increase of at least 10% as compared to a reference level, for example an increase of at least about 20%, or at least about 30%, or at least about 40%, or at least about 50%, or at least about 60%, or at least about 70%, or at least about 80%, or at least about 90% or up to and including a 100% increase or any increase between 10-100% as compared to a reference level, or at least about a 2-fold, or at least about a 3-fold, or at least about a 4-fold, or at least about a 5-fold or at least about a 10-fold increase, at least about a 20-fold increase, at least about a 50-fold increase, at least about a 100-fold increase, at least about a 1000-fold increase or more as compared to a reference level.

[0116] The term “pharmaceutically acceptable” can refer to compounds and compositions which can be administered to a subject (e.g., a mammal or a human) without undue toxicity.

[0117] As used herein, the term “pharmaceutically acceptable carrier” can include any material or substance that, when combined with an active ingredient, allows the ingredient to retain biological activity and is substantially non-reactive with the subject’s immune system (unless desired). Examples include, but are not limited to, any of the standard pharmaceutical carriers such as a phosphate buffered saline solution, emulsions such as oil/water emulsion, and various types of wetting agents. The term “pharmaceutically acceptable carriers” excludes tissue culture and bacterial culture media.

[0118] As used herein, the term “comprising” means that other elements can also be present in addition to the defined elements presented. The use of “comprising” indicates inclusion rather than limitation.

[0119] As used herein the term “consisting essentially of” refers to those elements required for a given embodiment.

The term permits the presence of additional elements that do not materially affect the basic and novel or functional characteristic(s) of that embodiment of the invention.

[0120] The term “consisting of” refers to compositions, methods, and respective components thereof as described herein, which are exclusive of any element not recited in that description of the embodiment.

[0121] Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular.

[0122] Other than in the operating examples, or where otherwise indicated, all numbers expressing quantities of ingredients or reaction conditions used herein should be understood as modified in all instances by the term “about.” The term “about” when used in connection with percentages can mean $\pm 1\%$.

BRIEF DESCRIPTION OF THE FIGURES

[0123] FIGS. 1A-1H. FIG. 1A, *Gluconoacetobacter* feeding suppresses the developmental delay phenotype of spg-7(ad2249), nduf-7(et7), and gas-1(fc21) mutants. FIG. 1B, *Gluconoacetobacter* feeding increases ATP content in wild-type animals compared to worms fed on *E. coli* OP50-1 as assessed using a strain that expresses Luciferase in all somatic cells. FIG. 1C, *Gluconoacetobacter* feeding increases ATP content in wildtype animals compared to worms fed on *E. coli* OP50-1 as assessed by measuring the endogenous ATP levels using biochemical method. FIG. 1D, *Gluconoacetobacter* feeding restores the ATP content in spg-7(ad2249) animals to wildtype levels. FIG. 1E, *Gluconoacetobacter* feeding restores the ATP content in nduf-7(et7) animals to wildtype levels. FIG. 1F, *Gluconoacetobacter* feeding increases the mitochondrial membrane potential in spg-7(ad2249), nduf-7(et7), and gas-1(fc21) mutants animals compared to the corresponding mutant worms fed on *E. coli* OP50-1 as assessed using MitoTracker Red CMXRos. FIG. 1G, Animals fed on *Gluconoacetobacter* spp and treated with varying doses of paraquat had significantly higher levels of bioluminescence, a surrogate for ATP levels, compared to animals fed on *E. coli* OP50 and treated with paraquat. FIG. 1H, Animals fed on *Gluconoacetobacter* spp and treated with varying doses of rotenone had significantly higher levels of bioluminescence, a surrogate for ATP levels, compared to animals fed on *E. coli* OP50 and treated with rotenone.

[0124] FIGS. 2A-2K. FIG. 2A, Worms fed *E. coli* OP50 and treated with either MPP+ or 6-OHDA or paraquat induced mitochondrial stress reporter hsp-6::gfp expression while worms fed on *Gluconoacetobacter* spp failed to induce the GFP expression. FIG. 2B, Feeding spg-7(ad2249); tbb-6::gfp animals on *Gluconoacetobacter* spp suppressed the activation of tbb-6::gfp expression. FIG. 2C, *Gluconoacetobacter* feeding suppresses hphd-1::gfp expression. FIG. 2D, *Gluconoacetobacter* feeding increases B0303.3::gfp expression. FIG. 2E, *Gluconoacetobacter* feeding increases prx-11::gfp expression. FIG. 2F, tcer-1(RNAi) suppresses the induction of cpt-2::gfp in worms fed on *Gluconoacetobacter* spp. FIG. 2G, nhr-49(RNAi) suppresses the induction of acox-1.2::gfp in worms fed on *Gluconoacetobacter* spp. FIG. 2H, tcer-1(RNAi) suppresses the induction of ech-1.2::gfp in worms fed on *Gluconoacetobacter* spp. FIG. 2I, nhr-49(RNAi) suppresses the induction of acdh-7::gfp in worms fed on *Gluconoacetobacter* spp. FIG. 2J, nhr-49(RNAi) suppresses the induction of pmp-4::gfp in worms

fed on *Gluconoacetobacter* spp. FIG. 2K, nhr-49(RNAi) suppresses the induction of prx-6::gfp in worms fed on *Gluconoacetobacter* spp.

[0125] FIGS. 3A-3H. FIG. 3A, tcer-1(tm1452) and nhr-49(nr2041) animals fed on *Gluconoacetobacter* spp feeding grew slower compared to wildtype worms fed on *Gluconoacetobacter* spp. FIG. 3B, RNAi of tcer-1, nhr-49, and prx-5 suppressed the *Gluconoacetobacter* spp feeding induced increase in bioluminescence phenotype. FIG. 3C, nhr-49(RNAi) suppressed the accelerated developmental progression phenotype of spg-7(ad2249) and nduf-7(et19) mutant worms fed on *Gluconoacetobacter* spp. FIG. 3D, prx-5(RNAi) suppressed the accelerated developmental progression phenotype of spg-7(ad2249) and nduf-7(et19) mutant worms fed on *Gluconoacetobacter* spp. FIG. 3E, nhr-49(RNAi) exacerbated the reduced mitochondrial membrane potential of spg-7(ad2249) mutants as assessed using Mito CMXRos. FIG. 3F, Representative picture showing the fused mitochondria phenotype in the muscle of worms fed on *Gluconoacetobacter* spp compared to the worms fed on *E. coli* OP50 as assessed using a mitochondrially targeted TOM20-RFP fusion protein. FIG. 3G, Representative picture showing increased punctate structures in the intestine of worms fed on *Gluconoacetobacter* spp indicating an increase in peroxisomes as assessed using the peroxisomally targeted GFP-DAF-22 fusion protein. FIG. 3H, Worms fed on *Gluconoacetobacter* spp had slightly lower levels of H₂O₂ indicating that *Gluconoacetobacter* spp as assessed using transgenic worms expressing HyPer sensor.

[0126] FIGS. 4A-4I. FIG. 4A, Worms fed on *Gluconoacetobacter* spp had significantly higher levels of bioluminescence, a surrogate for ATP levels, compared to animals fed on *E. coli* OP50 as the worms age. FIG. 4B, Dilution of *Gluconoacetobacter* spp with *E. coli* induces significantly higher levels of bioluminescence, a surrogate for ATP levels. FIG. 4C, *Gluconoacetobacter* strains including *Gluconoacetobacter hansenii* (MGH isolate), the canonical *Gluconoacetobacter hansenii* ATCC 23769, and non-cellulose producing mutant of *G. hansenii* ATCC 23769 induces significantly higher levels of bioluminescence, a surrogate for ATP levels compared worms fed on *E. coli* OP50. FIG. 4D, Members of Acetobacteraceae induce significantly higher levels of bioluminescence, a surrogate for ATP levels, compared worms fed on *E. coli* OP50. FIG. 4E, Worms fed on mutants in membrane-bound PQQ-dependent glucose dehydrogenase, cytochrome o ubiquinol oxidase subunit II, and ubiquinol-cytochrome c reductase iron-sulfur subunit have lower levels of bioluminescence, a surrogate for ATP levels, compared to worms fed on wildtype *G. oxydans*. FIG. 4F, ATP production was decreased in worms fed on deletion mutations in Membrane-bound PQQ-dependent glucose dehydrogenase, FAD-dependent Membrane-bound dehydrogenase, and Membrane-bound PQQ-dependent dehydrogenase 3 compared to worms fed on wildtype *G. oxydans*. FIG. 4G, PGC-1 α gene expression was increased in human primary dermal fibroblasts exposed to *Gluconoacetobacter* spp compared to mock-treated cells as assessed by RT-PCR. FIG. 4H, TFAM gene expression was increased in human primary dermal fibroblasts exposed to *Gluconoacetobacter* spp compared to mock-treated cells as assessed by RT-PCR. FIG. 4I, mtDNA copy numbers were increased in human primary dermal fibroblasts exposed to *Gluconoacetobacter* spp compared to mock-treated cells.

[0127] FIGS. 5A-5I. FIG. 5A, *Gluconacetobacter* spp, *Acetobacter* spp, and *Gluconobacter* spp suppresses the developmental delay phenotype of spg-7(ad2249). FIG. 5B, *Gluconacetobacter* spp, *Acetobacter* spp, and *Gluconobacter* spp accelerates the developmental growth rate of wildtype animals compared to the worms fed on *E. coli* OP50-1. FIG. 5C, Representative picture showing the accelerated developmental growth of wildtype worms fed on *Gluconacetobacter* spp. FIG. 5D, Wildtype worms fed on *Gluconacetobacter* spp are slightly longer compared to the worms fed on *E. coli* OP50-1. *Gluconacetobacter* spp feeding reduces the time taken for animals to reach egg-laying adult stage from L1-larval stage. FIG. 5E, *Gluconacetobacter* spp feeding reduces the time taken for animals to reach egg-laying adult stage from L4-larval stage. FIG. 5F, *Gluconacetobacter* spp induces increase in bioluminescence, a surrogate for ATP production, within 6 hours of initiating feeding with maximal levels by 18 hours of feeding spg-7(ad2249) mutants fed on *Gluconacetobacter* spp had significantly increased mitochondrial membrane potential compared to animals fed on *E. coli* OP50 as assessed using TMRE dye. FIG. 5G, Animals fed on *Gluconacetobacter* spp and treated with rotenone or paraquat or sodium azide had significantly higher levels of bioluminescence, a surrogate for ATP levels, compared to animals fed on *E. coli* OP50 and treated with rotenone or paraquat or sodium azide. FIG. 5H, spg-7(ad2249) mutants fed on *Gluconacetobacter* spp. had significantly increased mitochondrial membrane potential compared to animals fed on *E. coli* OP50 as assessed using TMRE dye. FIG. 5I, Animals fed on *Gluconacetobacter* spp and treated with rotenone or paraquat or sodium azide had significantly higher levels of bioluminescence, a surrogate for ATP levels, compared to animals fed on *E. coli* OP50 and treated with rotenone or paraquat or sodium azide.

[0128] FIGS. 6A-6L. FIG. 6A, Animals fed on *Gluconacetobacter* spp were more resistant to rotenone or paraquat than animals fed on *E. coli* OP50. FIG. 6B, Feeding spg-7(ad2249); tbb-6::gfp animals on *Gluconacetobacter* spp suppressed the activation of tbb-6::gfp expression. FIGS. 6C-6G, *Gluconacetobacter* feeding increases cpt-2::gfp (FIG. 6C), acox-1.2::gfp (FIG. 6D), pmp-4::gfp (FIG. 6E), acdh-7::gfp (FIG. 6F), ech-1.2::gfp (FIG. 6G) expression compared to worms fed on *E. coli* OP50. FIG. 6H, Representative figure showing the increased expression of B0303.3::gfp in worms fed on *Gluconacetobacter* compared to fed on *E. coli* OP50. FIG. 6I, Representative figure showing the increased expression of acdh-7::gfp in worms fed on *Gluconacetobacter* compared to fed on *E. coli* OP50. FIG. 6J, Representative figure showing the increased expression of acox-1.2::gfp in worms fed on *Gluconacetobacter* compared to those fed on *E. coli* OP50. FIG. 6K, Representative figure showing the increased expression of cpt-1::gfp in worms fed on *Gluconacetobacter* compared to fed on *E. coli* OP50. FIG. 6L, Representative figure showing no change in expression of rpt-3::gfp in worms fed on *Gluconacetobacter* compared to fed on *E. coli* OP50.

[0129] FIGS. 7A-7H. FIG. 7A, Representative figure showing the increased expression of icl-1::gfp in worms fed on *Gluconacetobacter* compared to worms fed on *E. coli* OP50. FIG. 7B, *Gluconacetobacter* feeding does not affect fat-7::gfp expression. FIG. 7C, Representative figure showing that *Gluconacetobacter* feeding does not affect UbV-gfp expression while treatment with Bortezomib stabilizes

UbV-gfp. FIG. 7D, Representative figure showing that *Gluconacetobacter* feeding does not induce autophagy while starvation induces autophagy as assessed using the lgg-1::GFP strain. FIG. 7E, Intestine-specific overexpression of nhr-49 was sufficient to accelerate growth rate of nhr-49(nr2041) mutants fed on *Gluconacetobacter* spp. FIG. 7F, Representative figure showing that the nhr-49 gene expression was induced in the worms fed on *Gluconacetobacter* spp as assessed using a nhr-49::GFP transcriptional fusion. FIG. 7G, Representative figure showing that TCER-1::GFP expression is increased in the intestinal cell nuclei compared to worms fed on *E. coli* OP50. FIG. 7H, Representative figure showing that *Gluconacetobacter* feeding induces mitochondrial fusion in the hypodermis of worms compared to worms fed on *E. coli* OP50 as assessed using Rhodamine 6G dye, which stains mitochondria.

[0130] FIGS. 8A-8H. FIG. 8A, Representative figure showing that *Gluconacetobacter* feeding does not restore the peroxisomal targeting defect in prx-5(RNAi) treated worms. FIG. 8B, Representative figure showing that *Gluconacetobacter* feeding does not affect the golgi morphology or distribution as assessed by a transgenic strain that targets GFP to the golgi. FIG. 8C, Representative figure showing that *Gluconacetobacter* feeding does not affect the gut lysosomal morphology or distribution as assessed by a transgenic strain that targets GFP to the lysosomes. FIG. 8D, Representative figure showing that *Gluconacetobacter* feeding does not affect the Endoplasmic Reticulum (ER) morphology or distribution as assessed by a transgenic strain that targets GFP to the ER. FIG. 8E, Worms fed on *Gluconacetobacter* spp and treated with H₂O₂ had lower levels of H₂O₂ compared to the worms fed on *E. coli* OP50 as assessed using the Amplex red assay. FIG. 8F, Glutathione intracellular concentrations were not affected in worms fed on *Gluconacetobacter* spp compared to worms fed on *E. coli* OP50. FIG. 8G, Representative figure showing the Halo formation around colonies in CaCO₃ media plates produced by different members of Acetobacteraceae. FIG. 8H, *Gluconobacter oxydans* transposon mutants failed to suppress the slow developmental growth phenotype of *C. elegans* spg-7(ad2249).

[0131] FIGS. 9A-9E. FIG. 9A, Representative figure showing the *Gluconobacter oxydans* transposon mutants that failed to suppress the slow developmental growth phenotype of wildtype animals. FIG. 9B, Mutants in membrane-bound PQQ-dependent glucose dehydrogenase, cytochrome o ubiquinol oxidase subunit II, ubiquinol-cytochrome c reductase iron-sulfur subunit, and carbon-nitrogen hydrolase fail to form clear halo around media plates containing CaCO₃. FIG. 9C, Deletion alleles in various dehydrogenases and found that spg-7(ad2249) mutant worms fed on deletion mutations in Membrane-bound PQQ-dependent glucose dehydrogenase, FAD-dependent Membrane-bound dehydrogenase, and Membrane-bound PQQ-dependent dehydrogenase 3 grew slower compared to worms fed on wildtype *G. oxydans* FIG. 9D, Exposure of *E. coli* OP50 to human primary dermal fibroblasts results in significant reduction in the ATP production while *Gluconacetobacter* spp does not affect ATP production compared to the mock-treated cells. FIG. 9E, *Gluconacetobacter* spp does not reduce the cellular ATP production even at higher doses of the bacteria.

[0132] FIGS. 10A-10B. FIG. 10A, Transgenic worms expressing human α -synuclein and GFP in the dopaminergic

neurons were fed with *E. coli* op50 or EBT405 from first larval stage and subjected to Rotenone treatment at the L4-larval stage. Arrows show the axons of CEP neurons. In the rotenone treated animals, the CEP neurons are degenerating or lost. FIG. 10B, Quantification of the CEP neuron loss. Three independent trials were performed and each trial consisted of 20-30 animals.

[0133] FIG. 11 Panneuronal expression of human $A\beta_{1-42}$ is sufficient to induce UPR^{mt} reporter phsp-6::gfp in the intestine. Representative fluorescence micrographs of strains either not expressing human $A\beta_{1-42}$ (control strain on the left) and strain expressing the human $A\beta_{1-42}$ (Center and right images) fed on either negative control bacteria or *Gluconoacetobacter* EBT405 bacteria. All the strains contain phsp-6::gfp and phsp-4::mcherry reporters. While only basal levels of phsp-6::gfp expression is observed in the control strain (Left image), the GFP is activated in the gut of strain carrying human $A\beta_{1-42}$ (Center image); however, in the strain fed on the *Gluconoacetobacter* EBT405 bacteria, the phsp-6::gfp expression is reduced. The ER unfolded protein response UPR^{ER} reporter phsp-4::mcherry is only weakly activated in the spermatheca (white arrowheads). Scale bar on the left image is 200 μ M.

[0134] FIG. 12 Panneuronal expression of human aggregative tau mutant is sufficient to induce UPR^{mt} stress reporter phsp-6::gfp in the intestine. Representative fluorescence micrographs of strains either not expressing human tau mutant (control strain on the left) and strain expressing the human tau mutant (Center and right images) fed on either negative control bacteria or EBT405 bacteria. All the strains contain phsp-6::gfp, phsp-4::mcherry reporters and a pharyngeal marker (white arrows), pmyo-2::mcherry. While only basal levels of phsp-6::gfp expression is observed in the control strain (Left image), the GFP is activated in the gut of strain carrying human tau mutant (Center image); however, in the strain fed on the *Gluconoacetobacter* EBT405 bacteria, the phsp-6::gfp expression is reduced. This reduction is specific because the expression of the pharyngeal marker is not affected. The UPR^{ER} reporter phsp-4::mcherry is only weakly activated in the spermatheca (white arrowheads). Scale bar on the left image is 200 μ M.

[0135] FIG. 13 shows a schematic representation of the HTS strategy.

[0136] FIG. 14 shows data indicating that EBT 405 feeding reduces $A\beta$ aggregates. Representative fluorescence micrographs of CL2331 strain expressing the $A\beta_{3-42}$ peptide conjugated with GFP fed on negative control bacteria or EBT405 and imaged 60 hours post L1 stage. Arrows indicate amyloid deposits. Scale bar, 200 μ M

DETAILED DESCRIPTION

[0137] Provided herein are methods and compositions for the treatment or prevention of diseases and/or disorders involving or characterized by mitochondrial dysfunction and/or peroxisomal dysfunction. Such compositions comprise one or more bacterial species from the Acetobacteriaceae family or a bacterial extract or component derived therefrom.

Mitochondrial Function/Dysfunction and Production of Intracellular ATP

[0138] Mitochondria are intracellular organelles present in most eukaryotic cells, which primarily function to produce

energy, in the form of ATP, through oxidative phosphorylation for use in a multitude of cellular processes. For example, energy derived from the metabolism of glucose or fatty acids is converted to ATP, which is then used to drive various energy-requiring biosynthetic reactions and other metabolic activities. Mitochondria have their own genomes, separate from nuclear DNA, comprising circular, double-stranded DNA with about 16,000 base pairs in human cells. Each mitochondrion may have multiple copies of its genome, and individual cells may have hundreds of mitochondria. Mitochondria play a role in the tricarboxylic acid (TCA) cycle, heme synthesis, β -oxidation of fatty acids, amino acid metabolism, and the like. Moreover, functions for maintaining calcium homeostasis, an active oxygen production system, and transport systems for metabolites, ions, proteins, and the like are present in mitochondria. Hence, mitochondria play an important role in both catabolic and anabolic reactions in eukaryotic cells.

[0139] It has been shown that mitochondrial dysfunction can contribute to a variety of diseases. Some mitochondrial diseases are due to mutations or deletions in the mitochondrial genome. Mitochondria divide and proliferate with a faster turnover rate than their host cells, and their replication is under control of the nuclear genome. Thus, if a threshold proportion of mitochondria in a cell are defective, and if a threshold proportion of such cells within a tissue have defective mitochondria, symptoms of tissue or organ dysfunction can result. Practically any tissue can be affected, and a large variety of symptoms may be present, depending on the extent to which different tissues are involved.

[0140] Diseases of the mitochondria are apparent most often in organs that have a high energy demand, such as the brain, heart, liver, skeletal muscles, kidney and the endocrine and respiratory systems. Even a mild impairment in mitochondrial function can be reflected by symptoms of disease. Symptoms of a mitochondrial disease can include loss of motor control, muscle weakness and pain, seizures, visual/hearing problems, cardiac diseases, liver diseases, gastrointestinal disorders, swallowing difficulties and more. A mitochondrial disease can be inherited or can be due to spontaneous mutations, either of which can lead to altered functions of the proteins or RNA molecules normally residing in the mitochondria.

[0141] It is important to note that the bacterial compositions, extracts and/or components described herein will have the greatest effect in cells, tissues, organs, etc. of a subject who retains at least some mitochondrial activity, even if the activity is impaired. While the bacterial compositions, extracts and/or components described herein can induce both mitochondrial and peroxisomal biogenesis (thereby enhancing total cellular metabolic function), the benefit of the compositions described herein is thought to be due to an enhancement of existing mitochondrial function through the electron transport chain. That is, the compositions described herein do not provide an exogenous gene for repair of mitochondrial dysfunction, but rather will enhance the existing function of mitochondria by either increasing the number of mitochondria and/or the function of the electron transport chain within each mitochondrion. This enhancement of function is likely due, in part, to the metabolites generated by the bacteria as described herein.

[0142] Disorders associated with impaired mitochondrial function also include, for example, metabolic disorders, neurodegenerative disorders, aging related disorders and

chronic inflammatory disorders. Mitochondrial disorders can also include diseases with inherited and/or acquired mitochondrial dysfunction, such as Charcot-Marie-Tooth Disease Type 2A2, Mitochondrial Encephalopathy Lactic Acidosis and Stroke (MELAS), Leigh syndrome, Barth syndrome, Leber's optic neuropathy, fatty acid oxidation disorders, inherited forms of deafness and blindness, metabolic abnormalities induced by exposure to toxic chemicals and/or drugs (e.g., cisplatin induced deafness).

[0143] To reiterate, the methods and compositions described herein are not intended to cure diseases or disorders with impaired mitochondrial function but rather are useful in reducing at least one symptom of disease that is associated with impaired mitochondrial function. That is, while the compositions described herein cannot correct an inherited mitochondrial defect, they can be used to enhance mitochondrial function that is present in the cell despite the presence of disease. This can be achieved, for example, by increasing mitochondrial number or increasing flux through the electron transport chain (e.g., at complex I and complex III). Thus, in one embodiment, a subject to be treated with the compositions and methods described herein retains at least 10% mitochondrial and/or peroxisomal function as (prior to treatment) as compared to a non-affected subject. In other embodiments, the subject retains at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% mitochondrial and/or peroxisomal function (prior to treatment) as compared to a non-affected individual.

[0144] It is also contemplated herein that individuals who retain normal mitochondrial and/or peroxisomal function can be treated to enhance mitochondrial function to enhance performance, for example, athletic performance.

[0145] In certain embodiments, the methods and compositions described herein can increase mitochondrial and/or peroxisomal function by at least 10% (e.g., as assessed by measuring ATP production) compared to the mitochondrial and/or peroxisomal function prior to such treatment. In some embodiments, mitochondrial and/or peroxisomal function is increased by at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 99%, at least 1-fold, at least 2-fold, at least 5-fold, at least 10-fold, at least 20-fold, at least 50-fold, at least 100-fold, at least 1000-fold or more as compared to the mitochondrial and/or peroxisomal function prior to such treatment.

Deficiencies in the Electron Transport Chain:

[0146] Many diseases have been found to involve a mitochondrial deficiency such as a Complex I, II, III or IV impairment or an enzyme deficiency e.g. pyruvate dehydrogenase impairment. Complex I, II, III and IV are protein complexes embedded in the inner membrane of the mitochondrion. They are known as the respiratory chain and function by coupling electron transfer between an electron donor (such as NADH) and an electron acceptor (such as O_2) with the transfer of H^+ ions. The resulting electrochemical proton gradient over the inner membrane is used to generate chemical energy in the form of adenosine triphosphate (ATP) by oxidation of glucose, pyruvate and NADH, which all are produced in the cytosol of the cell. This process of cellular respiration, also known as aerobic respiration, is dependent on the presence of oxygen. When oxygen is limited, the glycolytic products will be metabolized by

anaerobic fermentation, a process that is independent of the mitochondria. The production of ATP from glucose is about 13 times higher during aerobic respiration compared to fermentation.

[0147] Complexes I-IV carry electrons along four series of reactions, known as the Electron Transport Chain, resulting in energy production. A fifth group (Complex V) is an ATP synthase enzyme that generates ATP. Together, the electron transport chain and the ATP synthase form the respiratory chain and the entire process is known as oxidative phosphorylation or OXPHOS.

Complex I Deficiency:

[0148] Complex I, the first step in this chain, is the most common site for mitochondrial abnormalities, representing as much as one third of the respiratory chain deficiencies. Often presenting at birth or in early childhood, Complex I deficiency is usually a progressive neuro-degenerative disorder and is responsible for a variety of clinical symptoms, particularly in organs and tissues that require high energy levels, such as brain, heart, liver, and skeletal muscles. A number of specific mitochondrial disorders have been associated with Complex I deficiency including: Leber's hereditary optic neuropathy (LHON), mitochondrial myopathy, encephalopathy, lactic acidosis and stroke (MELAS), myoclonic epilepsy with ragged red fibers (MERRF), and Leigh Syndrome (LS).

[0149] To date, three major forms of Complex I deficiency have been identified: i) Fatal infantile multisystem disorder—characterized by poor muscle tone, developmental delay, heart disease, lactic acidosis, and respiratory failure; ii) Myopathy (muscle disease)—starting in childhood or adulthood, and characterized by weakness or exercise intolerance; and iii) Mitochondrial encephalomyopathy (brain and muscle disease)—beginning in childhood or adulthood and involving variable symptom combinations, which can include: eye muscle paralysis, pigmentary retinopathy (retinal color changes with loss of vision), hearing loss, sensory neuropathy (nerve damage involving the sense organs), seizures, dementia, ataxia (abnormal muscle coordination), and involuntary movements. This form of Complex I deficiency may cause Leigh Syndrome and MELAS. Most cases of Complex I deficiency result from autosomal recessive inheritance (combination of defective nuclear genes from both the mother and the father). Less frequently, the disorder is maternally inherited or sporadic and the genetic defect is in the mitochondrial DNA.

[0150] In one embodiment, the compositions and methods described herein increase intracellular ATP levels by at least 10%, at least 20%, at least 25%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98%, at least 99%, at least 1-fold, at least 2-fold, at least 5-fold, at least 10-fold, at least 20 fold, at least 100-fold, at least 1000-fold or more compared to the level of intracellular ATP in a substantially similar cell or cell population, with the exception that the reference cells are not contacted with the compositions or metabolites described herein. In one embodiment, the compositions and methods described herein restore intracellular ATP levels in a subject having mitochondrial dysfunction relative to a subject (or group of subjects) not treated with such compositions and methods.

Metabolic Disorders:

[0151] Metabolic disorders include, for example, type II diabetes, obesity, hyperglycemia, glucose intolerance, insulin resistance (i.e., hyperinsulinemia, metabolic syndrome, syndrome X), hypercholesterolemia, hypertension, hyperlipoproteinemia, hyperlipidemia (e.g., dyslipidemia), hypertriglyceridemia, cardiovascular disease, atherosclerosis, peripheral vascular disease, kidney disease, ketoacidosis, thrombotic disorders, nephropathy, diabetic neuropathy, diabetic retinopathy, sexual dysfunction, dermatopathy, dyspepsia, hypoglycemia, metabolic syndrome, cancer or edema. While the methods and compositions described herein may not address the underlying cause(s) of such diseases or disorders, the promotion of mitochondrial function provided by such methods and compositions can ameliorate those aspects of these diseases or disorders that are related to mitochondrial dysfunction or insufficiency.

[0152] In one embodiment of the methods and compositions provided herein, the subject is suffering from or is susceptible to developing a metabolic disorder. Subjects suffering from or at risk of developing a metabolic disorder are identified by methods known in the art. For example, diabetes can be diagnosed by measuring e.g., fasting blood glucose levels or insulin or by a glucose tolerance test. Normal adult glucose levels are 60-126 mg/dl. Normal insulin levels are 7 mU/mL $\pm$ 3 mU. Glucose intolerance is diagnosed by a two-hour glucose level of 140 to 199 mg per dL (7.8 to 11.0 mmol) on the 75-g oral glucose tolerance test. Insulin resistance is diagnosed by a fasting serum insulin level of greater than approximately 60 pmol/L. Hypoglycemia can be diagnosed by a blood glucose level lower than 2.8 to 3.0 mmol/L (50 to 54 mg/dl). Hypertension can be diagnosed by a blood pressure consistently at or above 140/90. Cardiovascular disease can be diagnosed, at least in part, by measuring cholesterol levels (e.g., LDL, HDL, VLDL etc.). For example, LDL cholesterol above 137 or total cholesterol above 200 is indicative of cardiovascular disease. Hyperglycemia is diagnosed by a blood glucose level higher than 10 mmol/l (180 mg/dl). Obesity is diagnosed for example, by body mass index. Alternatively, waist circumference (estimates fat distribution), waist-to-hip ratio (estimates fat distribution), skinfold thickness (if measured at several sites, estimates fat distribution), or bioimpedance (based on principle that lean mass conducts current better than fat mass (i.e., fat mass impedes current), estimates % fat) is measured. The parameters for normal, overweight, or obese individuals are as follows: Underweight: BMI<18.5; Normal: BMI 18.5 to 24.9; Overweight: BMI=25 to 29.9. Overweight individuals are characterized as having a waist circumference of >94 cm for men or >80 cm for women and waist to hip ratios of $\geq$ 0.95 in men and $\geq$ 0.80 in women. Obese individuals are characterized as having a BMI of 30 to 34.9, being greater than 20% above "normal" weight for height, having a body fat percentage >30% for women and 25% for men, and having a waist circumference >102 cm (40 inches) for men or 88 cm (35 inches) for women. Individuals with severe or morbid obesity are characterized as having a BMI of $\geq$ 35.

Neurodegenerative Disease:

[0153] Neurodegenerative disorders include diseases such as dementia, Alzheimer's disease, Parkinson's disease, and Huntington's disease.

[0154] In one embodiment, the neurodegenerative disorder is Parkinson's disease. Parkinson's disease (PD) is a neurodegenerative movement disorder characterized by resting tremor, rigidity, bradykinesia, and postural instability. PD symptoms are classically attributed to dopamine depletion and the degeneration of dopaminergic neurons in the substantia nigra pars compacta (SNc). However, additional neuronal circuits are affected, and non-motor symptoms are often present, suggesting a systemic pathology. There is compelling evidence that mitochondrial dysfunction is a primary event in the disease process. It has been reported that PD-related mutations and mitochondrial dynamics have a reciprocal relationship. PD-related mutations can perturb mitochondrial dynamics, and the consequences of these mutations can be modulated by mitochondrial dynamics.

[0155] In one embodiment, effective treatment of Parkinson's disease is determined by a reduction in the dose of pharmacological treatments, such as L-DOPA, required to maintain adequate control of symptoms of Parkinson's disease. In another embodiment, efficacy of treatment is monitored using the Unified Parkinson's Disease Rating Scale (UPDRS) as known in the art.

[0156] Alzheimer's disease, amyotrophic lateral sclerosis (ALS), Huntington's disease, Friedrich's ataxia, hereditary spastic paraplegia, neurodegeneration with brain iron accumulation (NBIA), and optic atrophy type I are also each associated with various mitochondrial activity deficits (see e.g., Lin et al. Nature 443:787-795 (2006)).

Inflammatory Disease:

[0157] Chronic inflammatory diseases include disease such as celiac disease, vasculitis, lupus, chronic obstructive pulmonary disease (COPD), irritable bowel disease, atherosclerosis, arthritis, and psoriasis. While the methods and compositions described herein may not address the underlying cause(s) of such diseases or disorders, the promotion of mitochondrial function provided by such methods and compositions can ameliorate those aspects of these diseases or disorders that are related to mitochondrial dysfunction or insufficiency.

Aging:

[0158] Aging is a progressive accumulation of metabolic and physiologic changes associated with an increasing susceptibility to disease. In the membrane hypothesis of aging (MHA), aging is related to decreasing effectiveness of cellular protective and reparative mechanisms secondary to damage from oxygen radicals. This yields biochemical and metabolic errors which progressively accumulate, resulting in cell aging and ultimately death. Therefore, the MHA suggests that reactive oxygen species (ROS) induced cell membrane structural damage is the primary mediator in cellular aging.

[0159] ROS are a normal byproduct of oxidative phosphorylation, and are also formed under conditions of ischemia, hypoperfusion and in response to environmental contaminants. Among the many detrimental activities of ROS, or free oxygen radicals, is direct damage to mitochondrial DNA (mtDNA). Progressive accumulation of mtDNA damage renders cells unable to conduct oxidative phosphorylation reactions effectively, thereby leading to a bioenergetically deficient cell. Over time, mitochondrial DNA damage accumulates and leads to cellular dysfunction with subse-

quent organ failure, aging and ultimately death. Additionally, there is evidence of a reduction in the oxidant-protective enzymes superoxide dismutase and catalase associated with aging. Thus not only are there increases in the deleterious effects of ROS, but there is a reduction in the enzymes and mitochondrial metabolites necessary for protection from ROS and for effective mitochondrial function.

Peroxisomal Disorders

[0160] In some embodiments, the subject has a peroxisomal disorder, with or without the presence of mitochondrial dysfunction.

[0161] Peroxisomes are intracellular organelles that are expressed in all eukaryotic cells except erythrocytes and play an important role in intracellular metabolism. Peroxisomes are dynamic organelles that change in shape and numbers based on the specific metabolic needs of different tissues and cell types. Peroxisomes are capable of catabolizing very long chain fatty acids to shorter fatty acids that can then enter the β -oxidation cycle. (VLCFA: carbon chain >22). Peroxisomes also play a role in intracellular antioxidant activity.

[0162] Peroxisomes are also involved in the synthesis of lipids such as bile acids, docosahexanoic acid, an omega 3 fatty acid, and plasmalogens, a specialized class of membrane phospholipids. Although a single cell may be able to survive without peroxisomes in certain conditions, this organelle is essential for the proper function and viability of tissues and organs. The importance of peroxisomes is most evident by the group of diseases resulting from the lack of functioning peroxisomes called Peroxisome Biogenesis Disorders or "PBD" (Weller et al., 2003).

[0163] In some embodiments, the methods and compositions described herein are not intended to cure peroxisomal diseases or disorders but rather are useful in reducing at least one symptom of disease that is associated with impaired peroxisomal function. That is, while the compositions described herein cannot correct an inherited peroxisomal defect, they can be used to enhance existing peroxisomal function. This can be achieved, for example, by increasing peroxisomal number or increasing flux through the metabolic pathways of the peroxisome (e.g., metabolism of very long chain fatty acids (VLCFA)). Thus, in one embodiment, a subject to be treated with the compositions and methods described herein retains at least 10% of the peroxisomal function as compared to the peroxisomal function of a non-affected subject. In other embodiments, the subject retains at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 99% peroxisomal function as compared to a non-affected individual.

[0164] Peroxisome Biogenesis Disorders are a continuum of disorders that are divided into three phenotypes based on the severity of the disorder. Zellweger Syndrome (ZS) is the most severe form; followed by neonatal adrenoleukodystrophy (NALD), an intermediate form, and infantile Refsum disease (IRD), which is the least severe. Children with ZS present early in the neonatal period with profound hypotonia, facial dysmorphisms, liver dysfunction, seizures, and rarely survive past their first year of life. At a biochemical level, laboratory tests show profound deficiencies in peroxisomal metabolism and cells from these patients show near absence of functional cellular peroxisomes. NALD and IRD are more difficult to distinguish from each other, from the

majority of cases, and patients can live through adulthood. Both may present soon after birth, but are often diagnosed later by developmental delays, mild liver dysfunction, and hearing and visual impairment leading to their classification as deaf-blind children. A proportion of NALD-IRD children will develop a leukodystrophy that represents destruction of normal myelin. They may also develop adrenal gland insufficiency and osteopenia, the latter causing pathological fractures. Cells from patients with NALD-IRD contain more functional peroxisomes than ZS patients, and laboratory tests show higher residual peroxisome metabolism. Clinical features result from deficiency of products made by the peroxisome and toxicity of substrates that accumulate (such as very long chain fatty acids).

[0165] These autosomal recessive diseases are derived from dysfunction of peroxisomes and current treatment is primarily focused on supportive care, dietary management, symptomatic therapy and treatment strategies involving the use of pharmacological induction of peroxisomes. Administration of bacteria, or an extract or metabolite preparation, as described herein to a subject with a peroxisome deficiency can improve peroxisome activities or function and thereby provide therapeutic benefit.

Engineered Bacteria

[0166] In some embodiments, one or more of the compositions or bacterial extracts as described herein comprises an engineered microbe(s). For example, engineered microbes include microbes harboring one or more introduced genetic changes, such change being an insertion, deletion, translocation, or substitution, or any combination thereof, of one or more nucleotides contained on the bacterial chromosome or on an endogenous plasmid, wherein the genetic change can result in the alteration, disruption, removal, or addition of one or more protein coding genes, non-protein-coding genes, gene regulatory regions, or any combination thereof, and wherein such change can be a fusion of two or more separate genomic regions or can be synthetically derived. The engineered microbe(s) can be produced using techniques including but not limited to site-directed mutagenesis, transposon mutagenesis, knock-outs, knock-ins, polymerase chain reaction mutagenesis, chemical mutagenesis, ultraviolet light mutagenesis, transformation (chemically or by electroporation), phage transduction, or any combination thereof.

[0167] In one embodiment, the bacterium is from the Acetobacteriaceae family of gram negative bacteria. Members of the Acetobacteriaceae family are generally grouped according to their ability to oxidize sugars or ethanol to produce acetic acid during fermentation. In one embodiment, the bacterium is an *Acetobacter* spp., which is distinguished from other members of the Acetobacteriaceae family, including *Gluconobacter* spp., and *Gluconoacetobacter* spp.

[0168] In another embodiment, the bacterium is a *Gluconobacter* spp. Non-limiting examples of *Gluconobacter* spp. include *Gluconobacter albidus*, *Gluconobacter asaii*, *Gluconobacter cerevisiae*, *Gluconobacter cerinus*, *Gluconobacter frateruii*, *Gluconobacter japonicus*, *Gluconobacter kanchanaburiensis*, *Gluconobacter kondonii*, *Gluconobacter nephelii*, *Gluconobacter oxydans*, *Gluconobacter sphaericus*, *Gluconobacter thailandicus*, *Gluconobacter uchimurae*, and *Gluconobacter wancheriae*.

[0169] In another embodiment, the bacterium is a *Gluconacetobacter* spp. Non-limiting examples of *Gluconacetobacter* spp. include *Gluconoacetobacter aggeris*, *Gluconoacetobacter asukensis*, *Gluconoacetobacter azotocaptans*, *Gluconoacetobacter diazotrophicus*, *Gluconoacetobacter entanii*, *Gluconoacetobacter europaeus*, *Gluconoacetobacter hansenii*, *Gluconoacetobacter intermedius*, *Gluconoacetobacter johannae*, *Gluconoacetobacter kakiaceti*, *Gluconoacetobacter kombuchae*, *Gluconoacetobacter liquefaciens*, *Gluconoacetobacter maltaceti*, *Gluconoacetobacter medellinensis*, *Gluconoacetobacter nataicola*, *Gluconoacetobacter oboediens*, *Gluconoacetobacter rhaeticus*, *Gluconoacetobacter sacchari*, *Gluconoacetobacter saccharivorans*, *Gluconoacetobacter sucrofermentans*, *Gluconoacetobacter swing-sii*, *Gluconoacetobacter takamatsuzukensis*, *Gluconoacetobacter tumulicola*, *Gluconoacetobacter tumulisoli*, and *Gluconoacetobacter xylinus*. In another embodiment of this aspect and all other aspects provided herein, the bacterium is *Gluconobacter* EBT 405.

[0170] In some embodiments, the bacterium is *Gluconobacter albidus*, *Gluconobacter cerinus*, *Gluconobacter frateruii*, *Gluconobacter japonicus*, *Gluconobacter kon-*

donii, *Gluconobacter nephelii*, *Gluconobacter oxydans*, *Gluconoacetobacter diazotrophicus*, *Gluconoacetobacter hansenii*, *Gluconoacetobacter saccharivorans*, *Acetobacter aceti* or *Acetobacter malorum*.

[0171] In some embodiments, the bacterium is not a member of the Acetobacteriaceae family but is genetically modified to produce similar metabolites (e.g., gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid). For example, in some embodiments, a bacterium as described herein is genetically modified to express, e.g., one or more of: membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II. Methods of introducing heterologous genetic material to bacterial cells by transformation or transduction are well known, and include, as non-limiting examples, chemical transformation, particle bombardment, and electroporation. The approach best suited to a given bacterium will be known to those of ordinary skill in the art.

[0172] The following sequences are provided for selected metabolic pathway enzymes that can be expressed in an engineered bacterium.

SEQ ID NO: 1

```
>mGDH nucleotide sequence
atgagcacaatttcccgccagggtctctgggccctgataacggccgcgggtattcgcgctttgcggcgcc
gatccttacggttgccggcgcatgggtcgctgccatcgccggccctctttattatgtcatccttgccc
tggaacttctcgccacgggtttcctctcattccggcgcaatccggctgcccctctatctgttcgcagtc
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SEQ ID NO: 2

```
>mGDH protein sequence
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LLPFVSRQIGGTRTTVLPLAGAVGVAVLALFASLFTDPHDISGELPTQIANASPADPD
NVPASEWHAYGRTOAGDRWSPLNQINATNVSNLKVAWHIHTKDMMSNDPGEATNEAT
PIEFNNTLYMCSLHQLFVAVDGTGNVKNVYDPLQLINPFGQHLTCRGVSFHETPANA
MDSGDNPAPTDCAKRIILPVNDGRLVEVDADTGKTCSGFGNGEIDLRLVNPQYPTFG
QYEPTSPPVITDKLIIANSAITDNGSVKQASGATQAFDVYTGKRVWVFDASNPDPNQL
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PDESHPVFHPNSPNSWISSYDANLNLVYIPMGVGTDPQWGGDRTKDSERFAPGIVAL
NADTGKLAWFYQTVHHDLDWMDVPSQPSLVDTQKDGTLVPAIYAPTKTGDIFVLDRR
TGKEIVPAPEPTVPQGAAPGDHTSPTQPMQLTLRPKNPLNDSDIWGGTIFDQMFCSI
YFHTLRYEGPFTPPSLKGSILFPGLGMFEWGGGLAVDPQRQVAFANPISLFPVSQLVP
RGPGNPLWPEGNAKTGGTGLQHNYGIPYAVNLHPFLDPVLLPFGIKMPCRTPPWGY
VAGIDLTKNKVVWQHNRNGLRDSMYGSSLPILPPIKIGVPSLGGPLSTAGNLGFLTA
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SEQ ID NO: 3

>ubiquinol-cytochrome c reductase iron-sulfur subunit nucleotide sequence
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SEQ ID NO: 4

> ubiquinol-cytochrome c reductase iron-sulfur subunit protein sequence
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AILQKLRDPESCIFQQPKDATNWHRSVSPDIGVMIGICTHLGCVPTFDAPTQAEPAKG
YLCPCHGSQFDSAGRAYRNAPAPYNLPVPPVTMISDTHLRIGESKNDPDFDIANIQQI

SEQ ID NO: 5

>TonB-dependent receptor nucleotide sequence
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SEQ ID NO: 6

>TonB-dependent receptor protein sequence
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 LGKSRVLVLIDGKRAAIDGSSSCFDLNTINIHQVASVEILKDGGSSELYGADAVSGVIN
 IKLKHNLDDANLTVRGGITDRGDGQSGMISGYKGNFDHGRGNVTVSGSYMTQSGIRQ
 NSRDWANPVVSGLIAPGGSPTYGSSIPTAGRFITDTADNVPNGDGSFHNFSKKDRYNY
 GNDQSLTNSLQDATLSFDAHYDVNRHFTPYGNFLYSHRNSNTQMAPIPVSGSIYPSTL
 PVAITIPGSAPYNSLGEDATMYKRMGEWGDVRSQTATDITYTAKIGASGDI THGWKYDL
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 QLAYHPDALVESGQTLTNSASYTGGGFNVTEGYLEGKATLLHNAFLAKDLTIDGQGRY
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 SGQVGTYSLSLTPIVAANCAKQGINSSNFVSASSSQVPTLFGGNPKLKPETGRITYFTGT
 TVTPRWIPGLSASVEYWHYTLKNMISYLSQYIMNQCYTGANTSYCNDITRVGSTNQL
 NSVTALYGNIGGLKTSGLDFDLDIRVTSRDVLTLSNNFQQLVSYLQQLNELGGKWN
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SEQ ID NO: 7

>carbon-nitrogen hydrolase nucleotide sequence
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SEQ ID NO: 8

>carbon-nitrogen hydrolase protein sequence
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SEQ ID NO: 9

>ubiquinol oxidase subunit II nucleotide sequence
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SEQ ID NO: 10

>ubiquinol oxidase subunit II protein sequence
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 LGAISWYSTHAYDPYRPLQTADNVKPLNVQVSLDWKWLFIYPDLGIATINQLDVPTN
 TPLNPFQITSDTVMTSFFIPRLGSMIYSMPGQQTQLHLLATESGDYLGAEASFSGRGS
 DMKFRITLAMAPEEFSAWVEKVKSGSENLDITYPKYAAPQEAAPVQYFAHVQPDLDG
 IVAKYNNNGMMVEKTTGKVMHMQSASSAAPSDTGMKE

[0173] In order to facilitate transfer of genetic material (e.g., a plasmid, DNA etc.) to a bacterium, artificial cell competence can be induced by exposing the bacterium to particular conditions. For example, one method of inducing

cell competence is by incubating the bacterium in a solution with divalent cations (e.g., calcium chloride) to partially disrupt the membrane and then heat-shocking the host cells to induce them to take up e.g., plasmid DNA. An alternative

method for inducing cell competence is electroporation, where the cells are exposed to an electric field, which can generate small holes in the cell membrane such that plasmid DNA can enter the cell.

[0174] The plasmid-supplied nucleic acid (e.g., DNA) can be stably integrated into the genome or can be maintained episomally, e.g., on a plasmid or other episomal vector. In some embodiments, a sequence directing the expression of the one or more enzymes described herein can be placed under the control of naturally-occurring regulatory elements in the cell. In other embodiments, constructs for the expression of at least one of: membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II will generally include regulatory elements, including, promoters, enhancers, etc. that direct the expression of the encoded sequences. A gene under the control of a set of regulatory elements is generally referred to as “operably linked” to those elements. Typically, an expression vector comprises a transcription promoter, a transgene encoding sequence, and a transcription terminator.

[0175] An expression vector, or a vector, as described herein, is a nucleic acid molecule encoding a gene that is expressed when the molecule is introduced to a host-cell. Typically, an expression vector comprises a transcription promoter, a gene encoding sequence, and a transcription terminator. Gene expression in an expression vector is usually placed under the control of a promoter, and such a gene is said to be “operably linked to” the promoter. Similarly, a regulatory element and a core promoter are operably linked if the regulatory element modulates the activity of the core promoter.

[0176] In some embodiments it may be useful to include in the transformed cells a positive marker that permits the selection of cells positive for the plasmid-supplied nucleic acid sequence(s) in vitro. The positive selectable marker may be a gene that upon being introduced into the host cell expresses a dominant phenotype permitting positive selection of cells carrying the gene. Genes of this type are known in the art, and include, for example, an antibiotic resistance gene (e.g., resistance to blasticidin, mycophenolic acid, puromycin, zeocin, actinomycin, ampicillin, kanamycin, neomycin, polymyxin B sulfate, or streptomycin), or an enzyme that converts a substrate to a colored product (e.g., blue/white screening by expression of B-galactosidase, which converts X-gal into galactose and an insoluble blue pigment) etc. Other selection tools can include e.g., radioactive nucleic acid probes, and labeled antibodies that are specific to the protein expressed by the transgene.

Prebiotics

[0177] A prebiotic is an ingredient or compositions that is not digestible by a bacterial host, e.g., a human, but that is fermentable by, and selectively promotes the growth and/or activity of beneficial bacteria in the host. Such selective fermentation that allows specific changes, both in the composition and/or activity of the gastrointestinal microbiota, that confers neutral or positive benefits upon host well-being and health. Prebiotics can include complex carbohydrates, amino acids, peptides, or other nutritional components useful for the survival, colonization and persistence of the bacterial composition.

[0178] Suitable prebiotics are usually plant-derived complex carbohydrates, oligosaccharides or polysaccharides. Generally, prebiotics are indigestible or poorly digested by humans and serve as a food source for bacteria. Prebiotics, which can be used in the pharmaceutical dosage forms, and pharmaceutical compositions and methods provided herein include, without limitation, galactooligosaccharides (GOS), trans-galactooligosaccharides, fructooligosaccharides, or oligofructose (FOS), inulin, oligofructose-enriched inulin, lactulose, arabinoxylan, xylooligosaccharides (XOS), manooligosaccharides, gum guar, gum Arabic, tagatose, amylose, amylopectin, xylan, pectin, combinations thereof, and the like.

[0179] In some embodiments, the prebiotic comprises a mixture of one or more non-digestible oligosaccharides, non-digestible polysaccharides, free monosaccharides, non-digestible saccharides, starch, or non-starch polysaccharides. Suitable oligosaccharides and their production methods are further described in Laere K J M, “Degradation of structurally different non-digestible oligosaccharides by intestinal bacteria: glycosylhydrolases of *Bi. Adolescentis* (2000) PHD thesis, Wageningen Agricultural University, Wageningen, The Netherlands.

Culture and Storage of Bacteria

[0180] For banking, the strain(s) included in the bacterial composition can be (1) isolated directly from a specimen or taken from a banked stock, (2) optionally cultured on a nutrient agar or broth that supports growth to generate viable biomass, and (3) the biomass optionally preserved in multiple aliquots in long-term storage.

[0181] In embodiments using a culturing step, the agar or broth contains nutrients that provide essential elements and specific factors that enable growth. An example would be a medium composed of 20 g/L glucose, 10 g/L yeast extract, 10 g/L soy peptone, 2 g/L citric acid, 1.5 g/L sodium phosphate monobasic, 100 mg/L ferric ammonium citrate, 80 mg/L magnesium sulfate, 10 mg/L hemin chloride, 2 mg/L calcium chloride, and 1 mg/L menadione. A variety of microbiological media and variations are well known in the art (e.g. R. M. Atlas, *Handbook of Microbiological Media* (2010) CRC Press). Medium can be added to the culture at the start, can be added during the culture, or can be intermittently/continuously flowed through the culture. The strains in the bacterial composition can be cultivated alone, as a subset population of a bacterial composition, or as an entire population of different species or strains comprising a bacterial composition. As an example, a first strain can be cultivated together with a second strain in a mixed continuous culture, at a dilution rate lower than the maximum growth rate of either cell to prevent the culture from washing out of the cultivation.

[0182] The inoculated culture is incubated under favorable conditions for a time sufficient to build biomass. For bacterial compositions for human use this is often at normal body temperature (37° C.), pH, and other parameter with values similar to the normal human niche. The environment can be actively controlled, passively controlled (e.g., via buffers), or allowed to drift. For example, for anaerobic bacterial compositions (e.g., gut microbiota), an anoxic/reducing environment can be employed. This can be accomplished by addition of reducing agents/factors such as cysteine to the broth, and/or stripping it of oxygen. As an

example, a culture of a bacterial composition can be grown at 37° C., pH 7, in the medium above, pre-reduced with 1 g/L cysteine-HCl.

[0183] When the culture has generated sufficient biomass, it can be preserved for banking or storage. The organisms can be placed into a chemical milieu that protects from freezing (adding ‘cryoprotectants’), drying (‘lyoprotectants’), and/or osmotic shock (‘osmoprotectants’), dispensing into multiple (optionally identical) containers to create a uniform bank, and then treating the culture for preservation. Containers are generally impermeable and have closures that assure isolation from the environment. Cryopreservation treatment is accomplished by freezing a liquid at ultra-low temperatures (e.g., at or below -80° C.). Dried preservation removes water from the culture by evaporation (in the case of spray drying or ‘cool drying’) or by sublimation (e.g., for freeze drying, spray freeze drying). Removal of water improves long-term bacterial composition storage stability at temperatures elevated above cryogenic. If the bacterial composition comprises spore forming species and results in the production of spores, the final composition can be purified by additional means such as density gradient centrifugation and preserved using the techniques described above. Bacterial composition banking can be done by culturing and preserving the strains individually, or by mixing the strains together to create a combined bank. As an example of cryopreservation, a bacterial composition culture can be harvested by centrifugation to pellet the cells from the culture medium, the supernatant decanted and replaced with fresh culture broth containing 15% glycerol. The culture can then be aliquoted into 1 mL cryotubes, sealed, and placed at -80° C. for long-term viability retention. This procedure achieves acceptable viability upon recovery from frozen storage.

[0184] Organism production can be conducted using similar culture steps to banking, including medium composition and culture conditions. It can be conducted at larger scales of operation, especially for clinical development or commercial production. At larger scales, there can be several subcultivations of the bacterial composition prior to the final cultivation. At the end of cultivation, the culture is harvested to enable further formulation into a dosage form for administration. This can involve concentration, removal of undesirable medium components, and/or introduction into a chemical milieu that preserves the bacterial composition and renders it acceptable for administration via the chosen route. For example, a bacterial composition can be cultivated to a concentration of 10^{10} CFU/mL, then concentrated 20-fold by tangential flow microfiltration; the spent medium may be exchanged by diafiltering with a preservative medium consisting of 2% gelatin, 100 mM trehalose, and 10 mM sodium phosphate buffer. The suspension can then be freeze-dried to a powder and titrated.

[0185] After drying, the powder can be blended to an appropriate potency, and mixed with other cultures and/or a filler such as microcrystalline cellulose for consistency and ease of handling, and the bacterial composition formulated as provided herein.

[0186] In one embodiment, one or more of the populations of bacterial cells in the composition are co-cultured.

[0187] In some embodiments, the bacteria are not grown in a medium consisting of 0-5% yeast extract broth. In other embodiments, the bacteria are grown in 0-5% yeast extract broth with glucose or another metabolizable monosaccha-

ride (e.g., fructose) or disaccharide (e.g., sucrose) in order to optimize production of the metabolites as described herein. In some embodiments, the concentration of glucose or metabolizable monosaccharide or disaccharide is 0.5-5%. In one embodiment, the medium comprises 3% yeast extract broth.

[0188] In some embodiments, corn steep liquor is added to the bacterial medium to optimize metabolite production (e.g., 0.5-5% corn steep liquor). Corn steep liquor is a byproduct of wet corn milling. Its components are soluble proteins, amino acids, carbohydrates, organic acids (e.g., lactic acid), vitamins, and minerals. It is sometimes combined with other ingredients in corn gluten feed and widely used in complete feeds for dairy and beef cattle, poultry, swine, and pet foods. Some corn steep liquor is used in the production of acetic acid, food acids, and fermentation processes. Some corn steep liquor is used in the pharmaceutical industry in the production of intravenous solutions and drugs, most notably antibiotics (penicillin). In certain embodiments, the concentration of corn steep liquor is 0.5-4%, 0.5-3%, 0.5-2%, 0.5-1%, 2-4%, 3%, 2-5%, 3-5%, 4-5%, 1-4%, or any range therebetween. In one embodiment the concentration of corn steep liquor is 3%.

Dosage, Administration and Formulations

[0189] In some embodiments, cells over a range of, for example, 2.5×10^5 , or more, e.g., 1×10^6 , 1×10^7 , 1×10^8 , 5×10^8 , 1×10^9 , 5×10^9 , 1×10^{10} , 5×10^{10} , 1×10^{11} or more can be administered in a composition comprising a bacterium or collection of bacteria as described herein. The dosage range for the bacteria depends upon the potency, and includes amounts large enough to produce the desired effect, e.g., reduction in at least one symptom of a disease associated with mitochondrial dysfunction in a treated subject. The dosage should not be so large as to cause unacceptable adverse side effects. Generally, the dosage will vary with the type of illness, and with the age, condition, and sex of the patient. The dosage can be determined by one of skill in the art and can also be adjusted by the individual physician in the event of any complication.

[0190] For use in the various aspects described herein, an effective amount of cells in a composition as described herein comprises at least 1×10^5 bacterial cells, at least 1×10^6 bacterial cells, at least 1×10^7 bacterial cells, at least 1×10^8 bacterial cells, at least 1×10^9 bacterial cells, at least 1×10^{10} bacterial cells, at least 1×10^{11} bacterial cells, at least 1×10^{12} bacterial cells or more. In some embodiments of the aspects described herein, the bacterial cells (e.g., isolated from rotting fruit) are expanded or maintained in culture prior to administration to a subject in need thereof. In one embodiment, the bacterial strain(s) is/are obtained from a microbe bank. In some embodiments, two or more bacterial strains are administered together, e.g., in a single admixture. However, it is specifically contemplated herein that two or more bacterial strains can be administered as separate dosage forms or sub-mixtures or sub-combinations of the strains. Thus, for a consortium of e.g., three members, the consortium can be administered, for example, as a single preparation including all three members (in one or more dosage units, e.g., one or more capsules) or as two or more separate preparations that, in sum, include all members of the given strains. While administration as a single admixture is preferred, a potential advantage of the use of e.g., individual units for each strain, is that the actual strains administered to

any given subject can be tailored, if necessary, by selecting the appropriate combination of, for example, single species dosage units that together comprise the desired consortium.

[0191] In some embodiments, the compositions described herein can be administered in a form containing one or more pharmaceutically acceptable carriers. Suitable carriers are well known in the art and vary with the desired form and mode of administration of the composition. For example, pharmaceutically acceptable carriers can include diluents or excipients such as fillers, binders, wetting agents, disintegrators, surface-active agents, glidants, lubricants, and the like. Typically, the carrier may be a solid (including powder), liquid, or combinations thereof. Each carrier is preferably “acceptable” in the sense of being compatible with the other ingredients in the composition and not injurious to the subject. The carrier can be biologically acceptable and inert (e.g., it permits the composition to maintain viability of the biological material until delivered to the appropriate site).

[0192] Oral compositions can include an inert diluent or an edible carrier. For the purpose of oral therapeutic administration, the active compound can be incorporated with excipients and used in the form of tablets, lozenges, pastilles, troches, or capsules, e.g., gelatin capsules. Oral compositions can also be prepared by combining a composition of the present disclosure with a food. In some embodiments, the bacterium/bacteria can be formulated in a food item. Some non-limiting examples of food items to be used with the methods and compositions described herein include: popsicles, cheeses, creams, chocolates, milk, meat, drinks, pickled vegetables, kefir, miso, sauerkraut, etc. In other embodiments, the food items can be juices, refreshing beverages, tea beverages, drink preparations, jelly beverages, and functional beverages; alcoholic beverages such as beers; carbohydrate-containing foods such as rice food products, noodles, breads, and pastas; paste products such as fish, hams, sausages, paste products of seafood; retort pouch products such as curries, food dressed with a thick starchy sauce, and Chinese soups; soups; dairy products such as milk, dairy beverages, ice creams, and yogurts; fermented products such as fermented soybean pastes, fermented beverages, and pickles; bean products; various confectionery products including biscuits, cookies, and the like, candies, chewing gums, gummies, cold desserts including jellies, cream caramels, and frozen desserts; instant foods such as instant soups and instant soy-bean soups; and the like. It is preferred that food preparations not require cooking after admixture with the bacterial strain(s) to avoid killing the microbes.

[0193] In one embodiment a food used for administration is chilled, for instance, iced flavored water. In certain embodiments, the food item is not a potentially allergenic food item (e.g., not soy, wheat, peanut, tree nuts, dairy, eggs, shellfish or fish). Pharmaceutically compatible binding agents, and/or adjuvant materials can be included as part of the composition. The tablets, pills, capsules, troches and the like can contain any of the following ingredients, or compounds of a similar nature: a binder such as microcrystalline cellulose, gum tragacanth or gelatin; an excipient such as starch or lactose, a disintegrating agent such as alginic acid, primogel, or corn starch; a lubricant such as magnesium stearate or sterotes; a glidant such as colloidal silicon dioxide; a sweetening agent such as sucrose or saccharin; or a flavoring agent such as peppermint, methyl salicylate,

orange flavoring, or other suitable flavorings. These are for purposes of example only and are not intended to be limiting.

[0194] Formulations suitable for oral administration may be provided as discrete units, such as tablets, capsules, cachets, syrups, elixirs, prepared food items, microemulsions, solutions, suspensions, lozenges, or gel-coated ampules, each containing a predetermined amount of the active compound; as powders or granules; as solutions or suspensions in aqueous or non-aqueous liquids; or as oil-in-water or water-in-oil emulsions.

[0195] The compositions described herein can also be prepared in the form of suppositories (e.g., with conventional suppository bases such as cocoa butter and other glycerides) or retention enemas for rectal delivery. Formulations suitable for rectal administration include gels, creams, lotions, aqueous or oily suspensions, dispersible powders or granules, emulsions, dissolvable solid materials, douches, and the like. The formulations are preferably provided as unit-dose suppositories comprising the active ingredient in one or more solid carriers forming the suppository base, for example, cocoa butter. Suitable carriers for such formulations include petroleum jelly, lanolin, polyethyleneglycols, alcohols, and combinations thereof. Alternatively, colonic washes with the rapid recolonization deployment agent of the present disclosure can be formulated for colonic or rectal administration. The compositions can be prepared with carriers that will protect the bacteria against rapid elimination from the body, such as a controlled release formulation, including implants. Biodegradable, biocompatible polymers can be used, such as ethylene vinyl acetate, polyanhydrides, polyglycolic acid, collagen, polyorthoesters, and polylactic acid. Such formulations can be prepared using standard techniques. The materials can also be obtained commercially from, for instance, Alza Corporation and Nova Pharmaceuticals, Inc. Liposomal suspensions can also be used as pharmaceutically acceptable carriers. These can be prepared according to methods known to those skilled in the art.

[0196] In some embodiments, a composition can be encapsulated or microencapsulated (e.g., enteric-coated formulations). For instance, when the composition is to be administered orally, the dosage form is formulated so the composition is not exposed to conditions prevalent in the gastrointestinal tract before the small intestine, e.g., high acidity and digestive enzymes present in the stomach. An enteric coating can be stable at low pH (such as in the stomach) and can dissolve at higher pH (for example, in the small intestine). Material that can be used in enteric coatings includes, for example, alginic acid, cellulose acetate phthalate, plastics, waxes, shellac, and fatty acids (e.g., stearic acid, palmitic acid). Enteric coatings are described, for example, in U.S. Pat. Nos. 5,225,202, 5,733,575, 6,139,875, 6,420,473, 6,455,052, and 6,569,457, all of which are herein incorporated by reference in their entirety. The enteric coating can be an aqueous enteric coating. Examples of polymers that can be used in enteric coatings include, for example, shellac (trade name EmCoat 120 N, Marcoat 125); cellulose acetate phthalate (trade names AQUACOAT™, AQUACOAT ECD™, SEPIFILM™, KLUCEL™, and ETOLOSE™); polyvinylacetate phthalate (trade name SURETERIC™); and methacrylic acid (trade name EUDRAGIT™). The encapsulation of compositions for therapeutic use is known in the art. Encapsulation can

include hard-shelled capsules, which can be used for dry, powdered ingredients, or soft-shelled capsules. Capsules can be made from aqueous solutions of gelling agents such as animal protein (e.g., gelatin), plant polysaccharides or derivatives like carrageenans and modified forms of starch and cellulose. Other ingredients can be added to a gelling agent solution such as plasticizers (e.g., glycerin and or sorbitol), coloring agents, preservatives, disintegrants, lubricants and surface treatment.

[0197] In one embodiment, an enteric coated probiotic composition as described herein is administered to a subject. In another embodiment, an enteric coated probiotic and prebiotic composition is administered to a subject.

[0198] Formulations of a bacterial composition or bacterial extract can be prepared by any suitable method, typically by uniformly and intimately admixing the bacterial cells with liquids or finely divided solid carriers or both, in the required proportions and then, if necessary, shaping the resulting mixture into the desired shape. In some embodiments, the bacterial strain(s) as described herein is/are combined with one or more additional probiotic organisms prior to treatment of a subject. As used herein, the term “probiotic” refers to microorganisms that form at least a part of the transient or endogenous flora and thereby exhibit a beneficial prophylactic and/or therapeutic effect on the host organism. Probiotics are non-pathogenic under normal circumstances and include, but are not limited to, those designated “Generally Regarded as Safe (GRAS)” by the U.S. Food & Drug Administration.

[0199] A nutrient supplement comprising the bacterium or bacterial consortium as described herein can include any of a variety of nutritional agents, including vitamins, minerals, essential and nonessential amino acids, carbohydrates, lipids, foodstuffs, dietary supplements, short chain fatty acids and the like. Preferred compositions comprise vitamins and/or minerals in any combination. Vitamins for use in a composition as described herein can include vitamins B, C, D, E, folic acid, K, niacin, and like vitamins. The composition can contain any or a variety of vitamins as may be deemed useful for a particular application, and therefore, the vitamin content is not to be construed as limiting. Typical vitamins are those, for example, recommended for daily consumption and in the recommended daily amount (RDA), although precise amounts can vary. The composition can preferably include a complex of the RDA vitamins, minerals and trace minerals as well as those nutrients that have no established RDA, but have a beneficial role in healthy human or mammal physiology. The preferred mineral format can include, for example, those that are in either the gluconate or citrate form which are more readily metabolized by lactic acid bacteria. Similar considerations can be employed to favor other classes of bacteria as needed. In a related embodiment, the compositions described herein are contemplated to comprise one or more bacteria or a bacterial consortium as described herein in combination with a viable lactic acid bacteria in combination with any material to be absorbed, including but not limited to nutrient supplements, foodstuffs, vitamins, minerals, medicines, therapeutic compositions, antibiotics, hormones, steroids, and the like compounds where it is desirable to insure efficient and healthy absorption of materials from the gastrointestinal tract into the blood. The amount of material included in the composition can vary widely depending upon the material and the

intended purpose for its absorption, such that the composition is not to be considered as limiting.

[0200] The compositions as described herein can comprise from about 100 mg to about 100 g, alternatively from about 500 mg to about 50 g, and alternatively from about 1 g to about 40 g, of a prebiotic, per day or on a less than daily schedule.

[0201] Prior to administration of a bacterial composition, the patient may optionally have a pretreatment protocol with an antibiotic to prepare the gastrointestinal tract to receive the bacterial composition. In certain embodiments, the pretreatment protocol is advisable, such as when a patient has an acute infection with a highly resilient pathogen. In other embodiments, the pretreatment protocol is entirely optional, such as when a patient does not have an infection with a pathogen, or for example, when a pathogen causing an infection is not resilient, or the patient has had an acute infection that has been successfully treated but where the physician is concerned that the infection may recur. In these instances, the pretreatment protocol can enhance the ability of the bacterial composition to affect the patient’s microbiome and/or enhance the therapeutic outcome. In an alternative embodiment, the subject is not pre-treated with an antibiotic.

[0202] As one way of preparing the patient for administration of the therapeutic microbes, at least one antibiotic can be administered to alter the bacteria in the patient. As another way of preparing the patient for administration, a standard colon-cleansing preparation can be administered to the patient to substantially empty the contents of the colon, such as used to prepare a patient for a colonoscopy. By “substantially emptying the contents of the colon,” is meant removing at least 75%, at least 80%, at least 90%, at least 95%, or about 100% of the ordinary volume of colon contents. Antibiotic treatment can precede a colon-cleansing protocol.

[0203] If a patient has received an antibiotic for treatment of an infection, or if a patient has received an antibiotic as part of a specific pretreatment protocol, in one embodiment the antibiotic should be stopped in sufficient time to allow the antibiotic to be substantially reduced in concentration in the gut before a bacterial composition is administered. In one embodiment, the antibiotic may be discontinued 1, 2, or 3 days before the administration of the bacterial composition. In one embodiment, the antibiotic can be discontinued 3, 4, 5, 6, or 7 antibiotic half-lives before administration of the bacterial composition. If the pretreatment protocol is part of treatment of an acute infection, the antibiotic may be chosen so that the infection is sensitive to the antibiotic, but the constituents in the administered bacterial composition are not sensitive to the antibiotic.

[0204] Any of the preparations described herein can be administered once on a single occasion or on multiple occasions, such as once a day for several days or more than once a day on the day of administration (including twice daily, three times daily, or up to five times daily). Or the preparation can be administered intermittently according to a set schedule, e.g., once weekly, once monthly, or when the patient relapses from the primary illness. In another embodiment, the preparation can be administered on a long-term basis to assure the maintenance of a protective or therapeutic effect. Note that while description above and elsewhere herein refer to formulations for delivery of bacteria, formu-

lations comprising metabolites produced by such bacteria can be prepared and administered in a similar manner.

Excluded Bacteria:

[0205] As will be readily appreciated by one of skill in the art, a composition as described herein for treatment of disease will ideally not comprise one or more pathogenic bacteria. In one embodiment, a composition as described herein (e.g., a bacterial extract or a probiotic composition) does not include an organism conventionally classified as a pathogenic or opportunistic organism. It is possible that a function shared by all members of a given taxonomic group could be beneficial, e.g., for providing particular metabolites, yet for other reasons the overall effect of one or more particular members of the group is not beneficial and is, for example, pathogenic. Clearly, members of a given taxonomic group that cause pathogenesis, e.g., acute gastrointestinal pathologies, are to be excluded from the therapeutic or preventive methods and compositions described herein.

[0206] In one embodiment, the bacterial composition does not comprise at least one of *Acidaminococcus intestinalis*, *Lactobacillus casei*, *Lactobacillus paracasei*, *Raoultella* sp., and *Streptococcus mitis*. In another embodiment, the bacterial composition does not comprise any of these.

[0207] In another embodiment, the bacterial composition does not comprise at least one of *Barnesiella intestinihominis*, *Lactobacillus reuteri*, *Enterococcus hirae*, *Enterococcus faecium*, or *Enterococcus durans*, *Anaerostipes caccae*, *Clostridium indolis*, *Staphylococcus wamerei*, or *Staphylococcus pasteurii*, and *Adlercreutzia equolifaciens*. In another embodiment, the bacterial composition does not comprise any of these.

[0208] In another embodiment, the bacterial composition does not comprise at least one of *Clostridium botulinum*, *Clostridium cadaveris*, *Clostridium chauvoei*, *Clostridium clostridioforme*, *Clostridium cochlearium*, *Clostridium difficile*, *Clostridium haemolyticum*, *Clostridium hastiforme*, *Clostridium histolyticum*, *Clostridium indolis*, *Clostridium irregulare*, *Clostridium limosum*, *Clostridium malenominatum*, *Clostridium novyi*, *Clostridium oroticum*, *Clostridium paraputrificum*, *Clostridium perfringens*, *Clostridium piliforme*, *Clostridium putrefaciens*, *Clostridium putrificum*, *Clostridium septicum*, *Clostridium sordellii*, *Clostridium sphenoides*, and *Clostridium tetani*. In another embodiment, the bacterial composition does not comprise any of these.

[0209] In another embodiment, the bacterial composition does not comprise at least one of *Escherichia coli* and *Lactobacillus johnsonii*. In another embodiment, the bacterial composition does not comprise any of these.

[0210] In another embodiment, the bacterial composition does not comprise at least one of *Clostridium innocuum*, *Clostridium butyricum*, and *Blautia producta* (previously known as *Peptostreptococcus productus*). In another embodiment, the bacterial composition does not comprise any of these.

[0211] In another embodiment, the bacterial composition does not comprise at least one of *Eubacteria*, *Fusobacteria*, *Propionibacteria*, *Escherichia coli*, and *Gemmiger*.

[0212] In another embodiment, the compositions described herein do not comprise pathogenic bacteria in the Genera *Yersinia*, *Vibrio*, *Treponema*, *Streptococcus*, *Staphylococcus*, *Shigella*, *Salmonella*, *Rickettsia*, *Orientia*, *Pseudomonas*, *Neisseria*, *Mycoplasma*, *Mycobacterium*, *Listeria*, *Leptospira*, *Legionella*, *Klebsiella*, *Helicobacter*,

Haemophilus, *Francisella*, *Escherichia*, *Ehrlichia*, *Enterococcus*, *Coxiella*, *Corynebacterium*, *Chlamydia*, *Chlamydophila*, *Campylobacter*, *Burkholderia*, *Brucella*, *Borrelia*, *Bordetella*, *Bacillus*, multi-drug resistant bacteria, extended spectrum beta-lactam resistant Enterococci (ESBL), Carbapenem-resistant Enterobacteriaceae (CRE), or vancomycin-resistant Enterococci (VRE).

[0213] In other embodiments, the compositions described herein do not comprise pathogenic species or strains, such as *Aeromonas hydrophila*, *Campylobacter fetus*, *Plesiomonas shigelloides*, *Bacillus cereus*, *Campylobacter jejuni*, entero-aggregative *Escherichia coli*, enterohemorrhagic *Escherichia coli*, enteroinvasive *Escherichia coli*, enterotoxigenic *Escherichia coli* (such as, but not limited to, LT and/or ST), *Escherichia coli* 0157: H7, *Helicobacter pylori*, *Klebsiella pneumoniae*, *Lysteria monocytogenes*, *Plesiomonas shigelloides*, *Salmonella* spp., *Salmonella typhi*, *Salmonella paratyphi*, *Shigella* spp., *Staphylococcus* spp., *Staphylococcus aureus*, vancomycin-resistant enterococcus spp., *Vibrio* spp., *Vibrio cholerae*, *Vibrio parahaemolyticus*, *Vibrio vulnificus*, or *Yersinia enterocolitica*.

[0214] In one embodiment, the bacterial compositions or formulations as described herein do not comprise *Klebsiella pneumoniae*, *Proteus mirabilis*, *Enterobacter cloacae*, or *Bilophila wadsworthia*.

Efficacy Measurement

[0215] The term “effective amount” as used herein refers to the amount of a population of bacterial cells or their components or metabolites needed to alleviate at least one or more symptoms of a disease associated with mitochondrial dysfunction and relates to a sufficient amount of a composition to provide the desired effect. An effective amount as used herein also includes an amount sufficient to prevent or delay the development of a symptom of the disease, alter the course of a symptom of the disease (for example but not limited to, slow the progression of a symptom of the disease, such as myopathy), or reverse a symptom of the disease. It is understood that for any given case, an appropriate “effective amount” can be determined by one of ordinary skill in the art using routine experimentation. Given the intricacies of the body and the nature of cell establishment, the “effective amount” of cells may vary among different patients, however one can easily determine in hindsight if the amount of cells administered was indeed an “effective amount.” Thus, further treatments can be modified accordingly. Note that long-term colonization or establishment, while often desirable, is not necessary for effective treatment as regular administration can achieve effective treatment as well.

[0216] The efficacy of treatment can be determined by the skilled clinician. However, a treatment is considered “effective treatment,” as the term is used herein, if any one or all of the symptoms, or other clinically accepted symptoms or markers of a disease associated with mitochondrial dysfunction are reduced, e.g., by at least 10% following treatment with a composition comprising bacterial cells or a bacterial extract as described herein. Methods of measuring these indicators are known to those of skill in the art and/or described herein.

[0217] In one embodiment, effective treatment is determined by a reduction in the dose of a conventional pharmacological treatment required to maintain adequate control of symptoms of a disease associated with mitochondrial dysfunction.

[0218] In some embodiments, the subject is further evaluated using one or more additional diagnostic procedures, for example, by medical imaging, physical exam, laboratory test(s), clinical history, family history, genetic test, and the like. Medical imaging is well known in the art. As such, the medical imaging can be selected from any known method of imaging, including, but not limited to, ultrasound, computed tomography scan, positron emission tomography, photon emission computerized tomography, and magnetic resonance imaging.

[0219] The present invention may be as described in any one of the following numbered paragraphs:

[0220] 1. A composition comprising a measured amount of one or more bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0221] 2. The composition of paragraph 1, wherein the bacterium is from the Acetobacteriaceae family.

[0222] 3. The composition of paragraph 1 or 2, wherein the bacterium is *Gluconobacter* spp (e.g., *Gluconobacter* EBT 405), *Acetobacter* spp., *Gluconoacetobacter* spp., *Acidomonas* spp., *Ameyamaea* spp., *Asaia* spp., *Granulibacter* spp., *Kozakia* spp., *Neoasaia* spp., *Neokomagataea* spp., *Saccharibacter* spp., *Swaminathanian* spp., or *Tanticharoenia* spp.

[0223] 4. The composition of any one of paragraphs 1-3, wherein one or more of the nucleic acid sequences are exogenous nucleic acid sequences.

[0224] 5. The composition of any one of paragraphs 1-4, wherein the bacterium comprises and expresses nucleic acid sequences encoding each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0225] 6. The composition of any one of paragraphs 1-5, wherein the measured amount of the one or more bacterium is lyophilized.

[0226] 7. The composition of any one of paragraphs 1-6, wherein the measured amount is an amount effective to induce expression and/or activity of mitochondrial transcription factor A (TFAM), or peroxisome proliferator activated receptor gamma coactivator 1 (PGC) in a human cell upon administration to a subject in need thereof.

[0227] 8. The composition of paragraph 7, wherein the amount effective to induce expression and/or activity of TFAM, or PGC in a human cell is 1×10^6 bacteria.

[0228] 9. The composition of any one of paragraphs 1-8, wherein the composition further comprises one or more added bacterial metabolites selected from the group consisting of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0229] 10. The composition of any one of paragraphs 1-9, wherein the bacterium is viable, attenuated or heat-inactivated.

[0230] 11. The composition of any one of paragraphs 1-10, wherein the composition is formulated as a food,

a beverage, a feed composition, a probiotic, a nutritional supplement, or a pharmaceutical composition.

[0231] 12. The composition of any one of paragraphs 1-11, which further comprises a prebiotic.

[0232] 13. The composition of paragraph 12, wherein the prebiotic comprises a fructooligosaccharide, an inulin, an isomaltooligosaccharide, lactitol, lactosucrose, lactulose, a soy oligosaccharide, a transgalactooligosaccharide or a xylooligosaccharide.

[0233] 14. The composition of any one of paragraphs 1-13, further comprising a pharmaceutically acceptable carrier.

[0234] 15. The composition of any one of paragraphs 1-14, wherein the composition is formulated for oral administration.

[0235] 16. The composition of paragraph 15, wherein the composition is an enteric-coated formulation.

[0236] 17. A composition comprising a therapeutically effective amount of an extract or fraction derived from at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding an enzyme selected from the group consisting of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0237] 18. The composition of paragraph 17, wherein the at least one bacterium comprises and expresses each of the enzymes membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

[0238] 19. The composition of paragraph 17 or 18, wherein the extract or fraction comprises one or more metabolites selected from gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

[0239] 20. The composition of any one of paragraphs 17-19, wherein the extract or fraction is from bacterial cells cultured in a medium comprising: (i) standard Lysogeny broth (1% tryptone, 0.5% yeast extract, and 1% sodium chloride) containing 1% glucose, (ii) standard Hestrin-Schramm broth (D-glucose 2%, 0.5% yeast extract, 0.5% peptone, 0.27% disodium phosphate, 0.115% citric acid) or

[0240] (iii) CaCO_3 medium comprising 8% glucose, 0.5% yeast extract, 0.2% mannitol, 0.05% magnesium sulphate, and 10% calcium carbonate.

[0241] 21. The composition of any one of paragraphs 17-20, wherein the extract or fraction comprises metabolites or bacterial byproducts that promote ATP production in at least one cell type of a host mammal administered the composition.

[0242] 22. The composition of any one of paragraphs 17-21, wherein the extract or fraction does not comprise viable bacterial cells.

[0243] 23. The composition of any one of paragraphs 17-22, wherein the extract or fraction lacks detectable bacteria.

[0244] 24. The composition of any one of paragraphs 17-23, wherein the extract or fraction further comprises attenuated or heat-inactivated bacteria.

- [0245] 25. The composition of any one of paragraphs 17-24, wherein the bacterium is from the Acetobacteriaceae family.
- [0246] 26. The composition of paragraph 25, wherein the bacterium is *Gluconobacter* spp (e.g., *Gluconobacter* EBT 405), *Acetobacter* spp., *Gluconoacetobacter* spp., *Acidomonas* spp., *Ameyamaea* spp., *Asaia* spp., *Granulibacter* spp., *Kozakia* spp., *Neoasaia* spp., *Neokomagataea* spp., *Saccharibacter* spp., *Swaminathanian* spp., or *Tanticharoenia* spp.
- [0247] 27. The composition of any one of paragraphs 17-26, wherein one or more of the nucleic acid sequences is/are an exogenous nucleic acid sequence (s).
- [0248] 28. The composition of any one of paragraphs 17-27, wherein the composition is formulated as a food, a beverage, a feed composition, a nutritional supplement, or a pharmaceutical composition.
- [0249] 29. The composition of any one of paragraphs 17-28, wherein the composition further comprises a pharmaceutically acceptable carrier.
- [0250] 30. The composition of any one of paragraphs 17-29, wherein the composition is formulated for oral administration.
- [0251] 31. A method for increasing cellular ATP production in at least one cell type of a subject in need thereof, the method comprising administering to the subject a composition comprising an amount of at least one non-pathogenic bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, effective to increase cellular ATP production in at least one cell type.
- [0252] 32. The method of paragraph 31, wherein the at least one bacterium expresses each of membrane-bound of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0253] 33. The method of any one of paragraphs 31-32, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.
- [0254] 34. The method of any one of paragraphs 31-33, wherein the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.
- [0255] 35. The method of any one of paragraphs 31-34, wherein one or more of the nucleic acid sequences is exogenous to the bacterium.
- [0256] 36. The method of any one of paragraphs 31-35, wherein the activity of complex I and/or complex II of the mitochondrial electron transport chain is increased in the one or more cell types.
- [0257] 37. The method of any one of paragraphs 31-36, wherein the administering increases mitochondrial membrane potential.
- [0258] 38. The method of any one of paragraphs 31-37, wherein the subject is human.
- [0259] 39. The method of any one of paragraphs 31-38, wherein expression of peroxisome proliferator-activated receptor gamma coactivator 1-x (PGC-1x), and/or mitochondrial transcription factor A (TFAM) is increased.
- [0260] 40. The method of any one of paragraphs 31-39, wherein AMP-activated protein kinase (AMPK) phosphorylation levels, nuclear respiratory factor-2 (Nrf2) protein levels, PGC α mRNA levels TFAM mRNA levels and/or mitochondrial DNA replication is increased.
- [0261] 41. The method of any one of paragraphs 31-40, wherein the mitochondrial DNA copy numbers (mtDNA) is increased.
- [0262] 42. The method of any one of paragraphs 31-41, wherein the method increases the developmental growth rate of a subject carrying a mutations in mitochondrial respiratory Complex I NADH: ubiquinone reductase.
- [0263] 43. The method of any one of paragraphs 31-42, wherein the expression of at least one mitochondrial β -oxidation enzyme is increased.
- [0264] 44. The method of paragraph 43, wherein the at least one mitochondrial β -oxidation enzyme is B0303.3, cpt-2, cpt-1, ech-1.2, or acdh-7.
- [0265] 45. The method of any one of paragraphs 31-44, wherein the longevity of the subject is increased.
- [0266] 46. The method of any one of paragraphs 31-45, wherein mitochondrial biogenesis is maintained or increased.
- [0267] 47. The method of any one of paragraphs 31-46, wherein cellular ATP production is increased by at least 10% compared to the cellular ATP production prior to administration of the composition.
- [0268] 48. A method for making a bacterial extract, the method comprising culturing at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II in a medium comprising: (i) standard Lysogeny broth (1% tryptone, 0.5% yeast extract, and 1% sodium chloride) containing 1% glucose, (ii) standard Hestrin-Schramm broth (D-glucose 2%, 0.5% yeast extract, 0.5% peptone, 0.27% disodium phosphate, 0.115% citric acid) or (iii) CaCO₃ medium comprising 8% glucose, 0.5% yeast extract, 0.2% mannitol, 0.05% magnesium sulphate, and 10% calcium carbonate.
- [0269] 49. The method of paragraph 48, wherein the at least one bacterium comprises and expresses nucleic acid sequences encoding each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0270] 50. The method of paragraph 48 or 49, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.
- [0271] 51. A composition comprising an amount of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid effective to

- increase cellular ATP production in at least one cell type of a host mammal administered the composition.
- [0272] 52. A composition comprising an amount of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid effective to increase the expression and/or activity of mitochondrial transcription factor A (TFAM), or peroxisome proliferator activated receptor gamma coactivator 1 (PGC) in at least one cell type of a human subject administered the composition.
- [0273] 53. The composition of paragraph 51 or 52, wherein the composition comprises a bacterial extract or active fraction thereof.
- [0274] 54. The composition of paragraph 51, 52 or 53, wherein the composition is formulated as a food, a beverage, a feed composition, a nutritional supplement, or a pharmaceutical composition.
- [0275] 55. A method of treating Parkinson's disease, the method comprising administering to a subject having Parkinson's disease, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of Parkinson's disease.
- [0276] 56. The method of paragraph 55, wherein the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0277] 58. The method of paragraph 55 or 56, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0278] 59. The method of any one of paragraphs 55-58, wherein (i) the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid and/or (ii) the bacterium comprises *Gluconobacter* EBT 405.
- [0279] 60. The method of any one of paragraphs 55-59, wherein the at least one symptom is
- [0280] selected from the group consisting of: tremor, sleep disturbance, mobility impairment, involuntary movement, muscle rigidity, rhythmic muscle contraction, slow bodily movement, slow shuffling gait, fatigue, dizziness, impaired balance, restlessness, amnesia, confusion, dementia, cognitive impairment, impaired speech, anxiety, apathy, distorted or loss of sense of smell, urinary incontinence, reduced facial expression, weight loss and constipation.
- [0281] 61. A method of treating a mitochondrial electron transport chain disorder, the method comprising administering to a subject having a mitochondrial electron transport chain disorder, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of the mitochondrial electron transport chain disorder.
- [0282] 62. The method of paragraph 61, wherein the mitochondrial electron transport chain disorder comprises a disorder or impaired activity in Complex I and/or Complex II.
- [0283] 63. The method of paragraph 61 or 62, wherein the mitochondrial electron transport chain disorder is NADH dehydrogenase (NADH-CoQ reductase) deficiency, succinate dehydrogenase deficiency, Leigh Disease, mitochondrial DNA depletion, or mitochondrial insufficiency.
- [0284] 64. The method of any one of paragraphs 61-63, wherein the at least one symptom is selected from the group consisting of: myopathy, mitochondrial encephalomyopathy, failure to thrive, developmental delay, hypotonia, lethargy, respiratory failure, ataxia, myoclonus, lactic acidosis, seizures, fatigue, nystagmus, poor reflexes, difficulty eating or swallowing, breathing difficulties, ataxia, congenital myopathy, infantile myopathy and hepatopathy.
- [0285] 65. The method of any one of paragraphs 61-64, wherein the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0286] 66. The method of any one of paragraphs 61-65, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0287] 67. The method of any one of paragraphs 61-66, wherein the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0288] 68. A method of treating a peroxisomal disorder, the method comprising administering to a subject having a peroxisomal disorder, a composition comprising a therapeutically effective amount of at least one bacterium that comprises one or more nucleic acid sequences such that the bacterium expresses the following enzymes: membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or a bacterial extract thereof, thereby reducing at least one symptom of the peroxisomal disorder.
- [0289] 69. The method of paragraph 68, wherein the peroxisomal disorder is Zellweger syndrome spectrum (PBD-ZSD), or rhizomelic chondrodysplasia punctate type 1 (RCDP1).
- [0290] 70. The method of paragraph 68 or 69, wherein the PBD-ZSD is infantile Refsum disease, neonatal adrenoleukodystrophy, or Zellweger syndrome.
- [0291] 71. The method of any one of paragraphs 68-70, wherein the at least one symptom is selected from the group consisting of: skeletal and craniofacial dysmorphism, liver dysfunction, progressive sensorineural hearing loss and retinopathy.
- [0292] 72. The method of any one of paragraphs 68-71, wherein the at least one bacterium expresses each of

membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.

- [0293] 73. The method of any one of paragraphs 68-72, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0294] 74. The method of any one of paragraphs 68-73, wherein the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0295] 75. A method for increasing the biogenesis of cellular mitochondria or peroxisomes, the method comprising administering to a subject, a composition comprising an amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or a fraction or extract thereof, effective to increase the biogenesis of cellular mitochondria or peroxisomes.
- [0296] 76. The method of paragraph 75, wherein the size and/or number of peroxisomes is increased.
- [0297] 77. The method of paragraph 75 or 76, wherein mitochondrial activity is increased.
- [0298] 78. The method of paragraph 75, 76, or 77, wherein the at least one bacterium expresses each of encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0299] 79. The method of any one of paragraphs 75-78, wherein the at least one bacterium produces one or more of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0300] 80. The method of any one of paragraphs 75-79, wherein the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0301] 81. A method of treating Alzheimer's disease, the method comprising administering to a subject having Alzheimer's disease, a composition comprising a therapeutically effective amount of at least one bacterium that comprises and expresses one or more nucleic acid sequences encoding one or more of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction thereof, thereby reducing at least one symptom of Alzheimer's disease.
- [0302] 82. The method of paragraph 81, wherein the at least one bacterium expresses each of membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II.
- [0303] 83. The method of paragraph 81 or 82, wherein the at least one bacterium produces one or more of

gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid

- [0304] 84. The method of paragraph 81, 82 or 83, wherein the at least one bacterium produces each of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and/or 2,5-diketo-D-gluconic acid.
- [0305] 85. The method of any one of paragraphs 81-84, wherein the at least one symptom is selected from the group consisting of: cognitive decline, confusion, delusion, disorientation, forgetfulness, difficulty concentrating, inability to generate new memories, inability to do simple math, inability to recognize common items, aggression, agitation, irritability, meaningless repetition of own words, personality changes, restlessness, lack of restraint, wandering, anger, apathy, general discontent, loneliness, mood swings, depression, hallucination, paranoia, loss of appetite, inability to combine muscle movements and jumbled speech.
- [0306] 86. The method of any one of paragraphs 81-85, wherein the bacterium comprises *Gluconobacter* EBT 405.

Examples

[0307] The following provides non-limiting Examples demonstrating and supporting the technology as described herein.

Example 1: Results

[0308] Six microbes that suppress the slow developmental growth rate phenotype of three mitochondrial mutants of *C. elegans*: spg-7(ad2249)¹, gas-1(fc21)², and nduf-7(et19)³ were identified using a qualitative visual screen. spg-7 encodes a homolog of AFG3L2, which is a conserved m-AAA metalloprotease¹. Along with another m-AAA metalloprotease paraplegin, AFG3L2 is involved in the removal of abnormal proteins from the mitochondria and thus maintenance of the mitochondrial proteome. nduf-7 encodes NADH-ubiquinone oxidoreductase Fe—S, which is a component of complex I in the mitochondrial electron transport chain³. gas-1 encodes a NADH dehydrogenase [ubiquinone] iron-sulfur protein 2, which is a component of complex I in the mitochondrial electron transport chain². gas-1(fc21) mutant animals exhibit significantly decreased mitochondrial mass and membrane potential and thus, reduced respiratory capacity^{4,5}. spg-7(ad2249), gas-1(fc21), and nduf-7(et19) mutants grow at a significantly slower rate compared to the wildtype animals (FIG. 1A) when fed on *E. coli* OP50. *E. coli* OP50 is the standard microbial strain on which *Caenorhabditis elegans* are typically grown in the laboratory. A screen was performed to test whether any of the microbes that are naturally associated with *C. elegans* in the wild can modify the slow growth phenotype of the mitochondrial mutants. In the screen, ~200 strains of wild microbes were fed individually to synchronized L1-stage spg-7(ad2249), gas-1(fc21), and nduf-7(et19) mutants. Mutant animals that grew at a significantly faster rate compared to the animals fed on *E. coli* OP50 (FIG. 5A) were identified. In this screen, 6 microbes were identified that partially suppress the slow growth phenotype of the mitochondrial mutants. This study focuses on three of the microbes: *Gluconacetobacter* spp, *Acetobacter* spp, and *Gluconobacter* spp (FIG. 5A). Each of these microbes belong to the family Acetobacteraceae of the phylum Pro-

teobacteria and are characterized by the ability to oxidize sugars or ethanol and produce acetic acid during fermentation⁷. They are naturally present in sugary and acidic niches such as fruits⁸ and particularly in fermented beverages such as natural Apple cider vinegar⁹ and Kombucha¹⁰. Interestingly, it was found that these three bacteria (i.e., *Gluconobacter*, *Acetobacter*, and *Gluconoacetobacter*) were able to accelerate the developmental growth rate of even wildtype animals (FIGS. 5B-5D). Among the three species, *Gluconacetobacter* spp were comparatively better in accelerating the developmental growth rate (FIGS. 5A-5B). To determine specifically how *Gluconacetobacter* spp enhance developmental growth rate, the developmental acceleration rate was examined at each larval stage. It was found that *Gluconacetobacter* spp feeding specifically shortened the time taken for animals to reach egg-laying adult stage from L4-larval stage (FIGS. 5E-5F).

[0309] To test whether the enhancement of developmental growth rate was general for all mitochondrial mutants, mutations in other mitochondrial respiratory chain namely *mev-1(kn1)* and *isp-1(qm150)* were tested. *mev-1* encodes a subunit of succinate-coenzyme Q oxidoreductase in the complex II of mitochondrial electron transport chain¹¹. *isp-1* is an iron sulfur protein (*isp-1*) of cytochrome bc₁ complex (complex III)¹². *Gluconacetobacter* spp feeding failed to suppress the slow developmental growth rate of *mev-1(kn1)* and *isp-1(qm150)* mutants, indicating that enhancement of complex I and/or complex II activity is a likely mode-of-action. It was hypothesized that *Gluconacetobacter* spp feeding might enhance the developmental growth rate of wildtype and the mitochondrial mutants by increasing the mitochondrial ATP production. To test this hypothesis, a transgenic firefly luciferase-expressing *C. elegans* strain¹³ was used for rapid in vivo assessment of the ATP levels. Transgenic luciferase expressing strain fed on *Gluconacetobacter* spp had higher bioluminescence compared to the animals fed on *E. coli* OP50, indicating increased ATP content (FIG. 1B). This result was confirmed using an in vitro ATP determination assay (FIG. 1C). An increase in bioluminescence was observed as quickly as within 6 hours of initiating feeding with *Gluconacetobacter* spp (FIG. 5G). Further, it was found that the ATP levels were reduced in *spg-7(ad2249)* mutant fed on *E. coli* OP50 compared to the wildtype animals fed on *E. coli* OP50. In addition, *Gluconacetobacter* spp feeding was able to restore the ATP content of *spg-7(ad2249)* mutant (FIG. 1D). Similarly, it was found that *Gluconacetobacter* spp feeding was able to partially restore the ATP content of *nduf-7(et19)* mutant (FIG. 1E).

[0310] It was found that the mitochondrial membrane potential was significantly decreased in *spg-7(ad2249)*, *gas-1(fc21)*, and *nduf-7(et19)* mutants fed on *E. coli* OP50 compared to the wildtype animals fed on *E. coli* OP50 as assessed using Mitotracker™ CMXRos (FIG. 1F). Mitotracker™ CMXRos stains mitochondria and its accumulation in live cells is dependent upon membrane potential. *Gluconacetobacter* spp feeding was able to partially restore the mitochondrial membrane potential of *spg-7(ad2249)*, *gas-1(fc21)*, and *nduf-7(et19)* mutants (FIG. 1F). Using another dye, Tetramethylrhodamine, ethyl ester (TMRE), which is sequestered by active mitochondria, it was found that *spg-7(ad2249)* mutants fed on *Gluconacetobacter* spp had significantly increased dye levels compared to animals fed on *E. coli* OP50 (FIG. 5H) indicating that the mitochondrial membrane potential was increased.

[0311] Next, the effect of different mitochondrial poisons on the increased ATP production phenotype of animals fed on *Gluconacetobacter* spp was assessed. It was found that animals fed on *Gluconacetobacter* spp and treated with Rotenone or Paraquat or Sodium Azide had significantly higher levels of bioluminescence compared to animals fed on *E. coli* OP50 and treated with Rotenone or Paraquat or Sodium Azide (FIGS. 1G-1H; FIG. 5I). Both Paraquat (1,1'-dimethyl-4,4'-bipyridinium dichloride) and Rotenone are inhibitors of complex I in the mitochondrial electron transport chain while Sodium Azide and Antimycin are inhibitors of complex III.

[0312] It was found that animals fed on *Gluconacetobacter* spp were more resistant to Rotenone or Paraquat than animals fed on *E. coli* OP50 (FIG. 6A). However, animals fed on *Gluconacetobacter* spp were not more resistant to Sodium Azide or Antimycin compared to animals fed on *E. coli* OP50 (FIG. 6A).

[0313] The unfolded protein response of the mitochondria (UPR^{mt}) is initiated by mitochondrial stress and activates the expression of several nuclear genes, including the *hsp-6* mitochondrial chaperone¹⁴. Using a *hsp-6::gfp* transgenic animal strain, it was determined whether *Gluconacetobacter* spp affects UPR^{mt} induction. Transgenic *hsp-6::gfp* animals fed on *E. coli* OP50 or *Gluconacetobacter* spp were treated with either MPP+ or 6-OHDA or Paraquat and induction of GFP expression was assessed after 24 hours of treatment. MPTP [1-methyl-4-phenylpyridinium (MPP+)] and 6-hydroxydopamine (6-OHDA) are neurotoxins that induce ROS production and are used in animal models of Parkinson's Disease (PD). While worms fed *E. coli* OP50 and treated with either MPP+ or 6-OHDA or Paraquat induced GFP expression, worms fed on *Gluconacetobacter* spp failed to induce GFP expression (FIG. 2A). Genetic mutations in *gas-1(fc21)*, *nduf-7(et19)*, *mev-1(kn1)* and *isp-1(qm150)* induced *hsp-6::gfp* expression constitutively. Feeding these mutants on *Gluconacetobacter* spp did not suppress the *hsp-6::gfp* expression. While the induction of *hsp-6::gfp* during UPR^{mt} is mediated by the ATFS-1 transcription factor¹⁵, a subset of the genes upregulated in UPR^{mt} are activated in an ATFS-1 independent fashion. Recently, *tbb-6*, which encodes an uncharacterized β -tubulin, was shown to be upregulated independently of ATFS-1 following mitochondrial ETC disruption¹⁶. A *tbb-6::gfp* transgenic strain was generated and it was found this GFP was induced in the *spg-7(ad2249)* mutant (FIG. 6B; FIG. 2B). Feeding *spg-7(ad2249)*; *tbb-6::gfp* animals on *Gluconacetobacter* spp suppressed the *tbb-6::gfp* expression (FIG. 6B; FIG. 2B).

[0314] To address how *Gluconacetobacter* spp feeding improves mitochondrial function, a variety of transgenic worm strains were tested, including worms that carry a construct encoding a fusion protein comprising a GFP reporter protein and a gene product that is involved in stress response, mitochondrial activity, peroxisomal function, innate immune response, and different signaling pathways (Table 1; FIGS. 2C-2K; FIGS. 6C-6L; FIGS. 7A-7D).

TABLE 1

Transcriptional GFP fusion gene	Gene description
<i>cyp-35B1</i>	Cytochrome P450
<i>haf-7</i>	ABC-type multidrug transport system
<i>ugt-61</i>	UDP-glucuronosyl and UDP-glucosyl transferase

TABLE 1-continued

Transcriptional GFP fusion gene	Gene description
pgp-3	ABC-type multidrug transport system
mrp-2	ABC-type multidrug transport system
abch-1	ABC-type multidrug transport system
pmp-4	ABC-type multidrug transport system
ugt-58	UDP-glucuronosyl and UDP-glucosyl transferase
dhs-9	short-chain dehydrogenases/reductases family (SDR)
cyp-37A1	Cytochrome P450
dhs-19	short-chain dehydrogenases/reductases family (SDR)
cyp-34A9	Cytochrome P450
cyp-14A3	Cytochrome P450
ugt-8	UDP-glucuronosyl and UDP-glucosyl transferase
cyp-13a7	Cytochrome P450
pgp-4	ABC-type multidrug transport system
cyp-31a3	Cytochrome P450
pgp-6	ABC-type multidrug transport system
cyp-25a2	Cytochrome P450
ugt-22	UDP-glucuronosyl and UDP-glucosyl transferase
pgp-9	ABC-type multidrug transport system
pgp-5	ABC-type multidrug transport system
sod-3	Superoxide dismutase/daf-16 target
gst-4	Glutathione S-transferase/oxidative stress reporter
gcs-1	Gamma GlutamylCysteine Synthetase/oxidative stress reporter
hsp-60	Mitochondrial specific chaperone of GroE/Hsp 10/60 family
hsp-4	Heat shock response 70 (hsp70) protein/Endoplasmic reticulum stress reporter
hsp-6	Mitochondrion-specific chaperone of the DnaK/Hsp70family/mitochondrial stress reporter
nlp-29	Neuropeptide like protein/Anti-fungal innate immune response reporter
clec-60	C-type lectin family/innate immune response reporter
irg-1	Innate immune response reporter
hsp-16.2	Heat shock protein of hsp16/hsp20/alphaB-crystallin (HSP16) family/heat shock reporter
nhr-57	Nuclear hormone receptor; hypoxia stress reporter
rpt-3	triple A ATPase subunit of the 26S proteasome's 19S regulatory particle (RP) base subcomplex
fat-7	delta-9 fatty acid desaturase
tbb-6	human TUBB8 (tubulin beta 8 class VIII)
hphd-1	human ADHFE1 (alcohol dehydrogenase, iron containing 1)
B0303.3	3-Keto-coA thiolase beta-subunit of Trifunctional protein (HADHB)
prx-11	vertebrate peroxisomal membrane protein 11C
cpt-2	carntine palmitoyltransferase
acox-1.2	ACOX1 (acyl-CoA oxidase 1)
ech-1.2	enoyl-CoA hydratases/long-chain 3-hydroxyacyl-CoA dehydrogenases
acdh-7	ACADM (acyl-CoA dehydrogenase, C-4 to C-12 straight chain)
prx-6	Pex6 (peroxisomal biogenesis factor 6)
cpt-1	Carnitine Palmitoyltransferase I
icl-1	isocitrate lyase/malate synthase, an enzyme known to function in the glyoxylate cycle
lgg-1	ortholog of <i>Saccharomyces cerevisiae</i> Atg8p and mammalian MAP-LC3
UbV-gfp	a UFD model substrate in which a noncleavable ubiquitin is N-terminally fused to GFP (UbV-GFP)
lip1-1	human Lipases family including LIPF
hsp-70	heat-shock protein that is a member of the hsp70 family of molecular chaperones
ftn-2	ferritin heavy chain homolog
ftn-1	ferritin heavy chain homolog
fzo-1	GTPase orthologous to MFN1(Mitofusin)/FZO1
crtc-1	CREB-regulated transcriptional coactivator orthologous to mammalian CRTCs

TABLE 1-continued

Transcriptional GFP fusion gene	Gene description
cts-1	citrate synthase
acdh-1	Short-chain acyl-CoA dehydrogenase
acdh-2	short-chain acyl-CoA dehydrogenase
acdh-11	acyl-CoA dehydrogenase
ads-1	AGPS (alkylglycerone phosphate synthase)
agxt-1	Alanine-Glyoxylate aminotransferase (AGXT)
acox-1.6	ACOX1 (acyl-CoA oxidase 1)
nhr-64	NR2 subfamily of nuclear receptors that contains Drosophila and human HNF4
daf-22	ortholog of human sterol carrier protein SCP2
nhr-69	NR2 subfamily of nuclear receptors that contains Drosophila and human HNF4
spg-20	human SPG20 (spastic paraplegia 20 (Troyer syndrome))
cyp-29A2	CYP19A1 (cytochrome P450 family 19 subfamily A member 1)
dhs-28	human 17-BETA-HYDROXYSTEROID DEHYDROGENASE 4 (HSD17B4)
mtl-1	metallothioneins, small, cysteine-rich, metal-binding proteins
kap-1	kinesin-associated protein
C01H6.4	N,N-dimethylaniline monooxygenase activity
tmem-135	TMEM135 (transmembrane protein 135)
C53B4.3	Solute Carrier Family 22 (Organic Cation Transporter), Member 18
far-3	Fatty Acid/Retinol binding protein
irg-4	CUB-like domain-containing protein
irg-5	Infection Response Gene
lys-7	antimicrobial lysozyme encoded by the LYS4
abf-1	antibacterial factor ASABF
abf-3	antibacterial factor ASABF
spp-7	SPP-protein family, called caenopores
lys-1	putative lysozyme
dod-22	human EPHX1 (epoxide hydrolase 1)
F55G11.7	human EPHX1 (epoxide hydrolase 1)
K08D8.5	CUB-like domain-containing protein
clec-85	C-type LECTin
F10A3.4	Nematode-specific gene

[0315] In this screen, it was found that *Gluconacetobacter* spp feeding induced the expression of a subset of genes required for either mitochondrial function or peroxisomal activity. These include B0303.3 (ortholog of human HADHB [3-Ketoacyl-coA Thiolase beta subunit]) (FIG. 2D; FIG. 6H), prx-11 (ortholog of human PEX11G [peroxisomal biogenesis factor 11G]) (FIG. 2E), cpt-2 (ortholog of human CPT2 [Carnitine Palmitoyl transferase]) (FIG. 2F; FIG. 6C), acox-1.2 (ortholog of human ACOX1 [Acyl CoA oxidase 1]) (FIG. 2G; FIG. 6D; FIG. 6J), ech-1.2 (ortholog of human HADHA [3-Ketoacyl-coA Thiolase alpha subunit]) (FIG. 2H; FIG. 6G), acdh-7 (ortholog of human ACADM [Acyl-CoA dehydrogenase, medium-chain]) (FIG. 2I; FIG. 6F, FIG. 6I), pmp-4 (ortholog of human ABCD1 [peroxisomal membrane ABC transporter]) (FIG. 2J; FIG. 6E), prx-6 (ortholog of human PEX6 [peroxisomal biogenesis factor 6]) (FIG. 2K), cpt-1 (ortholog of human CPT1 [Carnitine Palmitoyl transferase]) (FIG. 6K), and icl-1 (encodes isocitrate lyase/malate synthase) (FIG. 7A). Expression of one gene, hphd-1 (an ortholog of Human ADHFE1 [Alcohol dehydrogenase]) was significantly reduced in worms fed on *Gluconacetobacter* spp (FIG. 2C). CPT (carnitine O-palmitoyltransferase) and ACADM (medium chain acyl CoA dehydrogenase) each play a pivotal role in mitochondrial fatty acid uptake and B-oxidation, two critical steps for energy production from fatty acids. Similarly, HADHA and HADHB encode the alpha and beta subunit of the mitochondrial hydroxyacyl-CoA dehydrogenase/3-ketoacyl-

CoA thiolase/enoyl-CoA hydratase (trifunctional protein), which catalyzes mitochondrial beta-oxidation of long chain fatty acids. ACOX1 catalyzes the desaturation of acyl-CoAs to 2-trans-enoyl-CoAs and is the first enzyme of the peroxisomal fatty acid beta-oxidation pathway. ICL-1 functions in the glyoxylate shunt, which converts isocitrate and acetyl-CoA to succinate, malate, and CoA using glyoxylate as an intermediate¹⁷. It was found that the *Gluconacetobacter* spp feeding induced *cpt-2::gfp* and that *ech-1.2::gfp* expression was dependent on *tcer-1* (ortholog of TCERG1 [Transcription elongation regulator 1])^{18,19} (FIGS. 2F & 2H) while *acox-1.2* (FIG. 2G), *acdh-7* (FIG. 2I), *pmp-4* (FIG. 2J), and *prx-6* (FIG. 2K), expression was dependent on *nhr-49* (encodes a nuclear hormone receptor with sequence similarity to HNF4 α but functionally resemble more like peroxisome proliferator-activated receptor alpha (PPAR α)²⁰. *tcer-1* (tm1452) and *nhr-49*(nr2041) animals fed on *Gluconacetobacter* spp feeding grew slower compared to wildtype worms fed on *Gluconacetobacter* spp (FIG. 3A) indicating that TCER-1 and NHR-49 activities promote accelerated developmental growth rate. Intestine-specific overexpression of *nhr-49* was sufficient to accelerate growth rate of *nhr-49*(nr2041) mutants fed on *Gluconacetobacter* spp (FIG. 7E). Further, a strong induction of *nhr-49::gfp* was observed in the worms fed on *Gluconacetobacter* spp (FIG. 7F). To test whether NHR-49 itself is sufficient to accelerate developmental growth rate, gain-of-function alleles were determined in *nhr-49*(et7), *nhr-49*(et8), *nhr-49*(et13)²¹. None of these strains had accelerated growth rate phenotype when fed on *E. coli* OP50, while they still showed accelerated growth when fed on *Gluconacetobacter* spp.

[0316] In addition, a strong nuclear localization of TCER-1::GFP was observed in the intestinal cells of worms fed on *Gluconacetobacter* spp compared to worms fed on *E. coli* OP50 (FIG. 7G).

[0317] Since genes associated with peroxisomal function were upregulated in response to *Gluconacetobacter* spp feeding, it was tested whether RNAi of peroxisomal genes affect the accelerated developmental growth rate phenotype. RNAi of *prx-5* (ortholog of PXR1 or PEX5 [human receptor for type I peroxisomal targeting signal protein]) or *prx-11* (ortholog of human PEX11G [peroxisomal biogenesis factor 11G]) suppressed the accelerated developmental growth rate phenotype of wildtype worms fed on *Gluconacetobacter* spp (FIG. 3A). It was found that RNAi of *tcer-1*, *nhr-49*, and *prx-5* suppressed the *Gluconacetobacter* spp feeding-induced increase in bioluminescence phenotype (FIG. 3B), indicating that these genes are required for increased ATP production upon *Gluconacetobacter* spp feeding. Further, RNAi of *nhr-49* or *prx-5* was able to suppress the accelerated developmental progression phenotype of *spg-7* (ad2249) and *nduf-7*(et19) mutant worms fed on *Gluconacetobacter* spp (FIGS. 3C-3D). Compared to wildtype worms, *spg-7*(ad2249) mutant have reduced mitochondrial membrane potential as assessed using Mito CMXRos and this reduction was further exacerbated by *nhr-49* RNAi (FIG. 3E). A partial reduction of mitochondrial membrane potential was observed in wildtype worms fed on *nhr-49* RNAi (FIG. 3E).

[0318] To examine the effect of *Gluconacetobacter* spp feeding on the mitochondrial morphology, a transgenic strain was used that expresses a fusion protein of *C. elegans* mitochondrial import receptor subunit TOMM-20 and monomeric red fluorescent protein (mRFP) and the expres-

sion is driven in body wall muscle cells by the myo-3 promoter²². In the transgenic worms fed on *Gluconacetobacter* spp, the mitochondria appear to be more fused compared to the worms fed on *E. coli* OP50 (FIG. 3F). Using Rhodamine 6G dye, which stains mitochondria, it was determined that the mitochondria in the hypodermis of worms fed on *Gluconacetobacter* spp appear more fused compared to worms fed on *E. coli* OP50 (FIG. 7H). Mitochondria are dynamic organelles that constantly undergo fusion and fission events. The elongated morphology is associated with increased ATP production efficiency²³ and decreased ROS formation²⁴. Thus, without wishing to be bound by theory, the increased number of fused mitochondria phenotype in worms fed on *Gluconacetobacter* spp may explain the increased ATP production phenotype. The mitochondrial fission and fusion dynamics are tightly regulated by *drp-1* (Ortholog of human DNML1 [dynamin-related protein]) and *fzo-1* (ortholog of human MFN1[Mitofusin]/FZO1GTPase)²⁵. Mutants in *fzo-1*(tm1133) or *drp-1* (tm1108) are extremely slow-growing (takes 6 days to reach adulthood from L1-stage) when fed on *E. coli* OP50 or on *Gluconacetobacter* spp. Without wishing to be bound by theory, these data indicate that the beneficial effects of *Gluconacetobacter* spp on mitochondrial function might be via regulation of mitochondrial dynamics.

[0319] Since *Gluconacetobacter* spp feeding induced the expression of peroxisomal biogenesis factors, peroxisomes were examined using a transgenic strain that expresses DAF-22 fusion protein that was tagged with green fluorescent protein (GFP) at its N terminus, under the control of *vha-6* intestinal promoter²⁶. In this strain, a punctate signal is observed, which denotes the peroxisomal targeting of the GFP-DAF-22 fusion protein. An increase in the punctate structures was observed in worms fed on *Gluconacetobacter* spp, indicating an increase in peroxisomes (FIG. 3G). The punctate peroxisomal structures were not observed in worms that were treated with RNAi of *prx-5* (ortholog of PXR1 or PEX5 [human receptor for type I peroxisomal targeting signal protein]), which is required for the targeting of GFP-DAF-22 fusion protein to the peroxisomes and fed on either *E. coli* OP50 or on *Gluconacetobacter* spp (FIG. 8A). This increase in peroxisomal numbers is specific, changes in the number or morphology of other subcellular structures such as Golgi, gut lysosomal granules, and Endoplasmic reticulum (FIG. 8B-8D) was not observed. Since peroxisomes are a major endogenous source of hydrogen peroxide (H₂O₂), it was examined whether *Gluconacetobacter* spp feeding increases H₂O₂. To measure endogenous H₂O₂ levels, transgenic *jrIs1*[Prpl-17::HyPer] worms expressing HyPer sensor²⁷ were fed on *Gluconacetobacter* spp and *E. coli* OP50. Compared to the worms fed on *E. coli* OP50, worms fed on *Gluconacetobacter* spp had slightly lower levels of H₂O₂ indicating that *Gluconacetobacter* spp does not increase endogenous H₂O₂ production (FIG. 3H). Even in HyPer sensor worms fed on *Gluconacetobacter* spp and treated with H₂O₂ had slightly lower level of H₂O₂ levels compared to the worms fed on *E. coli* OP50 (FIG. 3H) indicating an increased efficiency of H₂O₂ detoxification. Consistent with this observation, worms fed on *Gluconacetobacter* spp and treated with H₂O₂ had slightly lower levels of H₂O₂ compared to the worms fed on *E. coli* OP50 as assessed using the Amplex red assay (FIG. 8E). One of the major ways in which organisms handle ROS such as H₂O₂ is through glutathione oxidation²⁷. Using a transgenic

worm strain that expresses Grx1-roGFP2, which is a ratio-metric biosensor that detects glutathione redox potential²⁷, it was found that the glutathione intracellular concentrations were not affected in worms fed on *Gluconacetobacter* spp compared to worms fed on *E. coli* OP50 (FIG. 8F). Without wishing to be bound by theory, it is possible that worms fed on *Gluconacetobacter* spp have higher levels of peroxisomal catalases, which detoxifies H₂O₂.

[0320] Aging-related changes in mitochondria are associated with decline in mitochondrial activity²⁸. In aging animals, impairment of mitochondrial function such as decreased oxidative capacity, decreased ATP production, reduced oxidative phosphorylation, and increased ROS production are observed²⁹. Age-dependent decline in ATP production, metabolism, mitochondrial content and function were observed in *C. elegans*^{30,31}. The effect of *Gluconacetobacter* spp feeding on mitochondrial function was assessed in aging worms. It was found that in worms fed on *E. coli* OP50, ATP levels decrease significantly as the worms age (FIG. 4A). However, worms fed on *Gluconacetobacter* spp had significantly higher ATP content compared to worms fed on *E. coli* OP50 even as the worms age (FIG. 4A). The effects of *Gluconacetobacter* spp feeding is dominant because mixture of *Gluconacetobacter* spp with *E. coli* OP50 was sufficient to induce increased ATP production (FIG. 4B).

[0321] To test whether the phenotypes associated with *Gluconacetobacter* spp feeding are strain-specific, the ATP production was assessed in response to other *Gluconacetobacter* strains. It was found that other *Gluconacetobacter* strains including *Gluconacetobacter hansenii* (MGH isolate), the canonical *Gluconacetobacter hansenii* ATCC 23769, and non-cellulose producing mutant of *G. hansenii* ATCC 23769 each induced increased ATP production compared worms fed on *E. coli* OP50 (FIG. 4C).

[0322] Further, a variety of *Gluconobacter* and *Acetobacter* strains were isolated from rotting apples and grapes, by selecting for bacteria that produce acids and form a characteristic clear halo around media plates containing CaCO₃⁷ (FIG. 8G). All these microbes were able to induce increased ATP production compared to worms fed on *E. coli* OP50 (FIG. 4D) except for *G. kondanii* and *G. frateurii*, which were pathogenic to *C. elegans* (FIG. 4D).

[0323] One of the key questions that the inventors wanted to answer was how *Gluconacetobacter* spp feeding suppresses defects associated with mitochondrial mutants. To address this question, a forward genetic transposon screen for mutant microbial strains that failed to suppress the developmental growth phenotype of spg-7(ad2249) mutants was conducted. Since the *Gluconacetobacter* spp are very slow growing, the mutant screen was conducted on *Gluconobacter oxydans*. ~ 2000 *G. oxydans* transposon mutants were isolated and fed individually to synchronized L1-stage spg-7(ad2249) mutants. After 3 days of incubation, the worms were scored for percent of worms that reached L4-stage or adulthood. In this screen, 11 mutants were identified that failed to suppress the slow developmental growth phenotype of spg-7(ad2249) mutants (FIGS. 8H-8I). Further, these 11 mutants failed to accelerate the developmental growth rate of wildtype worms (FIG. 9A). In addition, in worms fed on mutants in membrane-bound PQQ-dependent glucose dehydrogenase (GOX0265; mGDH), Cytochrome o ubiquinol oxidase subunit II (GOX1911), and ubiquinol-cytochrome c reductase iron-sulfur subunit, the

ATP production was decreased compared to worms fed on Wildtype *G. oxydans* (FIG. 4E). However, in the worms fed on mutants in Carbon-Nitrogen hydrolase and TonB-dependent outer membrane receptor protein, the ATP levels were similar to the worms fed on Wildtype *G. oxydans* (FIG. 4E).

[0324] It was next tested whether the *G. oxydans* mutants affect the ability to produce acids using the CaCO₃ clearance assay. It was found that mutants in membrane-bound PQQ-dependent glucose dehydrogenase (GOX0265; mGDH), Cytochrome o ubiquinol oxidase subunit II (GOX1911), ubiquinol-cytochrome c reductase iron-sulfur subunit, and Carbon-Nitrogen hydrolase fail to form a clear halo around media plates containing CaCO₃ (FIG. 5B).

[0325] Six independent alleles of a gene encoding for Membrane-bound PQQ-dependent glucose dehydrogenase (Table 2), which oxidizes D-glucose to D-gluconate that can be further oxidized to two different keto-D-gluconates, 2-keto-D-gluconate and 5-keto-D-gluconate, as well as 2,5-di-keto-D-gluconate^{32,33} were isolated.

TABLE 2

	Alleles
Membrane bound glucose dehydrogenase	6
Cytochrome o ubiquinol oxidase	1
Ubiquinol-cytochrome C reductase iron-sulfur subunit	1
tonB-dependent receptor protein	2
Carbon nitrogen hydrolase	1

[0326] An independent deletion mutation in glucose dehydrogenase of *G. oxydans* isolated by another lab³⁴ also failed to suppress the slow developmental growth phenotype of spg-7(ad2249) mutants (FIG. 9C).

[0327] One of the characteristic attributes of acetic acid bacteria, including *G. oxydans*, is their ability to incompletely oxidize a variety of substrates including sugars³³. This incomplete oxidation is catalyzed by various membrane-bound dehydrogenases including Membrane-bound PQQ-dependent glucose dehydrogenase³⁴. Independent deletion alleles in various dehydrogenases were tested and it was found that spg-7(ad2249) mutant worms fed on deletion mutations in Membrane-bound PQQ-dependent glucose dehydrogenase, FAD-dependent Membrane-bound dehydrogenase, and Membrane-bound PQQ-dependent dehydrogenase 3 grew slower compared to worms fed on wildtype *G. oxydans* (FIG. 9C). FAD-dependent membrane-bound dehydrogenase is the D-gluconate-2-dehydrogenase, which oxidizes D-gluconate to 2-keto-gluconate³⁴. The function of Membrane-bound PQQ-dependent dehydrogenase 3 is not known. Further, it was found that ATP production was decreased in worms fed on deletion mutations in Membrane-bound PQQ-dependent glucose dehydrogenase, FAD-dependent Membrane-bound dehydrogenase, and Membrane-bound PQQ-dependent dehydrogenase 3 compared to worms fed on wildtype *G. oxydans* (FIG. 4F).

[0328] To determine whether *Gluconacetobacter* spp has similar biological effects in humans, human primary dermal fibroblasts were exposed to *Gluconacetobacter* spp and *E. coli* OP50. Exposure of *E. coli* OP50 to human primary dermal fibroblasts results in significant reduction in the ATP production while *Gluconacetobacter* spp does not affect ATP production compared to the mock-treated cells (FIG. 9D-9E). Since ATP levels are an indicator of metabolically active cells, which is a measure of cellular viability, the

reduction of ATP levels in cells exposed to *E. coli* OP50 is an indicator of cell death or reduced viability. Without wishing to be bound by theory, the reduced viability may be a result of infection. Interestingly, *Gluconacetobacter* spp does not reduce the cellular ATP production even at higher doses of the bacteria (FIG. S5E) indicating that *Gluconacetobacter* spp is not toxic or pathogenic to cells. PGC-1 α (Peroxisome proliferator-activated receptor gamma coactivator 1-alpha) is the master regulator of mitochondrial biogenesis and it regulates several genes involved in energy metabolism by its interaction with the nuclear hormone receptor, PPAR alpha³⁵. Using RT-QPCR, it was found that PGC-1 α gene expression was increased in human primary dermal fibroblasts exposed to *Gluconacetobacter* spp compared to mock-treated cells (FIG. 4H). PGC-1 α induces the transcription of nuclear respiratory factor 1, leading to the increased expression of mitochondrial transcription factor A (TFAM)^{36,37}. TFAM is the master regulator of mitochondrial phosphorylative oxidation (mOXPHOS) required for mitochondrial transcription and mitochondrial genome replication³⁸. Using RT-QPCR, it was found that TFAM gene expression was increased in human primary dermal fibroblasts exposed to *Gluconacetobacter* spp compared to mock-treated cells (FIG. 4H). Since TFAM induces the mitochondrial DNA replication, QPCR for mitochondrial DNA copy numbers (mtDNA) was conducted. It was found that the mtDNA copy numbers were increased in human primary dermal fibroblasts exposed to *Gluconacetobacter* spp compared to mock-treated cells (FIG. 4I). An increase in the PGC-1 α and TFAM expression, and mtDNA copy number reveals an overall enhancement of mitochondrial activity.

References for Example 1

- [0329] 1. Zubovych, I. O., Straud, S. & Roth, M. G. Mitochondrial dysfunction confers resistance to multiple drugs in *Caenorhabditis elegans*. *Mol Biol Cell* 21, 956-969 (2010).
- [0330] 2. Kayser, E. B., Morgan, P. G. & Sedensky, M. M. GAS-1: a mitochondrial protein controls sensitivity to volatile anesthetics in the nematode *Caenorhabditis elegans*. *Anesthesiology* 90, 545-554 (1999).
- [0331] 3. Rauthan, M., Ranji, P., Abukar, R. & Pilon, M. A Mutation in *Caenorhabditis elegans* NDUF-7 Activates the Mitochondrial Stress Response and Prolongs Lifespan via ROS and CED-4. *G3 (Bethesda)* 5, 1639-1648 (2015).
- [0332] 4. Dingley, S. et al. Mitochondrial respiratory chain dysfunction variably increases oxidant stress in *Caenorhabditis elegans*. *Mitochondrion* 10, 125-136 (2010).
- [0333] 5. Dancy, B. M., Sedensky, M. M. & Morgan, P. G. Effects of the mitochondrial respiratory chain on longevity in *C. elegans*. *Exp. Gerontol.* 56, 245-255 (2014).
- [0334] 6. McCormack, S. et al. Pharmacologic targeting of sirtuin and PPAR signaling improves longevity and mitochondrial physiology in respiratory chain complex I mutant *Caenorhabditis elegans*. *Mitochondrion* 22, 45-59 (2015).
- [0335] 7. *The Family Acetobacteraceae: The Genera Acetobacter, Acidomonas, Asaia, Gluconacetobacter, Gluconobacter, and Kozakia*. (Springer-Verlag, 2011). doi:10.1007/SpringerReference_3770
- [0336] 8. Swings, J. & De Ley, J. in *The Prokaryotes* 771-778 (Springer Berlin Heidelberg, 1981). doi:10.1007/978-3-662-13187-9_64
- [0337] 9. Trček, J., Mahnič, A. & Rupnik, M. Diversity of the microbiota involved in wine and organic apple cider submerged vinegar production as revealed by DHPLC analysis and next-generation sequencing. *Int. J. Food Microbiol.* 223, 57-62 (2016).
- [0338] 10. Coton, M. et al. Unraveling microbial ecology of industrial-scale Kombucha fermentations by metabarcoding and culture-based methods. *FEMS Microbiol. Ecol.* 93, (2017).
- [0339] 11. Ishii, N. et al. A mutation in succinate dehydrogenase cytochrome b causes oxidative stress and ageing in nematodes. *Nature* 394, 694-697 (1998).
- [0340] 12. Feng, J., Bussière, F. & Hekimi, S. Mitochondrial electron transport is a key determinant of life span in *Caenorhabditis elegans*. *Dev. Cell* 1, 633-644 (2001).
- [0341] 13. Lagido, C., McLaggan, D. & Glover, L. A. A Screenable In Vivo Assay for Mitochondrial Modulators Using Transgenic Bioluminescent *Caenorhabditis elegans*. *J Vis Exp* e53083-e53083 (2015). doi: 10.3791/53083
- [0342] 14. Haynes, C. M. & Ron, D. The mitochondrial UPR-protecting organelle protein homeostasis. *J. Cell. Sci.* (2010).
- [0343] 15. Pellegrino, M. W., Fiorese, C. J., Baker, B. M. & Haynes, C. M. Mitochondrial import efficiency of ATFS-1 regulates mitochondrial UPR activation. *Science* (2012).
- [0344] 16. Munkácsy, E. et al. DLK-1, SEK-3 and PMK-3 Are Required for the Life Extension Induced by Mitochondrial Bioenergetic Disruption in *C. elegans*. *PLOS Genet* 12, e1006133 (2016).
- [0345] 17. Erkut, C., Gade, V. R., Laxman, S. & Kurzchalia, T. V. The glyoxylate shunt is essential for desiccation tolerance in *C. elegans* and budding yeast. *Elife* 5, 12897 (2016).
- [0346] 18. Amrit, F. R. G. et al. DAF-16 and TCER-1 Facilitate Adaptation to Germline Loss by Restoring Lipid Homeostasis and Repressing Reproductive Physiology in *C. elegans*. *PLOS Genet* 12, e1005788 (2016).
- [0347] 19. Ratnappan, R. et al. Germline signals deploy NHR-49 to modulate fatty-acid β -oxidation and desaturation in somatic tissues of *C. elegans*. *PLOS Genet* 10, e1004829 (2014).
- [0348] 20. Van Gilst, M. R., Hadjivassiliou, H., Jolly, A. & Yamamoto, K. R. Nuclear hormone receptor NHR-49 controls fat consumption and fatty acid composition in *C. elegans*. *PLOS Biol* 3, e53 (2005).
- [0349] 21. Lee, K., Goh, G. Y. S., Wong, M. A., Klassen, T. L. & Taubert, S. Gain-of-Function Alleles in *Caenorhabditis elegans* Nuclear Hormone Receptor nhr-49 Are Functionally Distinct. *PLOS ONE* 11, e0162708 (2016).
- [0350] 22. Sapir, A. et al. Controlled sumoylation of the mevalonate pathway enzyme HMGs-1 regulates metabolism during aging. *Proc Natl Acad Sci USA* 111, E3880-9 (2014).
- [0351] 23. Tondera, D. et al. SLP-2 is required for stress-induced mitochondrial hyperfusion. *EMBO J* 28, 1589-1600 (2009).
- [0352] 24. Babbar, M. & Sheikh, M. S. Metabolic Stress and Disorders Related to Alterations in Mitochondrial Fission or Fusion. *Mol Cell Pharmacol* 5, 109-133 (2013).

[0353] 25. Rolland, S. G., Lu, Y., David, C. N. & Conradt, B. The BCL-2-like protein CED-9 of *C. elegans* promotes FZO-1/Mfn1,2- and EAT-3/Opal-dependent mitochondrial fusion. *J. Cell Biol.* 186, 525-540 (2009).

[0354] 26. Zhang, S. O. et al. Genetic and dietary regulation of lipid droplet expansion in *Caenorhabditis elegans*. *Proc Natl Acad Sci USA* 107, 4640-4645 (2010).

[0355] 27. Back, P. et al. Exploring real-time in vivo redox biology of developing and aging *Caenorhabditis elegans*. *Free Radic. Biol. Med.* 52, 850-859 (2012).

[0356] 28. Haigis, M. C. & Yankner, B. A. The aging stress response. *Mol. Cell* 40, 333-344 (2010).

[0357] 29. Balaban, R. S., Nemoto, S. & Finkel, T. Mitochondria, oxidants, and aging. *Cell* 120, 483-495 (2005).

[0358] 30. Brys, K., Castelein, N., Matthijssens, F., Vanfleteren, J. R. & Braeckman, B. P. Disruption of insulin signalling preserves bioenergetic competence of mitochondria in ageing *Caenorhabditis elegans*. *BMC Biol.* 8, 91 (2010).

[0359] 31. Gruber, J. et al. Mitochondrial changes in ageing *Caenorhabditis elegans*—what do we learn from superoxide dismutase knockouts? *PLOS ONE* 6, e19444 (2011).

[0360] 32. Matsushita, K., Toyama, H. & Adachi, O. Respiratory chains and bioenergetics of acetic acid bacteria. *Adv. Microb. Physiol.* 36, 247-301 (1994).

[0361] 33. Saichana, N., Matsushita, K., Adachi, O., Frébort, I. & Frébortova, J. Acetic acid bacteria: A group of bacteria with versatile biotechnological applications. *Biotechnol. Adv.* 33, 1260-1271 (2015).

[0362] 34. Peters, B. et al. Characterization of membrane-bound dehydrogenases from *Gluconobacter oxydans* 621H via whole-cell activity assays using multideletion strains. *Appl. Microbiol. Biotechnol.* 97, 6397-6412 (2013).

[0363] 35. Lin, J., Handschin, C. & Spiegelman, B. M. Metabolic control through the PGC-1 family of transcription coactivators. *Cell Metab.* 1, 361-370 (2005).

[0364] 36. Scarpulla, R. C. Metabolic control of mitochondrial biogenesis through the PGC-1 family regulatory network. *Biochim. Biophys. Acta* 1813, 1269-1278 (2011).

[0365] 37. Goffart, S. & Wiesner, R. J. Regulation and Co-Ordination of Nuclear Gene Expression During Mitochondrial Biogenesis. *Experimental Physiology* 88, 33-40 (2004).

[0366] 38. Jornayvaz, F. R. & Shulman, G. I. Regulation of mitochondrial biogenesis. *Essays Biochem.* 47, 69-84 (2010).

Example 2: Materials and Methods

[0367] *Caenorhabditis elegans* strains were maintained on NGM plates and OP50 *Escherichia coli* bacteria at 20° C. as described¹. The wild-type strain was N2 Bristol. The strains used are described in Table 3.

TABLE 3

Strain Name	Description
SAL150	denEx28 [F10A3.4::GFP + pha-1(+)]
SAL146	denEx24 [clec-85p::GFP + pha-1(+)]
SAL144	denEx22 [K08D8.5::GFP + pha-1(+)]
SAL143	denEx21 [F55G11.7::GFP + pha-1(+)]

TABLE 3-continued

Strain Name	Description
SAL139	denEx17 [dod-22::GFP + pha-1(+)]
SAL129	denEx14 [lys-1::GFP + pha-1(+)]
SAL117	denEx11 [spp-7::GFP + pha-1(+)]
SAL109	denEx6 [abf-3::GFP + pha-1(+)]
SAL107	denEx4 [abf-1::GFP + pha-1(+)]
SAL105	denEx2 [lys-7::GFP + pha-1(+)]
AY101	acIs101 [F35E12.5p::GFP + rol-6(su1006)]
SAL148	denEx26 [F08G5.6::GFP + pha-1(+)]
BC14852	sEx14852[rCesF15B9.1::GFP + pCeh36]
DM7065	raEx65 [C53B4.3::GFP + pha-1(+)+ rol-6]
BC15698	sEx15698 [rCes K02G10.3::GFP + pCeh36]
BC11949	sEx11949 [rCes C01H6.4::GFP + pCeh36]
BC16086	sEx16086 [rCes F08F8.3a::GFP + pCeh36]
CF2222	muEx336 [mtl-1::RFP + rol-6(su1006)]
BC15536	sEx15536 [rCes T19B10.1::GFP + pCeh36]
BC14979	sEx14979 [rCesF57B10.9::GFP + pCeh36]
SD1631	ccls4251 [(pSAK2) myo-3p::GFP::LacZ::NLS + (pSAK4) myo-3p::mitochondrial GFP + dpy-20(+)] I. stIs10656 [nhr-69p::HIS-24::mCherry + unc-119(+)].
VS11	hJls73 [vha-6p::GFP::daf-22 + C. briggsae unc-119(+)].
RW11514	stIs11514 [nhr-64::H1-wCherry + unc-119(+)].
BC14632	sEx14632 [rCes F59F4.1::GFP + pCeh36]
BC14386	sEx14386 [rCes T14D7.1::GFP + pCeh36]
BC11106	sEx11106 [rCes Y50D7A.7::GFP + pCeh36]
BC11207	sls10648 [rCes Y45F3A.3::GFP + pCeh36]
VL714	wwEx53 [acdH-2p::GFP]
VL717	wwEx54 [acdH-1p::GFP]
BC11314	sEx11314 [rCesT20G5.2::GFP + pCeh36]
AGD418	uthIs205 [crtc-1p::crtc-1::RFP::unc-54 3'UTR + rol-6(su1006)]
BC10214	sEx10214 [rCes ZK1248.14::GFP + pCeh36]
XA6900	qaEx6900 [ftn-1p::pes-10::GFP+his + pha-1(+)].
XA6901	qaEx6901 [ftn-2p::pes-10::GFP::his + lin-15(+)].
BC10060	sEx884 [rCesC12C8.1::GFP + pCeh36]
BC14666	sEx14666 [rCesF54F3.3::GFP + pCeh36]
PP563	unc-119(ed4)III; hHls64 [unc-119(+); sur-5::UbV-GFP]III
DA2123	adIs2122 [lgg-1p::GFP::lgg-1 + rol-6(su1006)]
BC10434	icl-1::gfp
BC14873	sEx10434 [rCes Y46G5A.17::GFP + pCeh36]
BC12611	sEx14873 [rCes F39G3.7::GFP + pCeh36]
BC15507	sls11096 [rCes T25G12.5::GFP + pCeh36]
BC14661	sEx15507 [rCesT08B2.7b::GFP + pCeh36]
BC11281	sEx14661[rCes F08A8.2::GFP + pCeh36]
BC14848	sEx11281 [rCes R07H5.2::GFP + pCeh36]
BC12923	sEx14848 [rCesC47B2.8::GFP + pCeh36]
BC14789	sls12801 [rCes B0303.3::GFP + pCeh36]
BC15777	sEx14789 [rCes Y38F1A.6::GFP + pCeh36]
GR2183	tbb-6::gfp
RW10592	sEx15777 [rCesF10D2.9::GFP + pCeh36]
CL2070	mgIs72 [rpt-3p::GFP + dpy-5(+)] II.
AU133	zuls178 [his-72(1kb 5' UTR)::his-72::SRPVAT::GFP::his-72 (1KB 3' UTR) + 5.7 kb XbaI - HindIII unc-119(+)].
IG274	stIs10024 [pie-1::H2B::GFP::pie-1 3' UTR + unc-119(+)].
SJ4100	stIs10520 [nhr-57::H1-wCherry + unc-119(+)].
SJ4005	dvIs70 [hsp-16.2p::GFP + rol-6(su1006)].
SJ4058	agIs17 [myo-2p::mCherry + irg-1p::GFP] IV.
LDH71	clec-60::gfp
CL2166	frIs7 [nlp-29p::GFP + col-12p::DsRed] IV.
CF1553	zcls13[hsp-6::GFP]
JG1	zcls4 [hsp-4::GFP] V
BC15680	zcls9 [hsp-60::GFP + lin-15(+)].
SD1444	ldIs3 [gcs-1p::GFP + rol-6(su1006)].
BC10034	dvIs19 [(pAF15)gst-4p::GFP::NLS] III.
BC13865	mulS84 [(pAD76) sod-3p::GFP + rol-6(su1006)].
BC14956	ajIs1 (pgp-5::gfp)
JG4	sEx15680 [rCesC47A10.1::GFP + pCeh36]
BC15565	gals240 [ugt-22p::his-24::mCherry + unc-119(+)]
BC14926	gals237 [cyp-25A2p::his-24::mCherry + unc-119(+)].
BC13846	sEx868 [rCesT21E8.1::GFP + pCeh36]
	sEx13865 [rCesY17G9B.3::GFP + pCeh36]
	sEx14956 [rCesF42E11.1::GFP + pCeh36]
	cyp-13A7::GFP
	sEx15565 [rCesH23N18.3::GFP + pCeh36]
	sEx14926 [rCesK09A11.4::GFP + pCeh36]
	sEx13846 [rCesB0213.15::GFP + pCeh36]

TABLE 3-continued

Strain Name	Description
BC14562	sEx14562[rCesTHF9.11::GFP + pCeh361]
BC15044	sEx15044 [rCes F01D5.9::GFP + pCeh361]
BC10338	sls10109 [rCesY32H12A.3::GFP + pCeh361]
BC10672	sls10263[rCesF35H8.6::GFP + pCeh361].
BC10366	sEx10366 [rCes T02D1.5::GFP + pCeh361]
BC10058	sEx10058 [rCes C56E6.5::GFP + pCeh36]
BC12028	sEx12028 [rCesF57C12.4::GFP + pCeh361]
BC10257	sEx10257 [rCesZK455.7::GFP + pCeh361]
BC11571	sEx11571 contains [rCes F39G3.1::GFP + pCeh36]
BC10027	sEx861 [rCes Y50E8A.16::GFP + pCeh361].
CY573	bvls5 [cyp-35B1p::GFP + gcy-7p::GFP]
STE68	nhr-49(nr2041) I
WBM409	nhr-49(nr2041) I; wbmEx149[ges-1p::3xHA::nhr-49(eDNA)::unc-54 3'UTR + myo-3p::mCherry::unc-54 3'UTR].
MH5239	prx-5(ku517) II
CF2167	teer-1(tm1452) II
CF2031	muEx306 [teer-1::GFP + odr-1::RFP]
DA2249	spg-7(ad2249) I
QC134	nduf-7(et19) I
CW152	gas-1(fc21) X
TK22	mev-1(kn1) III
MQ887	isp-1(qm150) IV
PE254	fels4 [sur-5p::luciferase::GFP + rol-6(su1006)] V
PS6187	pha-1(e2123) unc-119(ed4) III;syEx1155 [myo-3p::tommm-20::mRFP::3xMyc + Cbr-unc-119(+)]
VS11	hjl573 [vha-6p::GFP::daf-22 + C. briggsae unc-119(+)]
VS8	dhs-28(hj8) X
VS17	hjl59 [ges-1p::glo-1::GFP + unc-119(+)]
VS25	hjl514 [vha-6p::GFP::C34B2.10(SP12) + unc-119(+)]
JV1	unc-119(ed3) III;jrls1 [rpl-17p::HyPer + unc-119(+)].
JV2	unc-119(ed3) III;jrls2 [rpl-17p::Grx1-roGFP2 + unc-119(+)]

[0368] *E. coli* OP50, *Gluconobacter oxydans* DSM 2343 (621H) and its mutant derivatives were cultivated in a medium containing LB medium containing 1% glucose. The cultures were centrifuged and seeded onto NGM media without any antibiotics.

[0369] *Gluconacetobacter* strains were routinely grown in standard Hestrin-Schramm agar (D-glucose 20 g/L, yeast extract 5 g/L, peptone 5 g/L, disodium phosphate 2.7 g/L, citric acid 1.15 g/L, and agar 15 g/L) and incubated for 4 days at 30° C. For liquid culture, agar in the media was omitted. *E. coli* OP50 was also grown in the same media at 30° C. The liquid cultures were spotted onto NGM media plates without any antibiotics and left to grow at room temperature for at least 3 days.

Screening for Transcription Factors Responsible for Induction of Metabolic Genes in Response to *Gluconoacetobacter* Spp

[0370] To identify the transcription factor(s) that mediate the induction of metabolic genes in response to *Gluconacetobacter* feeding, the inventors conducted a cherry-picked RNAi screen of genes that were previously implicated in metabolism²⁻¹³.

[0371] Feeding RNAi clones corresponding to dsRNA-control, daf-16, pha-4, hlh-30, skn-1, hsf-1, daf-12, nhr-80, nhr-49, teer-1, sbp-1, nhr-69, and nhr-64 were grown in LB media with 25 µg/ml carbenicillin overnight and seeded onto RNAi agar plates containing 1 mM IPTG. The plates were

dried in a laminar flow hood and incubated at room temperature overnight to induce dsRNA expression. 200 synchronized L1 larvae each of pmp-4::gfp, cpt-2::gfp, ech-1.2::gfp, acox-1.2::gfp, acdh-7::gfp, and prx-6::gfp expressing animals were placed onto each of the RNAi-containing agar plates, allowed to develop at 20° C. for 4 days. When the animals reached adult-stage, the worms were bleached-prepped. 20 synchronized L1 larvae were then placed on *Gluconacetobacter* seeded plates and scored for failure to induce GFP expression. dsRNA-control RNAi treated worms fed on *Gluconacetobacter*-seeded plates were used as controls.

Screening for Microbes that Suppress the Slow Developmental Growth Rate Phenotype of Spg-7(Ad2249) Mutants [0372] ~200 Microbial strains were grown individually and seeded onto NGM media plates. ~ 20 Synchronized L1-stage spg-7(ad2249) mutant animals were placed on the 24-well plates and visually screened at 60 hours for the wells that contained animals that were fast growing compared to the worms fed on *E. coli* OP50. 6 microbial strains were recovered in the screen.

Isolation of *G. oxydans* Tn5 Transposon Mutants [0373] Tn5 transposon mutagenesis in *G. oxydans* was performed using vector pME9978 obtained from Prof. Stephan Heeb, University of Nottingham, UK. pME9978 was transformed into *E. coli* S17-1 and selected on gentamicin and ampicillin LB media plates. pME9978 present in *E. coli* S17-1 cells were transferred into *G. oxydans* through conjugation. Conjugation was performed by mixing *E. coli* S17-1 bearing pME9978 and *G. oxydans* in suspension, plating them on LB medium, and incubating at 30° C. for 4 hours. The culture was scrapped from the LB plate and resuspended in 10 ml of M9 buffer. The resultant cell suspension was serially diluted up to 10⁻⁶ and plated out on the selection media plates (LB media containing 1% glucose, 10 µg/ml-1 gentamicin, and 50 µg ml-1 cefoxitin). *G. oxydans* is naturally resistant cefoxitin while *E. coli* S17-1 is sensitive. Gentamicin-resistant *G. oxydans* colonies appeared after 4 days of incubation at 30° C.. ~2000 of these colonies were picked onto 96-well blocks filled with liquid selection media (LB media containing 1% glucose, 10 µg/ml-1 gentamicin, and 50 µg ml-1 cefoxitin), grown at 30° C. for 2 days and stored at -80° C.

Screening for *G. oxydans* Mutants that Failed to Suppress the Slow Developmental Growth Rate Phenotype of Spg-7 (Ad2249) Mutants

[0374] *G. oxydans* Tn5 mutants were grown in LB liquid media containing 1% glucose, 10 µg/ml⁻¹ gentamicin, and 50 µg ml⁻¹ cefoxitin and spotted onto 24-well NGM media plates. ~20 Synchronized L1-stage spg-7(ad2249) mutant animals were placed on the 24-well plates and visually screened at 60 hours for the wells that contained animals that were slow-growing compared to the worms fed on wildtype *G. oxydans*. 11 mutants were recovered in the screen.

Identification of Transposon Insertion Sites in the *G. oxydans* Tn5 Mutants

[0375] The transposon mutant bacteria were grown in 100 ml of LB liquid media containing 1% glucose, 10 µg/ml-1 gentamicin, and 50 µg ml-1 cefoxitin and centrifuged at 2500 g for 15 minutes. Genomic DNA isolation was performed using a commercial genomic DNA isolation kit (Qiagen). The isolated genomic DNA was subjected to restriction digestion using NcoI (does not cut pME9978 plasmid) and heat-inactivated. The restriction fragments

were self-ligated with T4 DNA ligase and electroporated into pir *E. coli* strain and selected on gentamicin containing LB media plates at 37° C. overnight. Since the transposon contains R6K_{ori} origin of replication, it can grow as independent plasmid in the pir *E. coli* strain. Multiple colonies from the gentamicin plates were pooled together and grown in gentamicin containing LB media plates at 37° C. overnight. The cultures were centrifuged and plasmid isolation was performed. The plasmid was sequenced to find the transposon insertion site using the following primers:

tpnRL17 (SEQ ID NO: 11):
5' -ACAAGCCAGGATGTAACG-3'

tpnRL13 (SEQ ID NO: 12):
5' -CAGCAACACCTTCTTCACGA-3'

Transposon Insertion Site in the Mutants

[0376] Glucose dehydrogenase_G7 (SEQ ID NO: 13 and SEQ ID NO: 34, respectively, in order of appearance):

CAGCTGACGCTTCGTCCGAAGAACCCGCTCTCCGATGCCGATATCTGGG
GTGGCACAATC...Tn5...GGCACAATCTTCGACCATGTTCTGCAG
CATCTACTTCCACTCCCTGCGCTACGAAGGT

[0377] Glucose dehydrogenase_G9 (SEQ ID NO: 14 and SEQ ID NO: 35, respectively, in order of appearance):

CTACATCCCGACCAAGACGGGCGATATCTTTGTCTCGACCGCCGTACG
GGCAAGGAAG...Tn5...GGCAAGGAAGTCGTCTGCTCTCTGAAA
CACCAGTTCAGGTGGTGCAGTCTCCGGGCGA

[0378] Cytochrome O Ubiquinol oxidase subunit_G1 (SEQ ID NO: 15 and SEQ ID NO: 36, respectively, in order of appearance):

AGCACCCACGCCCTACGACCCGTACCGCCCCCTTCAGACGGCTGACAACG
TCAAGCCCTG...Tn5...GACAGCGTCAAGCCACTGAATGTTCAAGGTG
GTCTCTCTCGACTGGAATGGCTGTTCATC

[0379] Glucose dehydrogenase_G6 (SEQ ID NO: 16 and SEQ ID NO: 37, respectively, in order of appearance):

CGCACCAACAAGTGGTCTGGCAGCATCGTAACGGTACGCTCCGTGACT
CCATGCACGGC...Tn5...ATGCACGGCAGCTCCCTGCCATCCCGCT
GCCTCCGATCAAGATCGGTGTTCCGAGCCTT

[0380] tonB_G2 (SEQ ID NO: 17 and SEQ ID NO: 38, respectively, in order of appearance):

TACAAGTGGTATCGATTGGGATTTTGATTATCGGATCCGTCTGACTAGC
ATGGATTCCAT...Tn5...GATTTCATCATGTCAGCAATAACCTGC
AGAACCTGGTCAATTTTCCAGCAGCAGAC

[0381] Ubiquinol-cytochrome C reductase iron-sulfur subunit_G3 (SEQ ID NO: 18 and SEQ ID NO: 39, respectively, in order of appearance):

CATCGCGTCATGATCGGGATCTGCACGCATCTGGGC...Tn5...TGC
GTGGCGACTTTCGACGCCCCGACGAGGCAGCAACCTGCCGGGAA

[0382] Glucose dehydrogenase_G14 (SEQ ID NO: 19 and SEQ ID NO: 40, respectively, in order of appearance):

CGATGCAGAAACCGGGCAGAAAGTGTCCGGCTTTGGCAACGACGGCGAA
CTGGAACCTG...Tn5...GAAGTGGAACTGCGCGAGCCGAACAGCC
TTACGTACGCCAGGCATGTATGAGCCGACG

[0383] Glucose dehydrogenase_promoter_G15 (SEQ ID NO: 20 and SEQ ID NO: 41, respectively, in order of appearance):

TCTTAATGCGCTTCTATGTGTTTGTCCGAAGGTCAAGTGC...Tn5...
AGGTCAAGTGTGTCACCAAATAACAGTCTTAGGAACATCATGACACAT
CTTCCCGGC

[0384] tonB_G11 (SEQ ID NO: 21 and SEQ ID NO: 42, respectively, in order of appearance):

GCAGTAATTTCTGTCTGCTTCTCCAGTCAGGTTCCGACCCCTGTTT
...Tn5...GGTGGTAATACAAAGCTGCGTCCAGAAGCAGGGCGTACCT
ACACCTTTGGT

[0385] Glucose dehydrogenase_G13 (SEQ ID NO: 22 and SEQ ID NO: 43, respectively, in order of appearance):

TGGCGATCGCTGGTCTCCGCTGAAGCAGATCAATTCGACCAATGTTGAG
AACCTCAAGGT...Tn5...CCTCAAGGTGGCATGGCACATCCACCA
AGGATCTGATGGGACCGAATGATCCGGGCGA

[0386] Carbon-Nitrogen Hydrolase_G5 (SEQ ID NO: 23 and SEQ ID NO: 44, respectively, in order of appearance):

AGACTCAATAGCGGAGCAGGACAAGAATGTCCAGTGAGCTTCACCC
GTTTGCTGCGT...Tn5...AAACGCTGCGGGCAGAAATCATCTCAA
CATCCGACGCCGAAG

Isolation of Acetobacteraceae from Fruits

[0387] Members of the Acetobacteraceae family are characterized by acid production resulting in clearing of CaCO₃ and formation of halo around the colonies on calcium carbonate-containing media². Samples of rotting apples, grapes, and oranges were crushed aseptically and mixed with 9 ml of M9 buffer and serially diluted up to 10⁻⁶ dilution. 100 µl of dilutions were spread on CaCO₃ medium (8% Glucose, 0.5% Yeast extract, 0.2% Mannitol, 0.05% Magnesium Sulphate, 10% Calcium carbonate, 100 µg/ml Cycloheximide, 10 µg/ml Nystatin).

[0388] Plates were incubated under aerobic conditions at 30° C. for 3-4 days. Colonies showing a clear halo were selected. Colony PCRs were conducted using the following primers:

27F (SEQ ID NO: 24):
AGAGTTTGATCMTGGCTCAG

1492R (SEQ ID NO: 25):
TACGGYTACCTTGTACGACTT

[0389] The conditions used for touchdown colony PCR were as follows: 95° C. for 10 minutes followed by 15 cycles of 30s at 95° C., 30s at 65° C. (step down of -1° C. for every cycle) and 1 minute at 72° C. This was followed by 40 cycles of 30s at 95° C., 30s at 50° C. and 1:30 minutes at 72° C. **[0390]** 16S rRNA sequencing on the isolates were conducted for identification of the isolates.

Microscopy

[0391] For high-magnification differential interference contrast and fluorescence images, worms treated with appropriate conditions were mounted onto agar pads and images were acquired using a Zeiss AXIO™ Imager Z1 microscope fitted with a Zeiss AxioCam™ HRc camera and Axiovision™ software. Fluorescent images were converted to 16-bit images, thresholded and quantified using ImageJ. Student's t test was used to determine statistical significance. GraphPad™ Prism 8.0 was used for these calculations and generating graphs. Low-magnification bright-field images were acquired using a Zeiss AxioZoom™ V16 Microscope, equipped with a Hamamatsu Orca™ flash 4.0 digital camera, and using Axiovision™ software. Immunofluorescence images (shown in FIG. 2) were acquired with an IX-70 microscope (Olympus, Waltham, MA) fitted with a cooled CCD camera (CH350; Roper Scientific) driven by the Delta Vision system (Applied Precision, Pittsburgh, PA). Images were deconvolved using the SoftWoRx 3.3.6 software (Applied Precision).

Measuring Endogenous Hydrogen Peroxide Levels Using HyPer Strain

[0392] To measure endogenous hydrogen peroxide levels, transgenic *jrIs1*[Prp1-17::HyPer] worms expressing a HyPer as described³ was used. Synchronized L1-stage transgenic *jrIs1*[Prp1-17::HyPer] animals were grown in *E. coli* OP50 and *G. hanseni* and about 1000 L4-stage worms were harvested in 96 microtiter well plates. For H₂O₂ treatment experiment, H₂O₂ was added to the worms in the 96-well itself and measurements were taken within 15 minutes. An excitation wavelength of either 490 nm or 405 nm was used to measure oxidized or reduced HyPer™ probe fluorescence respectively with an emission filter of 535 nm. The absorbance at 620 nm was used to normalize for worm numbers. The statistical significance of differences between conditions was determined by using unpaired t-test. GraphPad™ Prism 8.0 was used for these calculations.

Measuring Endogenous GSSG/2GSH Ratios

[0393] To measure in vivo GSSG/2GSH ratios, transgenic *jrIs2*[Prp1-17::Grx1-roGFP2] worms expressing a Grx1-roGFP2 as described³ were used. Synchronized L1-stage transgenic *jrIs2*[Prp1-17::Grx1-roGFP2] animals were grown in *E. coli* OP50 and *G. hanseni* and about 1000 L4-stage worms were harvested in 96 microtiter well plates after washing at least six times in M9 buffer. For H₂O₂ treatment experiment, H₂O₂ was added to the worms in the 96-well itself and measurements were taken within 15 minutes. An excitation wavelength of either 490 nm or 405 nm was used to measure reduced or oxidized Grx1-roGFP2 probe fluorescence respectively with an emission filter of 535 nm. The absorbance at 620 nm was used to normalize for worm numbers. The statistical significance of differences between conditions was determined by using unpaired t-test. GraphPad™ Prism 8.0 was used for these calculations.

Measuring Endogenous Hydrogen Peroxide Levels Using Amplex™ Red Assay

[0394] Measuring endogenous hydrogen peroxide levels using Amplex™ Red assay was performed as described⁴.

[0395] To measure ROS production, synchronized L1-stage wildtype worms were grown in *E. coli* OP50 and *G. hanseni* and about 1000 L4-stage worms were harvested in 96 microtiter well plates after washing at least six times in M9 buffer, followed by 3 washes with the reaction buffer supplied with the Amplex™ Red Hydrogen Peroxide/Peroxidase assay kit. Equal volume of the Amplex™ Red reaction buffer was then added to the wells, the absorbance at 540 nm was read after 3 hours with a plate reader (SpectraMax™ plate reader). For H₂O₂ treatment experiment, H₂O₂ was added to the worms in the 96-well itself and measurements were taken after 2 hours. The absorbance at 620 nm was used to normalize for worm numbers. All the values were normalized to the values of control untreated worms. The statistical significance of differences between conditions was determined by using unpaired t-test. GraphPad™ Prism 8.0 was used for these calculations.

Human mtDNA Copy Number Estimation

[0396] Human dermal primary Fibroblasts (Coriell GM25438) were grown in 6-well plates with EMEM containing 10% Fetal Bovine Serum, 1% MEM non-essential amino acids, and 1% Penicillin-Streptomycin until the cells were ~90% confluent in a 37° C., 5% CO₂ humidified incubator. The cells were treated with Mock-treatment or 10⁶ CFU of *G. hanseni*-treatment for 24 hours and the genomic DNA was isolated using a commercial DNA isolation kit (Qiagen). The quantity and purity of the isolated DNA was assayed using a Nanodrop™ 2000 spectrophotometer (Thermo Scientific, Wilmington, DE, USA) and the samples were stored at -70° C. until use. The relative mtDNA copy number was measured by qPCR as described⁵. The values were normalized to that of a single copy nuclear DNA gene as described. The primer sequences used were:

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L394,
5' - CACCAGCCTAACCAGATTTC - 3' (SEQ ID NO: 26)

H475,
5' - GGGTTGTATTGATGAGATTAGT - 3' (SEQ ID NO: 27)

HBG1F,
5' - GCTTCTGACACAACCTGTGTTCACTAGC - 3' (SEQ ID NO: 28)

HBG1R
5' - CACCAACTTCATCCACGTTACC - 3' (SEQ ID NO: 29)
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[0397] L394/H₄₇₅ were for the mtDNA content, and HBG1F/R primers were for nuclear β-globin gene⁵. qPCR was performed with an iCycler™ machine (Bio-Rad) using iQ SYBR Master Mix (Bio-Rad) and the assays were performed under the following conditions: denaturation at 95° C. for 10 minutes followed by 40 cycles of 10s at 95° C., 30s at 60° C. and 30s at 72° C. All assays were carried out in triplicate using 10 ng DNA template per reaction. All reactions were performed in triplicate and on 3 biological replicates. The relative mtDNA copy number was calculated using the ΔCt method.

RNA Isolation and Quantitative RT-PCR Analysis of PGC-1 and TEAM

[0398] Human primary Fibroblasts were grown in 6-well plates with EMEM containing 10% Fetal Bovine Serum, 1%

MEM non-essential amino acids, and 1% Penicillin-Streptomycin until the cells were ~90% confluent in a 37°C, 5% CO₂ humidified incubator. The cells were treated with Mock-treated or 10⁶ CFU of *G. hansenii*-treated for 24 hours. After incubation for 24 h in each condition, total cellular RNA was isolated from cells using commercial RNA isolation kit (Qiagen). Total RNA was DNase treated using the TURBO DNA-free kit (Applied Biosystems). cDNA was prepared using the First strand cDNA synthesis kit from Invitrogen. qRT-PCR assays were performed under the following conditions: denaturation at 95° C. for 10 minutes followed by 40 cycles of 10s at 95° C., 30 at 55° C. and 30s at 72° C. The primer sequence used^{6,7}:

htfamF (SEQ ID NO: 30):
TGTTCAATGGATAGGCAC

htfamR (SEQ ID NO: 31):
TCTGGGTTTCCAAAGCAAG

hPGC-1αF (SEQ ID NO: 32):
TGAAGACGGATTGCCCTCATT

hPGC-1αR (SEQ ID NO: 33):
GCTGGTGCCAGTAAGAGCTT

[0399] All reactions were done in triplicate and on 3 biological replicates. All the values are normalized to the nuclear β-globin gene as internal control as well as to the transcript levels in mock-treated cells.

ATP Determination Assay

[0400] ATP determination was performed as described⁸. To measure ATP levels, synchronized L1-stage worms were grown in *E. coli* OP50 and *G. hansenii* and were harvested after washing at least six times in M9 buffer and approximately 1000 worms in 100 μl of M9 were frozen in liquid nitrogen and stored at -80° C. until analysis. The samples were placed immediately in boiling water bath for 15 min. After adding twice the amount of distilled water, centrifugation was done at 14,000 rpm for 5 min and supernatant was transferred into a second tube. Equal volume of luciferase reagent (Promega™) was added and luminescence was measured using a plate reader (SpectraMax™ Microplate reader). Protein content was measured using BCA method. ATP concentrations were normalized to protein content of samples. All the values were normalized to wildtype fed on *E. coli* OP50.

ATP Determination Assay Using the Luciferase Strain

[0401] ATP determination assay using the Luciferase strain was described⁹. To measure in vivo ATP production, synchronized L1-stage Psur-5::luc+::gfp worms were grown in *E. coli* OP50 and *G. hansenii* and about 1000 adult-stage worms/well were harvested in 96 microtiter well plates after washing at least six times in M9 buffer. 100 μM D-luciferin was added to the wells and the luminescence was measured within 5 min in a SpectraMax™ microplate reader. GFP fluorescence was quantified in SpectraMax™ microplate reader using 485 nm excitation and a 520 nm emission filter. Background measurements were subtracted from readings. Luminescence values were normalized to GFP fluorescence of samples. All the values were normalized to wildtype fed on *E. coli* OP50. For the RNAi experiments, synchronized L1-stage Psur-5::luc+::gfp worms were grown in *E. coli*

HT115 strain carrying the corresponding the dsRNA until adulthood and egg-prepped. The L1-stage worms were then transferred to either *E. coli* OP50 or *G. hansenii*-seeded media plates. For the experiment using rotenone, paraquat, azide, and, antimycin, the worms grown on *E. coli* OP50 or *G. hansenii* were washed at least 6 times and transferred to *E. coli* OP50 plates containing the drugs. The *E. coli* OP50 plates with proper concentrations of drugs were made one day before use. All the assays were conducted on *E. coli* OP50 plates to avoid potential drug detoxification by *G. hansenii*.

Mitotracker CMXRos Staining

[0402] For the experiment using Mitotracker™ CMXRos, the worms grown on *E. coli* OP50 or *G. hansenii* were washed at least 6 times and transferred to *E. coli* OP50 plates containing the 10 μM of Mitotracker™ CMXRos. After 24 hours, worms were washed at least 6 times to remove the dye and dropped onto *E. coli* OP50 plates to clear off non-specific gut staining. After 2 hours, worms were washed at least 6 times and transferred to agar plates for imaging.

Tetramethylrhodamine Ethylester (TMRE) Staining

[0403] For the experiment using TMRE, the worms grown on *E. coli* OP50 or *G. hansenii* were washed at least 6 times and transferred to *E. coli* OP50 plates containing the 1 μM of TMRE. After 24 hours, worms were washed at least 6 times to remove the dye and dropped onto *E. coli* OP50 plates to clear off non-specific gut staining. After 2 hours, worms were washed at least 6 times and ~1000 worms were transferred to each well of 96-well plate and the fluorescence was quantified in SpectraMax™ microplate reader using 549 nm excitation and a 575 nm emission filter.

Optimization of Metabolite Production

[0404] It was discovered that when the bacteria are grown in either 0-5% yeast extract broth, the metabolites are not produced. Further, it was found that addition of glucose or other metabolizable monosaccharide sugars such as fructose or disaccharides such as sucrose to the 3% yeast extract medium is required for metabolite production. In addition, it was discovered that as little as 0.5% of such sugars is enough for optimal metabolite production. Thus, it is recommended that any medium used for metabolite production comprise at least 0.5-5% of a metabolizable sugar source.

[0405] Further experiments indicate that 3% corn steep liquor is sufficient for optimal metabolite production, thus 0.5-5% corn steep liquor is recommended for optimal production of metabolites. It is contemplated herein that when bacteria are grown on a commercial scale, 3% corn steep liquor may be the preferred method for fermentation.

References for Example 2

- [0406]** 1. Brenner, S. The genetics of *Caenorhabditis elegans*. Genetics 77, 71-94 (1974).
- [0407]** 2. The Family Acetobacteraceae: The Genera *Acetobacter*, *Acidomonas*, *Asaia*, *Gluconacetobacter*, *Gluconobacter*, and *Kozakia*. (Springer-Verlag, 2011). doi:10.1007/SpringerReference_3770
- [0408]** 3. Back, P. et al. Exploring real-time in vivo redox biology of developing and aging *Caenorhabditis elegans*. Free Radic. Biol. Med. 52, 850-859 (2012).

- [0409] 4. Chávez, V., Mohri-Shiomi, A., Maadani, A., Vega, L. A. & Garsin, D. A. Oxidative stress enzymes are required for DAF-16-mediated immunity due to generation of reactive oxygen species by *Caenorhabditis elegans*. *Genetics* 176, 1567-1577 (2007).
- [0410] 5. Chen, S. et al. Elevated mitochondrial DNA copy number in peripheral blood cells is associated with childhood autism. *BMC Psychiatry* 15, 50 (2015).
- [0411] 6 Zaccagnino, P. et al. An active mitochondrial biogenesis occurs during dendritic cell differentiation. *Int. J. Biochem. Cell Biol.* 44, 1962-1969 (2012).
- [0412] 7. Oral, E. A. et al. Inhibition of IKKE and TBK1 Improves Glucose Control in a Subset of Patients with Type 2 Diabetes. *Cell Metab.* 26, 157-170.e7 (2017).
- [0413] 8. Palikaras, K. & Tavernarakis, N. Intracellular Assessment of ATP Levels in *Caenorhabditis elegans*. *Bio Protoc* 6, (2016).
- [0414] 9. Lagido, C., McLaggan, D. & Glover, L. A. A Screenable In Vivo Assay for Mitochondrial Modulators Using Transgenic Bioluminescent *Caenorhabditis elegans*. *J Vis Exp* e53083-e53083 (2015). doi: 10.3791/53083

Example 3: Treatment of Parkinson's Disease

[0415] Parkinson's Disease (PD) is the most common motor-related disorder in middle or late life disease, affecting ~6.2 million people worldwide.¹ PD is characterized by accumulation of α -synuclein inclusions in the neurons and degeneration and/or loss of dopaminergic neurons. The cardinal clinical symptoms of PD includes slow movement, resting tremor, rigidity, and postural instability.² While a majority of PD cases are of unknown origin and sporadic, mutations in some genes have been associated with rare, familial forms of the disease. Several lines of evidence implicate defects in mitochondrial respiration in the etiology and pathogenesis of PD. First, MPTP, an inhibitor of complex I of the electron transport chain, can induce PD.^{3,4} Inhibition of complex I results in decreased mitochondrial ATP production, increased production of mitochondria-derived Reactive Oxygen Species (ROS), and activation of mitochondria-dependent apoptotic pathways. Second, post mortem studies of PD patients found elevated levels of oxidative stress markers/products⁵⁻⁷ in the dopaminergic neurons. Third, a reduction of mitochondrial complex I activity by 30% was observed in brain as well as peripheral tissues of PD patients.^{8,9} Fourth, neurotoxins such as rotenone, paraquat, and 6-hydroxydopamine (6-OHDA) induce mitochondrial dysfunction resulting in PD-associated phenotypes in animal models.¹⁰ Finally, PD-associated genes such as α -synuclein, LRRK2 (leucine-rich repeat kinase 2), parkin, PINK1, and DJ-1 affect mitochondrial dynamics, trafficking, autophagy, and quality control.^{11,12}

[0416] All cells require mitochondria for their energy demands including neurons, which are critically dependent on proper mitochondrial function. Neurons have high metabolic activity and they depend heavily on mitochondria for their bioenergetic demand. Several factors make the neurons in general, and dopaminergic neurons in particular, susceptible to degeneration; these include ROS (which result from dopamine metabolism and mitochondrial dysfunction), low endogenous antioxidant levels, and high levels of iron and calcium (which are known to promote ROS formation).¹³ Further, neuronal tissues contain high levels of polyunsaturated fatty acids, which are prone to lipid peroxidation and

the generation of toxic products.¹⁴ Regardless of whether a primary or secondary cause, mitochondrial dysfunction holds promise as a potential therapy target. Aging is the greatest risk factor for PD,¹⁵ thus with increasing average life expectancy worldwide,¹⁶ the number of people affected by PD will rise considerably in the near future. Thus, there is a significant clinical unmet need for new therapeutic approaches that not only can be used for slowing down PD, but also as preventive measures for the aging population.

[0417] Mitochondria are highly dynamic organelles required for cellular energy production, response to oxidative stress, and apoptosis. Thus, it is not surprising that mitochondrial dysfunction, characterized by a loss of efficiency in the electron transport chain and reduction in ATP synthesis, is a characteristic of aging, and aging-related neurodegenerative diseases.^{17,18} Despite having distinct pathological and clinical features, several other neurodegenerative disorders including Alzheimer's disease, Huntington's disease, and amyotrophic lateral sclerosis are associated with mitochondrial dysfunction-derived oxidative stress¹⁹ as well; this suggests a common mechanism that contributes to neuronal degeneration²⁰.

[0418] While most of the earlier studies on PD focus entirely on the brain pathologies, the gastrointestinal (GI) system is now recognized as an important source for PD pathogenesis.²¹⁻²³ GI symptoms, such as constipation, affects ~80% of PD-patients and idiopathic constipation is an important risk factor for PD.²⁴ In PD, constipation is associated with α -synuclein accumulation in the enteric nervous system,²⁵ gut inflammation, and increased gut permeability.²⁶ Further, the intestinal mucosal inflammation is thought to lead to synuclein accumulation in the enteric nerves, which can then spread in a prion-like fashion to the central nervous system via autonomic connections.²⁷⁻²⁹ Many of the GI tract changes are observed even before the onset of neuronal symptoms;³⁰ thus, PD pathogenesis might act primarily via the GI tract^{31,32}.

[0419] *C. elegans* as a model is well-positioned to play an important role in PD-associated discoveries.³³ *C. elegans* contain well-defined dopaminergic (DA) neurons that are visible under fluorescence microscopy and have a range of locomotor activities.³⁴ All orthologs of genes linked to familial PD are present in *C. elegans*, except α -synuclein. Ectopic overexpression of α -synuclein produces neurotoxic effects, which can be blocked by neuroprotective genes. A number of worm models of PD have been generated through either exposing worms to neurotoxins such as MPP+ or 6-OHDA or by introducing the mutations or human genes implicated in inherited forms of PD.³⁵⁻⁴¹ These worm models exhibit PD-associated phenotypes including degeneration or loss of dopamine neurons, low dopamine levels, defects in dopamine-dependent behaviors, and movement defects.

[0420] Overexpression of human α -synuclein under the control of a DA neuron-specific promoter results in age-dependent neurodegeneration.⁴² Using this strain, over 30 strains of microbes were screened for prospective neuroprotective potential. Expression of human α -synuclein in the dopaminergic neurons induces complete degeneration and loss of the neurons by day 10 of adulthood. It was observed that when combined with 1 uM of Rotenone, the dopaminergic neuron loss is achieved more rapidly in 4 days. Rotenone is a broad-spectrum insecticide that has been demonstrated to induce PD-like symptoms in animal models

accompanied by the selective destruction of DAergic neurons. In *C. elegans*, chronic exposure of rotenone caused dopaminergic neurodegeneration.⁴³ Rotenone acts by inhibiting mitochondrial respiratory complex I.⁴⁴ In the screen, the human alpha-synuclein animals were fed ad-libitum on a single strain of microbial lawn starting from the first larval stage and subjected to rotenone treatment at the fourth larval stage. After 60 hours of rotenone treatment, the animals were screened for survival of the anterior CEP (cephalic) dopaminergic neurons, which were visualized via co-expression of GFP driven in the dopaminergic neurons. While ~75% of the worms fed on the control *E. coli* op50 had lost the CEP neurons, worms fed on *Gluconoacetobacter* EBT405 decreased CEP neurodegeneration (FIGS. 10A-10B).

References for Example 3

- [0421] 1. GBD 2015 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990-2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet* 388, 1545-1602 (2016).
- [0422] 2. Pires, A. O. et al. Old and New Challenges in Parkinson's Disease Therapeutics. *Prog. Neurobiol.* (2017). doi: 10.1016/j.pneurobio.2017.04.006
- [0423] 3. Nicklas, W. J., Vyas, I. & Heikkila, R. E. Inhibition of NADH-linked oxidation in brain mitochondria by 1-methyl-4-phenyl-pyridine, a metabolite of the neurotoxin, 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine. *Life Sci.* 36, 2503-2508 (1985).
- [0424] 4. Ramsay, R. R., Salach, J. I., Dadgar, J. & Singer, T. P. Inhibition of mitochondrial NADH dehydrogenase by pyridine derivatives and its possible relation to experimental and idiopathic parkinsonism. *Biochem Biophys Res Commun* 135, 269-275 (1986).
- [0425] 5. Yoritaka, A. et al. Immunohistochemical detection of 4-hydroxynonenal protein adducts in Parkinson disease. *Proceedings of the National Academy of Sciences* 93, 2696-2701 (1996).
- [0426] 6. Floor, E. & Wetzel, M. G. Increased Protein Oxidation in Human Substantia Nigra Pars Compacta in Comparison with Basal Ganglia and Prefrontal Cortex Measured with an Improved Dinitrophenylhydrazine Assay. *J Neurochem* 70, 268-275 (2002).
- [0427] 7. Zhang, J. et al. Parkinson's disease is associated with oxidative damage to cytoplasmic DNA and RNA in substantia nigra neurons. *Am. J. Pathol.* 154, 1423-1429 (1999).
- [0428] 8. Schapira, A. H. V. et al. MITOCHONDRIAL COMPLEX I DEFICIENCY IN PARKINSON'S DISEASE. *The Lancet* 333, 1269 (1989).
- [0429] 9. Parker, W. D., Boyson, S. J. & Parks, J. K. Abnormalities of the electron transport chain in idiopathic Parkinson's disease. *Ann. Neurol.* 26, 719-723 (1989).
- [0430] 10. Tieu, K. A guide to neurotoxic animal models of Parkinson's disease. *Cold Spring Harb Perspect Med* 1, a009316-a009316 (2011).
- [0431] 11. Moore, D. J., West, A. B., Dawson, V. L. & Dawson, T. M. Molecular pathophysiology of Parkinson's disease. *Annu. Rev. Neurosci.* 28, 57-87 (2005).
- [0432] 12. Robinson, P. A. (2010). Understanding the molecular basis of Parkinson's disease, identification of biomarkers and routes to therapy. *Expert Review of Proteomics*, 7(4), 565-578.
- [0433] 13. Dias, V., Junn, E., & Mouradian, M. M. The role of oxidative stress in Parkinson's disease. *Journal of Parkinson's Disease*, 3 (4), 461-491 (2013).
- [0434] 14. Liu, X., Yamada, N., Maruyama, W., & Osawa, T. Formation of dopamine adducts derived from brain polyunsaturated fatty acids: mechanism for Parkinson disease. *Journal of Biological Chemistry*, 283(50), 34887-34895 (2008).
- [0435] 15. Collier, T. J., Kanaan, N. M., & Kordower, J. H. Ageing as a primary risk factor for Parkinson's disease: evidence from studies of non-human primates. *Nature Reviews. Neuroscience*, 12(6), 359-366 (2011).
- [0436] 16. GBD 2013 Mortality and Causes of Death Collaborators, *Lancet* 385, 117 (2015).
- [0437] 17. Federico, A., Cardaioli, E., Da Pozzo, P., Formichi, P., Gallus, G. N. and Radi, E. Mitochondria, oxidative stress and neurodegeneration. *J. Neurol. Sci.* 322, 254-262 (2012).
- [0438] 18. Winklhofer, K. F., and Haass, C. Mitochondrial dysfunction in Parkinson's disease. *Biochim. Biophys. Acta* 1802, 29-44 (2010).
- [0439] 19. Albers, D. S., and Flint Beal, M. Mitochondrial dysfunction and oxidative stress in aging and neurodegenerative disease. In *Advances in Dementia Research* (pp. 133-154) (2000).
- [0440] 20. Andersen, J. K. Oxidative stress in neurodegeneration: cause or consequence? *Nature Reviews. Neuroscience*, 10(7), S18-S25. (2004).
- [0441] 21. Mukherjee, A., Biswas, A., & Das, S. K. Gut dysfunction in Parkinson's disease. *World Journal of Gastroenterology*, 22(25), 5742-5752 (2016).
- [0442] 22. Pellegrini, C., Colucci, R., Antonioli, L., Barocelli, E., Ballabeni, V., Bernardini, N., et al. Intestinal dysfunction in Parkinson's disease: Lessons learned from translational studies and experimental models. *Neurogastroenterology & Motility*, 28(12), 1781-1791 (2016).
- [0443] 23. Kuo, Y.-M., Li, Z., Jiao, Y. N., Gaborit, A. K. Pani, B. M. Orrison, B. G. Bruneau, B. I. Giasson, R. J. Smeyne, M. D. Gershon, and R. L. Nussbaum. Extensive enteric nervous system abnormalities in mice transgenic for artificial chromosomes containing Parkinson disease-associated α -synuclein gene mutations precede central nervous system changes. 19, 1633-1650 (2010).
- [0444] 24. Noyce, A. J., Bestwick, J. P., Silveira-Moriyama, L., Hawkes, C. H., Giovannoni, G., Lees, A. J and Schrag, A. Meta-analysis of early nonmotor features and risk factors for Parkinson disease. 72, 893-901 (2012).
- [0445] 25. Cersosimo, M. G and Benarroch, E. E. Pathological correlates of gastrointestinal dysfunction in Parkinson's disease. 46, 559-564 (2012).
- [0446] 26. Devos, D., Lebouvier, T., Lardeux, B. M., Biraud, Rouaud, T., Pouclet, H., E. Coron, Bruley des Varannes, S., Naveilhan, P., Nguyen, J.-M., Neunlist, M., and Derkinderen, P. Colonic inflammation in Parkinson's disease. 50, 42-48 (2013).
- [0447] 27. Braak, H., Rüb, U., Gai, W. P and Del Tredici, K. Idiopathic Parkinson's disease: possible routes by which vulnerable neuronal types may be subject to neuroinvasion by an unknown pathogen. 110, 517-536 (2003).

- [0448] 28. Hawkes, C. H., Del Tredici, K and Braak, H. Neuropathol. Parkinson's disease: a dual-hit hypothesis. 33, 599-614 (2007).
- [0449] 29. Hawkes, C. H., Del Tredici, K and Braak, H. Parkinson's disease: the dual hit theory revisited. 1170, 615-622 (2009).
- [0450] 30. Verbaan, D., Marinus, J., Visser, M., van Rooden, S. M., Stiggelbout, A. M., and van Hilten, J. J. Patient-reported autonomic symptoms in Parkinson disease. *Neurology* 69, 333-341 (2007).
- [0451] 31. Shannon, K. M., Keshavarzian, A., Dodiya, H. B., Jakate S., and Kordower, J. H. Is alpha-synuclein in the colon a biomarker for premotor Parkinson's disease? Evidence from 3 cases. *Mov. Disord.* 27, 716-719 (2012).
- [0452] 32. Kiebert, K and Wunderle, K. B. Parkinson's disease: evidence for environmental risk factors. 28, 8-13 (2013).
- [0453] 33. Caldwell, G. A., & Caldwell, K. A. Traversing a wormhole to combat Parkinson's disease. *Disease Models & Mechanisms*, 1(1), 32-36 (2008).
- [0454] 34. Nass, R., Hall, D. H., Miller, D. M. & Blakely, R. D. Neurotoxin-induced degeneration of dopamine neurons in *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences* 99, 3264-3269 (2002).
- [0455] 35. Lakso, M. et al. Dopaminergic neuronal loss and motor deficits in *Caenorhabditis elegans* overexpressing human alpha-synuclein. *J Neurochem* 86, 165-172 (2003).
- [0456] 36. Ved, R. et al. Similar patterns of mitochondrial vulnerability and rescue induced by genetic modification of alpha-synuclein, parkin, and DJ-1 in *Caenorhabditis elegans*. *J Biol Chem* 280, 42655-42668 (2005).
- [0457] 37. van Ham, T. J. et al. *C. elegans* model identifies genetic modifiers of alpha-synuclein inclusion formation during aging. *PLoS Genet* 4, e1000027 (2008).
- [0458] 38. Caldwell, G. A. & Caldwell, K. A. Traversing a wormhole to combat Parkinson's disease. *Dis Model Mech* 1, 32-36 (2008).
- [0459] 39. Harrington, A. J., Hamamichi, S., Caldwell, G. A. & Caldwell, K. A. *C. elegans* as a model organism to investigate molecular pathways involved with Parkinson's disease. *Dev. Dyn.* 239, 1282-1295 (2010).
- [0460] 40. Braungart, E., Gerlach, M., Riederer, P., Baumeister, R. & Hoener, M. C. *Caenorhabditis elegans* MPP+ model of Parkinson's disease for high-throughput drug screenings. *Neurodegener Dis* 1, 175-183 (2004).
- [0461] 41. Cao, S., Gelwix, C. C., Caldwell, K. A. & Caldwell, G. A. Torsin-mediated protection from cellular stress in the dopaminergic neurons of *Caenorhabditis elegans*. *J. Neurosci.* 25, 3801-3812 (2005).
- [0462] 42. Hamamichi, S. et al. Hypothesis-based RNAi screening identifies neuroprotective genes in a Parkinson's disease model. *Proc Natl Acad Sci USA* 105, 728-733 (2008).
- [0463] 43. Zhou, S., Wang, Z. & Klaunig, J. E. *Caenorhabditis elegans* neuron degeneration and mitochondrial suppression caused by selected environmental chemicals. *Int J Biochem Mol Biol* 4, 191-200 (2013).
- [0464] 44. Gao, H.-M., Liu, B. & Hong, J.-S. Critical role for microglial NADPH oxidase in rotenone-induced degeneration of dopaminergic neurons. *J. Neurosci.* 23, 6181-6187 (2003).

Example 4: Treatment of Alzheimer's Disease

[0465] AD is a fatal neurodegenerative disease characterized by progressive decline in memory and cognitive functions. Early-onset familial AD associated with the mutations in APP or γ -secretase gene accounts for less than 5% of the all cases while 95% of the sporadic or late-onset AD has unknown etiology. The pathological hallmarks of AD are accumulation of extracellular senile plaques and intracellular neurofibrillary tangles (NFTs) in the AD patient's brain. The senile plaques consist of β -amyloid peptide ($A\beta$) as the primary component while NFTs consist of abnormal fibrillar forms of microtubule-associated protein tau as the primary component². $A\beta$ accumulation and NFTs are thought to indicate neuronal dysfunction and impending neuronal demise².

[0466] Mitochondrial dysfunction and defects in energy metabolism have been consistently observed in human AD patients^{3,4}. $A\beta$ and tau pathologies are strongly associated with mitochondrial dysfunction in AD. $A\beta$ and tau directly affect mitochondrial function causing impairment of ATP production, increased reactive oxygen species (ROS) production, decreased oxygen consumption, and decreased mitochondrial complex I and IV function⁵. Mitochondrial dysfunction is found to be an early event in AD. In the case of sporadic AD, progressive increase in oxidative stress with advancing age is shown to cause $A\beta$ deposition and NFTs formation⁶. This could lead to a continuous cycle of events where the $A\beta$ and tau exacerbates mitochondrial dysfunction leading to rapidly progressive AD symptoms. Thus, therapeutic efforts aimed at restoring mitochondrial function or protecting mitochondria against $A\beta$ -induced damage or replenishing the mitochondrial numbers and function in the aging population is a promising route for identifying AD therapeutics.

[0467] *C. elegans* as a model is well-positioned to play an important role in AD-associated discoveries. Ortholog of genes associated with the cause of AD are conserved in *C. elegans* including amyloid precursor protein (APP) and tau. A number of worm models of AD have been generated by introducing the mutations or human genes implicated in AD⁷⁻⁹. These worm models exhibit AD-associated phenotypes including amyloid deposits, increased oxidative stress and progressive paralysis⁷⁻⁹. Further, impairment of electron transport chain function and mitochondrial dysfunction were found in the *C. elegans* AD models^{10,11}, similar to those observed in AD patients. Further, these models have been used successfully in identifying neuroprotective compounds⁹.

[0468] About one third of the drugs used in the clinic today were initially isolated from plants or microbes. Though chemical synthesis of compounds has taken over the pharmaceutical industry as the source for identifying novel therapeutics, bioprospecting natural sources like plants and microbes continues to play important role as therapeutic agents. Recent studies have started to explore the human microbiome as a source of drugs.¹² The human body is home to a society of benign, symbiotic, commensal and pathogenic microbes collectively known as the microbiome. These microbes can modulate host brain function and behavior via the gut-brain axis and production of several metabolites such as GABA, glutamate and serotonin.

[0469] Dysfunction in the microbiome-derived metabolite signaling can contribute to neurological disorders including AD.^{13,14} In summary, the microbiome presents an untapped

rich resource for mining novel neuroprotective compounds or live biotherapeutics that will be relevant for AD.

[0470] The inventors have developed a high-throughput screening (HTS) assay to rapidly prioritize “single” microbial species for identifying microbiome-based therapeutics for AD using *C. elegans* whole-animal model system. In the transgenic strains expressing AD-relevant human pathogenic A β or tau in all the neurons, it was found that mitochondrial unfolded protein response (as assessed by (UPRmt) reporter construct phsp-6::gfp) was turned on in many tissues, including the gut (FIGS. 11&12). hsp-6 encodes a mitochondrial matrix chaperone HSP70, which is specifically up-regulated in response to impaired mitochondrial structure or function¹⁵ and can be assessed using the phsp-6::gfp strain in which the promoter of hsp-6 gene is tagged with GFP. These data are consistent with previous studies that show expression of A β in *C. elegans* results in the accumulation of abnormal mitochondria¹⁶ and muscle-specific expression of A β induced mitochondrial stress genes including hsp-6.¹¹ Expression of A β or tau in the neurons using the neuronal-specific (unc-119) promoter induces mitochondrial stress reporter expression (UPRmt) primarily in the gut (FIGS. 11 & 12), this response is cell-non-autonomous in nature. Similarly, neuron to gut cell-non-autonomous induction of UPRmt has been observed by other labs and proposed to function through neuropeptide and serotonin signaling pathways in *C. elegans*.¹⁷⁻¹⁹

[0471] Interestingly, modulation of serotonin signaling is considered to be a potential symptomatic treatment for cognitive symptoms in AD.^{20,21} To determine whether the activation of UPRmt is a direct consequence of A β aggregates, transgenic worms expressing A β and phsp-6::gfp reporter were treated with Thioflavin T, which is an amyloid fibril-binding flavonoid shown to reduce A β aggregation in vivo and rescue the paralysis phenotype.²² It was found that in worms treated with Thioflavin T, the induction of UPRmt was reduced, indicating that the induction of UPRmt is directly due to A β aggregation. Thus, it is possible to screen for interventions that will reduce the UPRmt induction by modulating the A β aggregation or toxicity of A β on mitochondria.

[0472] For the screen, thirty microbial strains including the negative control bacteria *E. coli* OP50 were grown in appropriate culture media conditions and spotted onto standard nematode growth medium plates. ~30 synchronized L1-larval-stage transgenic worm expressing A β 1-42 or tau in the pan-neurons were dropped onto the seeded plates and scored visually after 4 days under a fluorescent microscope for changes in phsp-6::gfp expression (FIG. 13). To synchronize worms, a bleach-prepping method was used, which also created germ-free animals that were fed on monocultures. The inventors screened for changes in the expression of phsp-6::gfp reporter and found that worms fed on *Gluconobacter* EBT 405 bacteria showed significant reduction in the expression of phsp-6::gfp induced by A β 1-42 expressing or transgenic tau (FIGS. 11 & 12).

[0473] To evaluate the A β aggregation state in vivo, a transgenic strain expressing human A β 3-42 conjugated in the body wall muscle²³ was used; EBT405 bacteria were tested on this model and found a significant reduction of A β deposits compared to transgenic worms fed on negative control bacteria (FIG. 14). This strain will be used to test the hits from the HTS screen for reduction of A β deposits.

[0474] To evaluate the ability of *Gluconobacter* EBT 405 to suppress A β -induced paralysis phenotype, the inventors used a strain that expresses full-length human A β 1-42 peptide in body wall muscle cells²⁴. Shifting L4 or young adult animals from 20° C. to 25° C. causes paralysis in worms fed on the control bacteria while the time of onset of paralysis is significantly delayed in worms fed on *Gluconobacter* EBT 405.

References for Example 4

- [0475]** 1. Masters, C. L. et al. Alzheimer's disease. Nat Rev Dis Primers 1, 15056 (2015).
- [0476]** 2. Nelson, P. T. et al. Correlation of Alzheimer disease neuropathologic changes with cognitive status: a review of the literature. J. Neuropathol. Exp. Neurol. 71, 362-381 (2012).
- [0477]** 3. Halliwell, B. Oxidative stress and neurodegeneration: where are we now? J Neurochem 97, 1634-1658 (2006).
- [0478]** 4. Cheignon, C. et al. Oxidative stress and the amyloid beta peptide in Alzheimer's disease. Redox Biol 14, 450-464 (2018).
- [0479]** 5. Müller, W. E., Eckert, A., Kurz, C., Eckert, G. P. & Leuner, K. Mitochondrial dysfunction: common final pathway in brain aging and Alzheimer's disease—therapeutic aspects. Mol. Neurobiol. 41, 159-171 (2010).
- [0480]** 6. Moreira, P. I., Carvalho, C., Zhu, X., Smith, M. A. & Perry, G. Mitochondrial dysfunction is a trigger of Alzheimer's disease pathophysiology. Biochim. Biophys. Acta 1802, 2-10 (2010).
- [0481]** 7. Link, C. D. *C. elegans* models of age-associated neurodegenerative diseases: lessons from transgenic worm models of Alzheimer's disease. Exp. Gerontol. 41, 1007-1013 (2006).
- [0482]** 8. Lublin, A. L. & Link, C. D. Alzheimer's disease drug discovery: in vivo screening using *Caenorhabditis elegans* as a model for β -amyloid peptide-induced toxicity. Drug Discov Today Technol 10, e115-9 (2013).
- [0483]** 9. Chen, X., Barclay, J. W., Burgoyne, R. D. & Morgan, A. Using *C. elegans* to discover therapeutic compounds for ageing-associated neurodegenerative diseases. Chem Cent J 9, 65 (2015).
- [0484]** 10. Ng, L. F. et al. The mitochondria-targeted antioxidant MitoQ extends lifespan and improves healthspan of a transgenic *Caenorhabditis elegans* model of Alzheimer disease. Free Radic. Biol. Med. 71, 390-401 (2014).
- [0485]** 11. Sorrentino, V. et al. Enhancing mitochondrial proteostasis reduces amyloid- β proteotoxicity. Nature 552, 187-193 (2017).
- [0486]** 12. Donia, M. S. & Fischbach, M. A. HUMAN MICROBIOTA. Small molecules from the human microbiota. Science 349, 1254766-1254766 (2015).
- [0487]** 13. Sharon, G., Sampson, T. R., Geschwind, D. H. & Mazmanian, S. K. The Central Nervous System and the Gut Microbiome. Cell 167, 915-932 (2016).
- [0488]** 14. Hill, J. M., Bhattacharjee, S., Pogue, A. I. & Lukiw, W. J. The gastrointestinal tract microbiome and potential link to Alzheimer's disease. Front Neurol 5, 43 (2014).
- [0489]** 15. Yoneda, T. et al. Compartment-specific perturbation of protein handling activates genes encoding mitochondrial chaperones. J. Cell. Sci. 117, 4055-4066 (2004).

- [0490] 16. Fonte, V. et al. A glycine zipper motif mediates the formation of toxic β -amyloid oligomers in vitro and in vivo. *Mol Neurodegener* 6, 61 (2011).
- [0491] 17. Shao, L.-W., Niu, R. & Liu, Y. Neuropeptide signals cell non-autonomous mitochondrial unfolded protein response. *Cell Res* 26, 1182-1196 (2016).
- [0492] 18. Durieux, J., Wolff, S. & Dillin, A. The cell-non-autonomous nature of electron transport chain-mediated longevity. *Cell* 144, 79-91 (2011).
- [0493] 19. Berendzen, K. M. et al. Neuroendocrine Coordination of Mitochondrial Stress Signaling and Proteostasis. *Cell* 166, 1553-1563.e10 (2016).
- [0494] 20. Ferrero, H., Solas, M., Francis, P. T. & Ramirez, M. J. Serotonin 5-HT₆ Receptor Antagonists in Alzheimer's Disease: Therapeutic Rationale and Current Development Status. *CNS Drugs* 31, 19-32 (2017).
- [0495] 21. Claeyssen, S., Bockaert, J. & Giannoni, P. Serotonin: A New Hope in Alzheimer's Disease? *ACS Chem Neurosci* 6, 940-943 (2015).
- [0496] 22. Alavez, S., Vantipalli, M. C., Zucker, D. J. S., Klang, I. M. & Lithgow, G. J. Amyloid-binding compounds maintain protein homeostasis during ageing and extend lifespan. *Nature* 472, 226-229 (2011).
- [0497] 23. Link, C. D. et al. The beta amyloid peptide can act as a modular aggregation domain. *Neurobiol. Dis.* 32, 420-425 (2008).
- [0498] 24. McColl, G. et al. Utility of an improved model of amyloid-beta ($A\beta_{1-42}$) toxicity in *Caenorhabditis elegans* for drug screening for Alzheimer's disease. *Mol Neurodegener* 7, 57 (2012).

SEQUENCE LISTING

Sequence total quantity: 44
 SEQ ID NO: 1 moltype = DNA length = 2427
 FEATURE Location/Qualifiers
 misc_feature 1..2427
 note = Description of Artificial Sequence: Synthetic polynucleotide
 source 1..2427
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 1
 atgagcacaa ttctccggcc agggctctgg gccctgataa cgcccgcggt attcgcgctt 60
 tgcggcgcgga tctttaccgt tggcgggcgca tgggtcgctg ccatcgggcg cctcttttat 120
 tatgtcatcc ttggcctggc acttctcgcc acggctttcc tctcattccg gcgcaatccg 180
 gctgccctct atctgttcgc agtcgtcgtc ttcggaacgg tcatctggga actcaccgtt 240
 gtccgtctcg acatctgggc cctgatcccg cgctcggaca tcgtcatcat cctcggccatc 300
 tggctgctgc tgcggttcgt ctccgcgcag atcgggtggca cgcggacgac cgtcctgcgg 360
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 catgacatca gcggcgaaact gccgacgcag atcgcaaacg cctccccgcg cgaccgggac 480
 aacgttccgg ccagcgagtg gcacgcttat ggtcgtacgc agggccggtga ccgctggtcc 540
 ccgctgaacc agatcaatgc gacgaacgct agcaacctca aggtcgcatg gcataccac 600
 accaaggata tgatgaactc caacgacccg ggcgaaagcga cgaacgaagc gacgccgatc 660
 gagttcaaca acacgcttta tatgtgctca ctgcacaga agctgtttgc ggttgatggt 720
 gccaccggga ctctcgaaag ggtctacgat ccgaagctcc agatcaaccc tggcttccag 780
 catctgacct gccgtggcgt cagcttccac gaaacgcggg ccaatgccat ggattccgat 840
 ggcaatcctg ctccgacgga ctgcgccaa cgcacatccc tgcgggtcaa tgatggccgt 900
 ctggttgaag tggatggcga caccggcaag acctgctcgg gcttcggcaa caatggcgag 960
 atcgacctgc gcgttccgaa ccagccttac acgacgcctg gccagtacga gccgacgtcc 1020
 ccgcccgtca tcacagacaa gctgatcatc gccaacagcg ccatcaccga taacgggttcg 1080
 gtcaagcgat ctccggcgcg caccgaggca ttccgactct acaccggcaa gcgcgtctgg 1140
 gtgttcgatg cgtccaaacc ggatccgaac cagcttccgg atgagagcca cctgtctctc 1200
 caccgcaact cgccaaactc ctggatcgtg tcgtcctaag acgccaacct gaacctcgtg 1260
 tactccccga tgggcgtggg gactccccgac cagtggggcg gtgaccgcac gaaggattcc 1320
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 cagcttgctc cccgcggacc gggcaaccgg ctctggcctg aaggaaatgc caagggcacg 1920
 ggtggtgaaa ccggcctgca gcacaactat ggcatccggt atgcccgtcaa cctgcatccg 1980
 ttcttgatc ccgtgctgct gccgttcggc atcaagatgc cgtgcccgcac gcccccctgg 2040
 ggctatgtcg ccggtattga cctgaagacc aacaaggtcg tctggcagca ccgcaacggc 2100
 accctgcgtg actcgatgta tggcagctcc ctgcccagcc cgtgcccgcg gatcaagatc 2160
 ggtgtcccga gccctcgggg ccgcctctcc acggctggca atctcggtt cctgacggcg 2220
 tccatggatt actacatccg tgcgtacaac ctgacgacgg gcaaggtgct gtggcaggac 2280
 cgtctgcggg ctgggtgctca ggcaacccgg atcacctatg ccatcaacgg caagcagtac 2340
 atcgtgacct atgcaggcgg acacaactcg ttcccagccc gcattggcgca cgacatcatc 2400
 gcctacgccc tgcgccatca gaaatga 2427

SEQ ID NO: 2 moltype = AA length = 808
 FEATURE Location/Qualifiers
 REGION 1..808

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note = Description of Artificial Sequence: Synthetic
      polypeptide
source 1..808
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 2
MSTISRPGWL ALITAAVFAL CGAILTVGGA WVAAIGGPLY YVILGLALLA TAFLSFRRNP 60
AALYLFVAVV FGTVIWELTV VGLDIWALIP RSDIIVILGI WLLLPFVSRQ IGGTRTTVLP 120
LAGAVGVAVL ALFASLFTDP HDISGELPTQ IANASPADPD NVPASEWHAY GRTQAGDRWS 180
PLNQINATNV SNLKVAVHIH TKDMMNSNDP GEATNEATPI EFNNTLYMCS LHQKLFAVDG 240
ATGNVWVVDY PKLQINPGFQ HLTCRGVSFH ETPANAMSDS GNPAPTDCAK RIILPVNDGR 300
LVEVDADTGG TCSGFGNNGE IDLRVPNQPY TTPGQYEPTS PPVITDKLII ANSAITDNGS 360
VKQASGATQA PDVYTGRVW VFDASNPDN QLPDESHPVF HPNSPNSWIV SSYDANLNLV 420
YIPMGVGTDP QWGGDRTKDS ERFAPGIVAL NADTGKLAWF YQTVHDLWD MDVPSQPSLV 480
DVTQKDGTLV PAIYAPTKTG DIFVLDRTG KEIVPAPETP VPQGAAPGDH TSPTQPMSQL 540
TLRPKNPLND SDIWGGTIFD QMPCSIYFHT LRYEGFPTTP SLKGSILFPG DLGMFEWGG 600
AVDPQRQVAF ANPISLPFVS QLVPRGPGNP LWPEGNAGKT GGETGLQHNY GIPYAVNLHP 660
FLDPVLLPFG IKMPCRTPPW GYVAGIDLKT NKVVWQHRNG TLRDSMYGSS LPIPLPIKI 720
GVPSLGGPLS TAGNLGLTA SMDYYIRAYN LTGKVLWQD RLPAGAATP ITYAINKGQY 780
IVTYAGGHNS FPTRMGDDII AYALPDQK 808

SEQ ID NO: 3      moltype = DNA length = 657
FEATURE          Location/Qualifiers
misc_feature      1..657
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..657
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 3
atgacccagg acgatgcctc tctcatttct gatccaacgt cttccagaca ggaggagggg 60
tctcgccgca gggatgtgct ggcaacggta acagttgcc a tgggatgtgc aggcgcgtgt 120
gctgtggcct atccttttct ggacagcctg aatgggagcg gcgccaatca tgcgggggag 180
agcgacatcc tggacgtgga tctctctacc ctcaaagccg gccagcagat tgttgtgaca 240
tggcgggggt ggcctgtggt cgtgcagaga agaacgccag aaatgctgaa aacgcttcag 300
gatccggcaa tctgcagaaa actacgagat cctgagtcct gcattttcca acaacccaaa 360
gacgcaacaa actggcatcg ttccgtcagc cccgatatcg gcgtcatgat tgggatctgt 420
acccatctcg gctgcgtgcc gactttcgac gcccgcagcg aggcagaacc tgcggggaaa 480
tacctctgcc cctgtcatgg ttcacagttt gacagtgca gccgcgccta caggaacgcc 540
cctgcaccaa acaatctgcc ggttcgcgcc gtgacaatga tttccgatac gcatctgcga 600
attggagaaa gcaaaaaacga tctcgacttt gatattgcta acatccagca gatctga 657

SEQ ID NO: 4      moltype = AA length = 218
FEATURE          Location/Qualifiers
REGION          1..218
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source            1..218
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 4
MTQDDASLIS DPTSSRQEEG SRRRDVLATV TVAMGCAGAC AVAYPFLDSL NGTRANHVGE 60
SDILDVDLST LKAGQQIVVT WRGWPVFVQR RTPEMPLKTLQ DPAILQKLRD PESCIFQQPK 120
DATNWHRSVS PDIGVMIGIC THLGCVPTFD APTQAEPAGK YLCPCHGSQF DSAGRAYRNA 180
PAPYNLPVPP VTMISDTHLR IGESKNDPDF DIANIQQI 218

SEQ ID NO: 5      moltype = DNA length = 2940
FEATURE          Location/Qualifiers
misc_feature      1..2940
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..2940
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 5
atgatcgat atcgctcgcaa tacgttactt tgcgcacggt tgctggggat gggggcgctt 60
tcgagcgttg ctcatgcggc aaccgaaagc actcacacgt cctcacatgt gcgccacaag 120
actgtgactc atggtccggg gcgttctgct gcgacgcctg caacaacagc tccggtctcg 180
gtagctcgtc cggtggcagc tccgcagttt tctgcacctg tagcgggtcaa ccggtcccat 240
gccgttcgtt ccattgatcc cggaaacgag gaaagcgttg tggtcacggg gtcggctctg 300
agcactgtca acaatcagaa cgcgaacccg gtccagatcg tgaccagcaa gcagatcgag 360
cagactggca tcaaacactc tgggtgattc ctgcagcgcc tgcggtctgt cggttcttcg 420
ggtacgacga acagccaaac caacaatacg gcgggtgttt cttgcacgga tatccgtaac 480
cttggcaaaa gccgtgttct ggttctgatt gatggcaagc gtgcagcaat tgacggctcg 540
tcaagctgct ttgatctgaa caccatcaat attcaccagg tggcgagtgt cgaatcctc 600
aaggatggtg gttctgagct gtaggtgtgt gatgccgttt ccggtgtcat caacatcaag 660

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ctcaaacaca atctggatga cgcgaacctg acggttcgtg gaggeatcac tgaccgtgga 720
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aacgtgacgg tctccggatc ttacatgacc cagagcggga tccgtcagaa cagccgtgac 840
tgggccaacc cggttgtgtc tgggtctgatt gcaccgggtg gttcgcgcac ttacggatcg 900
tccatcccga cggcggtctg ttctcact gatactgcgg ataagtgtcc caatggtgat 960
ggatcggtcc acaatttcag caagaagat cgctacaact acggtaatga tcagagcctg 1020
acgaactcct tgcaggatgc cagctgtctg ttcatgacac attacgacgt gaaccgtcat 1080
ttcacgccgt atggcaactt cctgtattcg catcgtaact ctaatacgcg gatggcgccg 1140
attccgggtg ctggttagcat ctaccgcctc actctgccgg tagccatcac cattccagga 1200
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ggatgatcgg tcagtcagac tgctaccgac acatacacgg caaagattgg cgcttcgggt 1320
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aataaccatg ttgttcacat gccgtggaag aacggtggcg atctaggtat cgcgtgggt 1680
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gaaggtaaa ccacgctgct gcacaatgca ttcccttgca aggatctgac gattgacggt 1860
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aactgggac cggttcagga ctccgcttc cgtgcaacgc ttggcacgtc ctaccgtcag 1980
cctaacgtct atgagctgta tggcggtcag tctctcggt atgcatacgc aactgacca 2040
tgccagacgg gccaggtcgg cacatatggc agtctgacgc caattgtggc ggcaaatgc 2100
gccaaagcagg ggaattacag cagtaatttc gtgtctgctt cctccagtcg ggttccgacc 2160
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ctcaagaaca tgatttcgta ctgagcagt cagtacatca tgaaccagtg ctacacgggt 2340
gcaaacacgt catattgcaa tgacattacc cgcgttgga gcacgaacca gctaaactcc 2400
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atgttctacc aaaacggtac tggcaacccc cgcgttcgtg attatgcgac cgttggtctg 2640
cagcatgggt caattggcgt cagatatatg atgagctata tgggtggtat gcgttggaac 2700
gactcggaaa ctgatgtgac ccgttcagct tcgggcccga tcaagacgac tggcatcttc 2760
tctcatgatg ttacgggtgac ttatcggttg aaaaagtggg acttcgaagc tgggtgtgaac 2820
aaactgctcg acaagaagcc tcctttgtgt tctggtggga cagacaacag cgcggctgcc 2880
ctttatggca accctttacat gggacgtaac gtcttctcgc aggcaggcgt gaactcttga 2940

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SEQ ID NO: 6          moltype = AA length = 979
FEATURE              Location/Qualifiers
REGION              1..979
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
source              1..979
                    mol_type = protein
                    organism = synthetic construct

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SEQUENCE: 6
MIVSRRLTLL CASVLGMGAL SSVAAHAATES THTSSHVRHK TVTHGPVRSR ATPATTAPVS 60
VARPVAAPQF SAPVAVNRSH AVRSIDSGTQ ESVVVTGSAL STSNQANANP VQIVTSKQIE 120
QTGINLGDY LQRLPSVGSS GTTNSQTNNI AGVSTDIRN LGKSRVLVLI DGKRAAIDGS 180
SSCFDLNTIN IHQVASVEIL KDGGSELYGA DAVSGVINIK LKHNLDDANL TVRGGITDRG 240
DQSGMISGY KGWNFHDHGRG NVTVSGSYMT QSGIRQNSRD WANPVVSGLI APGGSPTYGS 300
SIPTAGRFTI DTADNVPNGD GSFHNFSSKD RYNYGNDQSL TNSLQDATLS FDAHYDVNRH 360
FTPYGNFLYS HRNSNTQMAP IPVSGSIYPS TLPVAITIPG SAPYNSLGED ATMYKRMGEW 420
GDRVSQTATD TYTAKIGASG DITHGWKYDL SYTYGWNQVM SQTSGVGNYS KLLQSYGLSA 480
EEPGNPDSAL VYNPSICTAA AGCTLSNPFN KLSQAADYS NYTSHDHYYY QLRDLNLRIN 540
NNHVHMPWK NGGDLGIALG MEHRGEQLAY HPDALVESGQ TLTNSASYTG GGFNVTEGYL 600
EGKATLLHNA FLAKDLTIDG QGRYSSYNTP GSTKNWKASI NWAPVQDIRF RATLGTSYRQ 660
PNVYELYGQ SLGYASATDP CDSGQVGTYG SLTPIVAANC AKQGINSSNF VSASSSQVPT 720
LPGGNPKLKP ETGRITYFTG TVTPRWIPGL SASVEYWHYT LKNMISYLS QYIMNQCYTG 780
ANTSYCNDIT RVGSTNQLNS VTALYGNIGG LKTSIGIDFDL DYRIRVTSRD VLTLSNPFQ 840
LVSYLQQLNEL GSKWYNAYGR MFYQNGTGNP RVRDYATVGW QHGAIGVTYM MSYMGMRWN 900
DSETDVTRSA SGRIKTPGIF SHDVTVTYRW KKWNFEGAVN NLLDKKPPFV SGTGDNASAA 960
LYGNLYMGRN VFLQAGVNF 979

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SEQ ID NO: 7          moltype = DNA length = 843
FEATURE              Location/Qualifiers
misc_feature          1..843
                    note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source              1..843
                    mol_type = other DNA
                    organism = synthetic construct

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```

SEQUENCE: 7
atgcgcgttg ccctgatcca gatggctcct tcggcgccgacc ggagtgccaa tacccttcaa 60

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gctcagcggc tggtttcaga agctgtcaaa gctcgggaagc cagatcttgt ggtgctgect 120
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cctgatccag gcgatgctgg aggtgtactc tacgaagcgt tgagggccat ggcccgggaa 240
cataatgtct gggttcacgg tggttcaatc ggagaacttg tagggcctga gtcgggcgac 300
aagcttgcca atacttcact cgttttcaac cctgatggcg aggaatgtgg gcgttacaga 360
aaaatccatc tcttcgatgt tattacaccc aatggggacg gctatcgtga aagcgataat 420
tatgtgcccc gggaagcgat cgaagtcgtc gatattgatg gcgtcccaac cggcctcgcg 480
atgtgctatg atttgaggtt tgctgagctg tttcttgac ttcgggctgc ggatgttgag 540
atgattgttc tgcccgcagc gtttacgcag caaacgggtg aagctcactg ggacattctt 600
gtccgtgctc gcgctattga gtctcagacg tgggtgatag cgtgtggaac aacgggctgg 660
catgtcgatg ggcaaggcaa tcagcgccag acctatggcc attccatgat cgtcagccca 720
tggggcgagg ttgttcttca attgggtagt gaagaaggct ggggggtggc tgatcttgat 780
atggatgagg ttcgacaggt gcgggagaga atgcctgtgc agataaacag gcggctgatt 840
tga 843

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SEQ ID NO: 8          moltype = AA length = 280
FEATURE              Location/Qualifiers
REGION               1..280
                     note = Description of Artificial Sequence: Synthetic
                        polypeptide
source               1..280
                     mol_type = protein
                     organism = synthetic construct

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SEQUENCE: 8
MRVALIQMAP  SADR SANILQ  AQRLVSEAVK  ARKPDLVVLP  EIWSCLGGSA  ATKQANAELL  60
PDPGDAGGVL  YEALRAMARE  HNVVWHGCSI  GELVGPESGD  KLANTSLVFN  PDGEECGRYR  120
KIHLFDVITP  NGDGYRESDN  YVPGAIEVV  DIDGVPTGLA  ICYDLRFAEL  FLALRAADVE  180
MIVLPAAFTQ  QTGEAHWDIL  VRARAIESQT  WVIACGTTGW  HVDGQGNQRQ  TYGHSIMIVSP  240
WGEVVLQLGS  BEGWGVADLD  MDEVQRVRE  MPVQINRRLI  280

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SEQ ID NO: 9          moltype = DNA length = 939
FEATURE              Location/Qualifiers
misc_feature         1..939
                     note = Description of Artificial Sequence: Synthetic
                        polynucleotide
source               1..939
                     mol_type = other DNA
                     organism = synthetic construct

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```

SEQUENCE: 9
atgatgaaag caggaccgat gaaaaaactc tggcgatata tcccagcgtt gccggcgctg 60
atgctatcgg gttgcacggt tgatctgctt cagccgcgcg gtcctgatcg agaaatgaac 120
cgcgacgtta tgggtggcaga atttgccatc atgatggcga ttgtggttcc gacctgtatc 180
gcaacgcctt attttgcttg gaagtatcgc gcttccaata cccaggccga atatctgcgc 240
acctgggata actcaacgaa gattgagtat gtcctctggg gcgtccctgc tctgatcatt 300
attgcccttg gcgcgatcag ctgggtacagc acctatgctt atgacctgta ccgcccgcctc 360
cagacggctg acaacgtcaa gcgcgtgaac gttcagggtg tctctctcga ctggaaatgg 420
ctgttcctat atccggatct ggggatcgcc acgatcaacc agctggatgt gccacgaac 480
acgcgcgtga acctccagat cactctgac actgtcatga cgtcgttctt catcccgcgt 540
ctgggatcaa tgatctactc catgccgggt cagcagacac agctgcatct tcttgcaact 600
gagtcgggtg actatctggg tgaagcttcc cagttcagtg gtcgcggttt ctctgacatg 660
aagttccgca cctccgccat ggcacctgaa gaattcagcg cctgggtcga gaaggtgaag 720
agcggcagcg aaaaacctcga tgacacgact tatccgaagt acgcgcgccg gcagggaagct 780
gcgcgggttc agtatttcgc gcatgtccag ccggatctct tcgacggcat cgtcgccaag 840
tacaacaatg gcgatgaggt tgagaagacg acgggcaagg tcatgcatat gcagtcgcgt 900
tccagcgtcg caccgtccga cactggcatg aaggaataa 939

```

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SEQ ID NO: 10         moltype = AA length = 312
FEATURE              Location/Qualifiers
REGION               1..312
                     note = Description of Artificial Sequence: Synthetic
                        polypeptide
source               1..312
                     mol_type = protein
                     organism = synthetic construct

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SEQUENCE: 10
MMKAGPMKKL  WRYLPALPAL  MLSGCTVDLL  QPRGPIAEMN  RDVMVAEFAI  MMAIVVPTCI  60
ATLYPAWKYR  ASNTQAEYLP  TWDHSTKIEY  VIWGPALII  IALGAISWYS  THAYDPYRPL  120
QTADNVKPLN  VQVVSLLDWK  LFIYPLDLIA  TINQLDVPTN  TPLNFQITSD  TVMTSFFIPR  180
LGSMIYMPG  QQTQLHLLAT  ESGDYLGEAS  QFSGRGFSDM  KFRTLMAPE  EFSAWVEKVK  240
SGSENLDTT  YPKYAAPQEA  APVQYFAHVQ  PDLFDGIVAK  YNNGMMVEKT  TGKVMHMQSA  300
SSAAPSDTGM  KE  312

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SEQ ID NO: 11         moltype = DNA length = 20
FEATURE              Location/Qualifiers
misc_feature         1..20
                     note = Description of Artificial Sequence: Synthetic primer

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source                1..20
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 11
aacaagccag ggatgtaacg                                20

SEQ ID NO: 12        moltype = DNA length = 20
FEATURE              Location/Qualifiers
misc_feature          1..20
                      note = Description of Artificial Sequence: Synthetic primer
source                1..20
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 12
cagcaacacc ttcttcacga                                20

SEQ ID NO: 13        moltype = DNA length = 60
FEATURE              Location/Qualifiers
misc_feature          1..60
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source                1..60
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 13
cagctgacgc ttcgtccgaa gaaccgcgtc tccgatgccg atatctgggg tggcacaatc 60

SEQ ID NO: 14        moltype = DNA length = 59
FEATURE              Location/Qualifiers
misc_feature          1..59
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source                1..59
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 14
ctacatcccg accaagacgg gcgatatctt tgtcctcgac cgccgtacgg gcaaggaag 59

SEQ ID NO: 15        moltype = DNA length = 59
FEATURE              Location/Qualifiers
misc_feature          1..59
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source                1..59
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 15
agcaccacag cctacgaccc gtaccgcccc cttcagacgg ctgacaacgt caagccctg 59

SEQ ID NO: 16        moltype = DNA length = 60
FEATURE              Location/Qualifiers
misc_feature          1..60
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source                1..60
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 16
cgcaccaaca aggtggtctg gcagcatcgt aacggtacgc tccgtgactc catgcacggc 60

SEQ ID NO: 17        moltype = DNA length = 60
FEATURE              Location/Qualifiers
misc_feature          1..60
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source                1..60
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 17
tacaagtggg atcgattggg attttgatta tcggatccgt ctgactagca tggattccat 60

SEQ ID NO: 18        moltype = DNA length = 37
FEATURE              Location/Qualifiers
misc_feature          1..37
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide

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source          1..37
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 18
catcggcgctc atgatcgcca tctgcacgca tctgggc          37

SEQ ID NO: 19      moltype = DNA length = 59
FEATURE           Location/Qualifiers
misc_feature       1..59
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

source            1..59
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 19
cgatgcagaa accgggcaga agtgttcgg ctttggcaac gacggcgaac tggaacctg  59

SEQ ID NO: 20      moltype = DNA length = 40
FEATURE           Location/Qualifiers
misc_feature       1..40
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

source            1..40
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 20
tcttaatgcg cttctatgtg tttgtccgaa ggtcaagtgc          40

SEQ ID NO: 21      moltype = DNA length = 47
FEATURE           Location/Qualifiers
misc_feature       1..47
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

source            1..47
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 21
gcagtaattt cgtgtctgct tctccagtc aggttccgac cctgttt          47

SEQ ID NO: 22      moltype = DNA length = 60
FEATURE           Location/Qualifiers
misc_feature       1..60
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

source            1..60
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 22
tggcgatcgc tgggtccgc tgaagcagat caattcgacc aatgttcaga acctcaaggt  60

SEQ ID NO: 23      moltype = DNA length = 60
FEATURE           Location/Qualifiers
misc_feature       1..60
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

source            1..60
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 23
agactcaata gcgcgagcac ggacaagaat gtcccagtg gcttcacccg ttgctgcgt  60

SEQ ID NO: 24      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Description of Artificial Sequence: Synthetic primer

source            1..20
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 24
agagtttgat cmtgggtcag          20

SEQ ID NO: 25      moltype = DNA length = 22
FEATURE           Location/Qualifiers
misc_feature       1..22
                   note = Description of Artificial Sequence: Synthetic primer

source            1..22

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	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 25		
tacggytacc ttgttacgac tt		22
SEQ ID NO: 26	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 26		
caccagccta accagatttc		20
SEQ ID NO: 27	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
misc_feature	1..22	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 27		
gggttgatt gatgagatta gt		22
SEQ ID NO: 28	moltype = DNA length = 27	
FEATURE	Location/Qualifiers	
misc_feature	1..27	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..27	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 28		
gcttctgaca caactgtgtt cactagc		27
SEQ ID NO: 29	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..23	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 29		
caccaacttc atccacgttc acc		23
SEQ ID NO: 30	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 30		
tggtcacaat ggataggcac		20
SEQ ID NO: 31	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 31		
tctgggtttt ccaaagcaag		20
SEQ ID NO: 32	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
misc_feature	1..21	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 32		
tgaagacgga ttgccctcat t		21
SEQ ID NO: 33	moltype = DNA length = 20	

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FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 33		
gctggtgccca gtaagagctt		20
SEQ ID NO: 34	moltype = DNA length = 60	
FEATURE	Location/Qualifiers	
misc_feature	1..60	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..60	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 34		
ggcacaatct tcgaccagat gttctgcagc atctacttcc actccctgcg ctacgaaggt		60
SEQ ID NO: 35	moltype = DNA length = 60	
FEATURE	Location/Qualifiers	
misc_feature	1..60	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..60	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 35		
gggcaaggaa gtcgtcgctg ctctctgaaac accagttcca ggtgggtcag ctccgggcga		60
SEQ ID NO: 36	moltype = DNA length = 60	
FEATURE	Location/Qualifiers	
misc_feature	1..60	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..60	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 36		
gacagcgtca agccactgaa tgttcaggtg gtctctctcg actggaaatg gctgttcac		60
SEQ ID NO: 37	moltype = DNA length = 60	
FEATURE	Location/Qualifiers	
misc_feature	1..60	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..60	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 37		
atgcacggca gctccctgcc catcccgctg cctccgatca agatcggtgt tccgagcctt		60
SEQ ID NO: 38	moltype = DNA length = 60	
FEATURE	Location/Qualifiers	
misc_feature	1..60	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..60	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 38		
ggattccatc actgtcagca ataacctgca gaacctggtc aatttttccc agcagcagac		60
SEQ ID NO: 39	moltype = DNA length = 47	
FEATURE	Location/Qualifiers	
misc_feature	1..47	
	note = Description of Artificial Sequence: Synthetic oligonucleotide	
source	1..47	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 39		
tgcgtgccga ctttcgacgc cccgacgcag gcagaacctg ccgggaa		47
SEQ ID NO: 40	moltype = DNA length = 60	

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FEATURE	Location/Qualifiers
misc_feature	1..60
	note = Description of Artificial Sequence: Synthetic
	oligonucleotide
source	1..60
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 40	
	gaactggaac tgcgcgagcc gaaccagcct tacgtcacgc caggcatgta tgagccgacg 60
SEQ ID NO: 41	
	moltype = DNA length = 58
FEATURE	Location/Qualifiers
misc_feature	1..58
	note = Description of Artificial Sequence: Synthetic
	oligonucleotide
source	1..58
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 41	
	aggtcaagtg ctgcaccaaa taacagctct aggaacatca tgagcacatc ttcccggc 58
SEQ ID NO: 42	
	moltype = DNA length = 51
FEATURE	Location/Qualifiers
misc_feature	1..51
	note = Description of Artificial Sequence: Synthetic
	oligonucleotide
source	1..51
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 42	
	ggtggtaata caaagctgcg tccagaagca gggcgtacct acacctttgg t 51
SEQ ID NO: 43	
	moltype = DNA length = 60
FEATURE	Location/Qualifiers
misc_feature	1..60
	note = Description of Artificial Sequence: Synthetic
	oligonucleotide
source	1..60
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 43	
	cctcaagtg gcatggcaca tccacaccaa ggatctgatg ggaccgaatg atccgggcga 60
SEQ ID NO: 44	
	moltype = DNA length = 45
FEATURE	Location/Qualifiers
misc_feature	1..45
	note = Description of Artificial Sequence: Synthetic
	oligonucleotide
source	1..45
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 44	
	aaacgctgcg ggcagaacaa tcattctcaac atccgcagcc cgaag 45

1-95. (canceled)

96. A method of treating dementia, the method comprising:

administering to a subject at risk of or suffering from dementia, a therapeutically effective amount of a composition, thereby reducing the severity of at least one symptom of dementia compared to a subject without the administration, and

wherein the composition comprises:

a measured amount of a bacterial species that comprises and expresses nucleic acid sequences encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction derived therefrom,

wherein the extract or fraction of the bacterial species comprises membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II expressed from the nucleic acid sequences, wherein the bacterial species is from the Acetobacteriaceae family.

97. The method of claim 96, wherein the bacterium is *Gluconobacter* spp., *Acetobacter* spp., *Gluconoacetobacter* spp., *Acidomonas* spp., *Ameyamaea* spp., *Asaia* spp., *Granulibacter* spp., *Kozakia* spp., *Neoasaia* spp., *Neokomagataea* spp., *Saccharibacter* spp., *Swaminathania* spp., or *Tanticharoenia* spp.

98. The method of claim 96, wherein one or more of the nucleic acid sequences are exogenous nucleic acid sequences.

99. The method of claim **96**, wherein the composition further comprises one or more added bacterial metabolites selected from the group consisting of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

100. The method of claim **96**, wherein the bacterial species expresses and/or produces one or more added bacterial metabolites selected from the group consisting of gluconic acid, 2-keto-gluconic acid, 5-keto-gluconic acid, and 2,5-diketo-D-gluconic acid.

101. The method of claim **96**, wherein the composition is formulated as a food, a beverage, a feed composition, a probiotic, a nutritional supplement, or a pharmaceutical composition.

102. The method of claim **96**, wherein the composition further comprises a prebiotic.

103. The method of claim **96**, wherein the composition further comprises a pharmaceutically acceptable carrier.

104. The method of claim **96**, wherein the step of administering comprises orally administering the composition.

105. The method of claim **103**, wherein the composition is formulated for oral administration.

106. The method of claim **104**, wherein the composition is an enteric-coated formulation.

107. The method of claim **96**, wherein the subject is human.

108. The method of claim **96**, wherein the administering increases one or more of expression of peroxisome proliferator-activated receptor gamma coactivator 1- α (PGC-1 α), mitochondrial transcription factor A (TFAM), AMP-activated protein kinase (AMPK) phosphorylation level, nuclear respiratory factor-2 (Nrf2) protein level, PGC-1 α mRNA level, TFAM mRNA level, mitochondrial DNA replication, mitochondrial DNA copy number (mtDNA), and expression of at least one mitochondrial β -oxidation enzyme, compared to a subject without the administration.

109. The method of claim **96**, wherein the at least one symptom is selected from the group consisting of impaired speech, visual/hearing problems, cardiovascular disease, diabetes, cognitive decline, confusion, delusion, disorientation, forgetfulness, difficulty concentrating, inability to generate new memories, inability to do simple math, inability to recognize common items, aggression, agitation, irritability, meaningless repetition of own words, personality changes, restlessness, lack of restraint, wandering, anger, apathy, general discontent, loneliness, mood swings, depression, hallucination, paranoia, loss of appetite, inability to combine muscle movements, and jumbled speech.

110. A method of treating amyotrophic lateral sclerosis (ALS), the method comprising:

administering to a subject at risk of or suffering from ALS, a therapeutically effective amount of a composition, thereby reducing the severity of at least one symptom of ALS compared to a subject without the administration, and

wherein the composition comprises:

a measured amount of a bacterial species that comprises and expresses nucleic acid sequences encoding membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II, or an extract or fraction derived therefrom,

wherein the extract or fraction of the bacterial species comprises membrane-bound PQQ-dependent glucose dehydrogenase (mGDH), ubiquinol-cytochrome c reductase iron-sulfur subunit, TonB-dependent receptor, carbon-nitrogen hydrolase, and ubiquinol oxidase subunit II expressed from the nucleic acid sequences, wherein the bacterial species is from the Acetobacteriaceae family.

* * * * *